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OF QUEENSLAND
A U S T R A L I A

Motion-aware labware grasping for chemical research laboratories

by

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Submitted for the degree of BE (Hons) as part
of the course METR4912.

06.2026

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June 9, 2026

Prof Michael Brünig
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Dear Professor Brünig,

In accordance with the requirements of the Degree of Bachelor of Engineering (Honours) in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled

“Motion-aware labware grasping for chemical research laboratories”

The thesis was performed under the supervision of Professor Ruth Knibbe and Associate Professor Jen Jen Chung. I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'B. Tran', with a stylized flourish extending to the right.

Benjamin Thanh Tran

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This thesis is composed of my original work, and contains no material previously published, written by another person, or generated by artificial intelligence (AI), except where clearly referenced and acknowledged either in the main text or the preliminary pages of this thesis. I have clearly stated the contribution of others to jointly authored works that I have included in my thesis.

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AI Model and date	Artificial Intelligence (AI) tool usage (tick all that apply)											
	Language Translation	Grammar/style/spelling	Topic exploration	Research questions	Literature review	Data analysis	Generate data	Generate text	Content Creation visual (formatting)	Content Creation (e.g. code)	Feedback	Other (provide details)
ChatGPT 4o 4/08/25 - 18/09/25			✓	✓	✓							
ChatGPT 4o & 5 25/08/25 - 10/06/26									✓	✓		

Generative artificial intelligence tools were used in portions of this thesis as part of the topic exploration and software creation. After completing an initial topic exploration and literature review, ChatGPT 4o was utilised to identify any potential gaps in the literature review or understanding of the topic. This was done only after already completing a manual analysis as to minimise the effect of potential bias being introduced. In this sense, its usage did inform the research question by providing guidance towards a valid research gap. During development, ChatGPT 4o & 5 were also used to interpret error logs, debug issues, identify sources of documentation

and answer syntactical questions. In some cases, it was also used to generate small chunks of code for the purpose of debugging or to rapidly ideate a new idea. At the end of the thesis, it was further used to format tables in LaTeX.

Acknowledgments

First, I would like to acknowledge my supervisors, Prof. Ruth Knibbe and A/Prof. Jen Jen Chung. Thank you for your constant guidance and insight throughout this project, and for all that I have learnt in the process.

Second, to my family, my partner, and my friends who have supported me through not just my thesis but also my entire studies. I have you to thank for much of what I have achieved.

Dedication

To my Doris,

None of this would be possible without your unwavering encouragement, support, and love. I am eternally grateful for all that you bring to my life.

As with everything I do, this thesis is dedicated to you.

Abstract

Chemical research laboratories pose a unique challenge to automation efforts. While commercial and clinical laboratories within life sciences can be readily automated to improve throughput, chemical research laboratories suffer from large experimental variation, poor equipment standardisation and highly dynamic environments. Labware grasping presents a significant challenge: laboratory automation is not flexible enough to handle poorly standardised labware, yet state-of-the-art grasping algorithms do not incorporate the domain constraints of chemical laboratories. Labware grasping requires side-on grasps and spill-free trajectories, both of which make it difficult to identify grasps which are feasible and trajectory-efficient. To address this gap, MALab-Grasp is presented. This novel grasping pipeline extends upon AnyGrasp, a state-of-the-art grasp planner, and aims to identify stable, feasible and trajectory-efficient grasps within laboratories. When compared to AnyGrasp and tested in simulated laboratory environments, MALab-Grasp improves grasping success by 30%, grasping stability by 59%, and identifies feasible grasps faster in 78% of scenarios. In addition, it shows an improved ability to identify trajectory-efficient grasps quickly, although the extent of this improvement is less than anticipated. Finally, MALab-Grasp shows promise in adapting its planning times to the complexity of the surrounding environment, avoiding the need for manual tuning.

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Chapter 1

Introduction

Autonomous systems have pervaded many global industries, providing revolutionary benefits in terms of scalability, precision and throughput [1]. This is not a recent phenomenon in laboratories, which contain many tasks such as pipetting [2] and cell culturing [1] which are well-suited targets of automation. Clinical and commercial laboratories – primarily in the pharmaceutical or life sciences domains – have employed automation for years, with some labs reaching nearly 30 years of complete autonomy [3]. Much of the speed and widespread nature of automation in these laboratories is attributed to conducive financial structures and highly standardised equipment and processes [1]. This widespread adoption has undoubtedly led to immeasurable improvements in large-scale drug manufacture, clinical analysis, and worker ergonomics.

The benefits of automation extend to research laboratories, and particularly those which are chemical in nature. As data-hungry and AI-driven tools for chemical analysis and research gain popularity, the research laboratories adopting them require significantly higher throughput than what is currently available [4]. Thus emerges the utility of automation, of which high throughput is its flagship strength. To compound this use case, autonomous laboratories exhibit improved efficiency, safety, reproducibility and precision, all while allowing researchers to dedicate more time on higher-level reasoning rather than manual and laborious tasks [1, 4].

However, despite this clear value proposition, automation within chemical research laboratories lags significantly behind its commercial and clinical counterparts [1, 5]. The foremost obstacle causing this is the high level of flexibility research environments demand. Automated systems built for research labs would need to be compatible with many different experimental procedures, environments, and types of labware [1]. This is in direct opposition with the typical strengths of automation, which thrives in highly structured contexts.

Of all research laboratories, the life sciences field exhibits the largest adoption of automation. Liquid dispensing and pipetting are well researched domains [2] and commercial systems are also available which automate a subset of life sciences processes [6]. In contrast, although some research papers propose autonomous systems targeted towards individual processes, there is no literature which develops autonomous systems that truly fulfil the flexible requirements of chemical research laboratories [4, 7].

A significant factor in this is the difficulty posed by the grasping of chemical labware. Unlike life sciences laboratories which make extensive use of standardised plates and vial arrays [8], chemical labware is highly unstandardised in research environments [1]. This variation in shape and size, in conjunction with its transparent and fragile nature, makes grasping of labware particularly challenging and predominately unexplored [1, 7].

Though object grasping is an established field, no algorithms exist which consider the context of chemical research laboratories. Labware must be grasped from side-on and its subsequent trajectories be spillage-free, introducing geometric constraints to the grasping problem which have yet to be explored in the literature. Furthermore, these constraints introduce regions of trajectory infeasibility and large disparities between the trajectory efficiency of grasps. Under these conditions, the notion of a path-optimal grasp exists.

This paper aims to address this vacant field of labware grasping by proposing MALab-Grasp: the Motion-Aware Laboratory Grasping pipeline. MALab-Grasp is an extension upon AnyGrasp, an existing state-of-the-art grasp planner, and is tailored towards the domain-specific requirements of chemical research laboratories. To identify high quality grasps for labware, MALab-Grasp exploits the symmetrical nature of the grasps and proposes a novel grasp scoring method. Following this, it utilises a Gaussian process-based strategy to identify path-optimal grasps in a temporally efficient manner. Finally, it proposes a trajectory planning approach with adaptive planning time to retain flexibility in simple and complex environments. This pipeline will be integrated into a robotic system to assess its performance; however, the core focus of the thesis is on the development of MALab-Grasp. The success of such a system would be a critical link in the chain towards developing total lab automation in chemical research laboratories, as labware grasping underpins all higher-level automation systems and provides the means by which autonomous subsystems can be combined.

This thesis will commence by providing further detail on the motivation of developing MALab-Grasp in Chapter 2, outlining the current state of laboratory automation

and its challenges. Subsequently, Chapter 3 will provide a background into laboratory grasping, the current state-of-the-art grasp planners and the attempts to develop motion-aware grasping algorithms. It will outline clear research gaps as to why existing grasp planners are insufficient for the labware grasping problem. Chapter 4 will provide auxiliary theory relevant to components of MALab-Grasp or the robotic system. In Chapter 5, a formal problem description is outlined, motivated by the research gaps identified. In this chapter, the aims and success criteria of the thesis are further formalised. Chapter 6 describes preliminary studies conducted to aid in the design of MALab-Grasp and motivate the inclusion of various features.

A detailed description and justification of MALab-Grasp is provided in Chapter 7, and it is integrated into a robotic system as outlined in Chapter 8. Finally, the experimental setup used to assess MALab-Grasp's performance is described in Chapter 9 and the results detailed in Chapter 10. This thesis concludes with a discussion of the results in Chapter 11 and general conclusions in Chapter 12.

Chapter 2

Motivation

This chapter outlines the key literature motivating the design a labware-grasping robot in this thesis. The motivation for automating chemical research laboratories is outlined in Section 2.1. In Section 2.2, the status of laboratory automation in fields adjacent to chemical research will be summarised. Section 2.3 will expand on this to explain why these systems struggle within chemical research laboratories and the domain-specific challenges from both a technical and a logistical perspective. In this section, the manipulation of chemical labware will be outlined as the key challenge in this domain. Finally, Section 2.4 will investigate the current literature in chemical labware handling and establish a clear research gap in labware grasping.

2.1 Benefits of Automation in Chemical Research Laboratories

There are a number of benefits chemical research laboratories can gain through the adoption of laboratory automation. Humans and robots possess fundamentally disparate strength profiles, with humans excelling at high-level reasoning and robots more suited to repetitive, structured tasks [9]. By supplementing humans in labour-intensive tasks such as sample preparation or pipetting, laboratory automation systems provide a means by which researchers can focus on high-level design and creativity [10, 4]. Moreover, laboratory automation stands to improve safety, ergonomics, precision, efficiency and reproducibility, among other factors [1, 4, 11].

In addition to improving existing processes within laboratories, automation reveals a new frontier of chemical research: data-driven chemical experimentation. Automated systems offer the opportunity to increase the throughput of laboratories by

orders of magnitude beyond what is possible in a traditional lab setting, with thousands of experiments possible per day [4]. With this level of throughput, laboratories can adopt data-driven tools for chemical analysis and exploration. One of the most common examples of this is determining the optimal environmental conditions or reagent proportions to maximise the yield of chemical reactions [1]. Holland & Davies [1] and Takahashi et al. [4] provide more comprehensive reviews on the benefits of automation in laboratories.

2.2 Laboratory Automation in Adjacent Fields

Automation has found practical success in many fields adjacent to chemical research. Industrial and clinical laboratories, particularly those in life sciences, have benefitted from autonomous systems to scale up production and throughput [1]. These systems heavily favour processes which have high repetitiveness, widely standardised equipment and known operating environments, making them well suited to single-process industrial plants or clinical labs which process samples in a highly structured way. Vaccine development, cell culturing [1] and pharmaceutical testing [5] are all examples of processes which are commonly automated today in this field. Being within life sciences, the majority of these systems are micro-plate based [8]. This standardisation of equipment and processes, coupled with more concentrated and sustained funding [1], lends itself to a reasonably well-developed automation landscape in life sciences and pharmaceutical applications [6]. In the commercial realm, products for liquid and solid dispensing are common, with a particular focus on the former due to its ubiquitous use in life sciences [2, 12].

Due to the structural and process rigidity of automation, laboratory automation exists predominately as a bespoke service offered by service providers and targeted towards specific laboratory processes [1]. Aside from bespoke solutions, several commercial products also exist, primarily taking the form of monolithic machines which can both prepare and analyse samples with high throughput. Examples of these include the Tecan Freedom, as shown in Fig. 2.1, which operates as a rigid workstation with a rail-based actuator. These systems are available for purchase and automate a range of common life sciences processes, including pipetting, autosampling, and analysis. Additionally, they provide the option to extend the base workstation with modular additions that extend functionality [6]. These systems are constrained largely to processes that use microplates, deep well plates or standard vial sizes, and ultimately act as standalone products which cannot interface with other systems [6].

Liquid dispensing has achieved a more generalised automation profile than other



Figure 2.1: A commercially available rail-based system for laboratory automation in life science laboratories [6]. It has been designed specifically to accommodate micro-plate arrays and procedures for life sciences, limiting its applicability to chemical research laboratories.

processes due to the relatively predictable and consistent nature of its operation. Kong et al. [2] provide a critical review of this profile within the life sciences field. In this field, there is a particular focus on small-volume handling at high throughput. Most liquid dispensers act as their own workstations, which solely handle liquid dispensing in a highly structured environment. To transfer samples to earlier or later parts of the process, manual transfers or conveyer belts are typically employed [2]. Alternately, for a workspace automating an entire process, many automated workstations allow modular accessories to be added to execute additional tasks [2, 13]. Given the large focus of laboratory automation on life sciences, most liquid dispensaries are designed to accommodate fixed formats of labware, such as microplate arrays [14].

2.3 Difficulties in Chemical Research Laboratory Automation

Despite the growing opportunity for automation to improve efficiency in laboratories, its widespread adoption is limited [1, 15]. The cost associated with introducing automation in laboratories can be within the millions [4], making them suitable for commercial enterprise but not research. Furthermore, the highly dynamic nature

2.3. DIFFICULTIES IN CHEMICAL RESEARCH LABORATORY AUTOMATION⁷

of research methodologies makes it difficult to predict which automated systems could be leveraged throughout the project lifecycle. Meanwhile, details of funding are typically required at project commencement [1].

Complementing the financial barrier is the need for high flexibility and adaptability in research automation [4]. Automation thrives within the industry or clinical sectors because it excels at scaling up standardised, fixed experiments. However research laboratories can have highly varied experimental procedures due to the nature of exploratory research [4]. Ultimately, this level of flexibility is not something which currently available automated systems provide as they remain largely bespoke and tailored towards a single reaction or synthesis pipeline [1, 4]. Research laboratories also demand flexibility in terms of form and an ability to adapt to a dynamically changing environment, especially if they second as a teaching space [1]. Unlike the large and highly structured nature of industrial laboratories, research labs are significantly more space-limited and still require a human-centric design [4].

The grasping and manipulation of chemical labware presents itself as one of the primary challenges in chemical research laboratory automation. Unlike clinical or commercial labs, there is very poor standardisation of labware in research [1]. Simultaneously, the demand for flexibility implies that an automated system may have to grasp objects of unknown or highly variable shape and size. This makes determining how to grasp difficult, as one cannot rely on preprogrammed grasping positions or known object geometries [7, 14]. To compound this, the fragile and transparent nature of glass makes the physical grasping and visual detection of labware challenging. Finally, the manipulation of labware must be heavily constrained to prevent spillage of labware contents [7].

Chemical laboratories are also largely unique in their requirements for labware grasping. Typically, life sciences laboratories utilise standardised vial or microplate arrays, even in research. This provides structure by which systems can grasp entire plates of samples, often avoiding grasping labware directly. When individual sample-level grasping is required, systems often employ a pose detection algorithm for the overall plate of samples. They then exploit the known layout of the plates and geometries of each vial to preprogramme grasping positions. These simplifications have led to the commercial products seen in Fig. 2.1. However, they also restrict these systems from being transferable to chemical research laboratories, which do not use the same plate-based labware. This culminates in a large research gap surrounding the handling of labware in chemical research environments.



Figure 2.2: Constrained labware manipulation by Yoshikawa et al. [7]. Orientation constraints are applied to the trajectories to prevent unintended spillage, however the position at which to grasp labware is presumed to be preprogrammed.

2.4 Current Literature in Labware Grasping

There is a limited body of work within chemical labware grasping. This section will outline the prevailing literature currently available.

2.4.1 Spill-Free Trajectories

Yoshikawa et al. [7] investigate the spill-free requirements of labware manipulation in their paper. They propose an autonomous system, shown in Fig. 2.2, for executing experiments in chemical research laboratories using a 7-degree of freedom Franka robot. To generate spill-free trajectories, orientation constraints are applied to the trajectory of the robot to ensure labware stays upright. They note that the constraining of the robot's acceleration and velocity is not necessary as labware is not typically filled to full capacity. To generate trajectories satisfying these constrained, probabilistic roadmaps (PRM) are used as the primary motion planner, with an additional degree of freedom added to increase the probability of success [7]. Yuan et. al [16] provide a different perspective, asserting that trajectories must be smooth to prevent spillage. To do so, they generate trajectories using RRT* and then apply low-pass filtering and cubic B-splines to smooth the path [16].

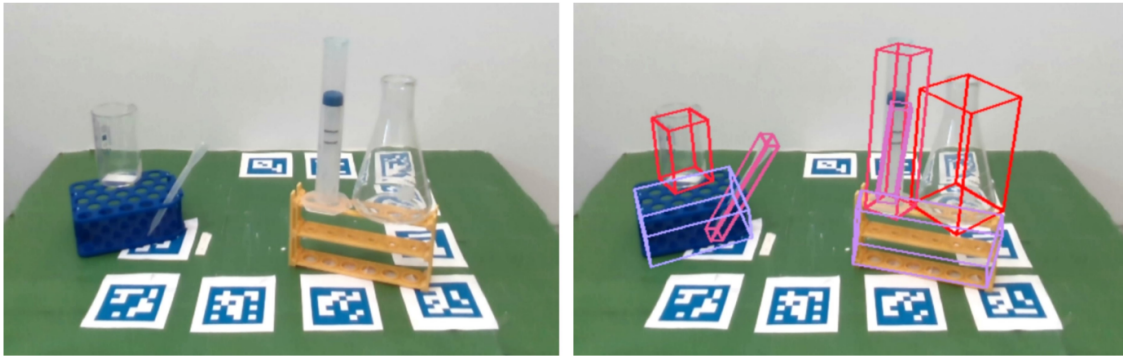


Figure 2.3: Transparent object pose detection [17].

2.4.2 Labware Detection

Some literature also exists surrounding the visual detection of labware. Transparent object detection is an emerging field of literature, with traditional vision systems struggling with the diffractive nature of transparent objects and their lack of colour and texture [17, 18, 19]. Makarova et al. [17] propose a system for detecting transparent labware, using a RealSense D435 camera and segmenting transparent regions via identifying abnormalities in the depth image. Following this, they use the CNN-based Ove6D network to identify object poses. They claim this method to be robust to occlusions, computationally efficient, and requiring of no model retraining. As an additional task, they identify the neck of each transparent object to facilitate easier grasping [17]. In a similar manner, Eppel et al. [20] present multiple deep learning methods by which to detect chemical labware.

As an alternative to transparent object detection, Yoshikawa et al. [7] rely on attaching fiducial markers to each piece of labware, with each marker mapping uniquely to an object with known geometry. This avoids the issue of transparency entirely, however requires that all labware has a marker attached to it and a known geometry. Thus, this solution is limited in terms of scalability and flexibility to adapt to new environments, where labware geometry may be unknown [7]. In their paper, Yoshikawa et al. [7] emphasise the need for further research into fiducial-free labware detection systems.

2.4.3 Adaptive Gripping for Labware

There is also a body of work which investigates the application of adaptive grippers towards labware handling. Traditional rigid graspers struggle to effectively handle fragile or irregularly shaped objects [21], leading to the emerging field of adaptive grasping. These grippers have inbuilt compliance which can reduce the force imparted



Figure 2.4: Cable-based adaptive gripper for labware grasping [24].

on target objects, while also being able to deform to a wide range of shapes [22, 23]. Thus, they have shown potential as flexible laboratory automation tools that do not depend on large-scale standardisation of labware [1].

Manes et. al [24] develop an adaptive gripper that exploits the compliance of a cable to grasp various types of chemical labware, as shown in Fig. 2.4. This approach demonstrated a high success rate of 99.2%, able to successfully grasp labware of variable shape and size without explicitly detecting its geometry. However, there are several limitations of a method such as this. First, the mechanism must first encircle each object from above, which may not be possible in some laboratory setups. Second, the niche nature of the gripper mechanism limits its versatility as it cannot grasp large objects such as trays [24]. Thus, a laboratory automation system may have to often swap this end effector with a more general gripper, increasing the complexity of the system. Third, the vertical position at which labware is grasped is preprogrammed.

Li et al. [25] and Cáceres-Suazo & Reyes-Duke [26] both implement a different paradigm of soft grippers and test their performance on labware. Both use a three-fingered gripper with a top-down approach, as shown Fig. 2.5, with the grippers able to deform around objects of variable shape. However, the top-down approach presents a severe drawback in chemical labware grasping. These methods introduce the possibility of cross-contamination between labware, as liquid residues on the rim of one object may be accidentally transferred to another via gripping. To compound this, common actions such as pipetting or pouring, shown in Fig. 2.2, would not be possible with this approach. This identifies a key constraint of grasping in laboratories: labware must be grasped from side-on.

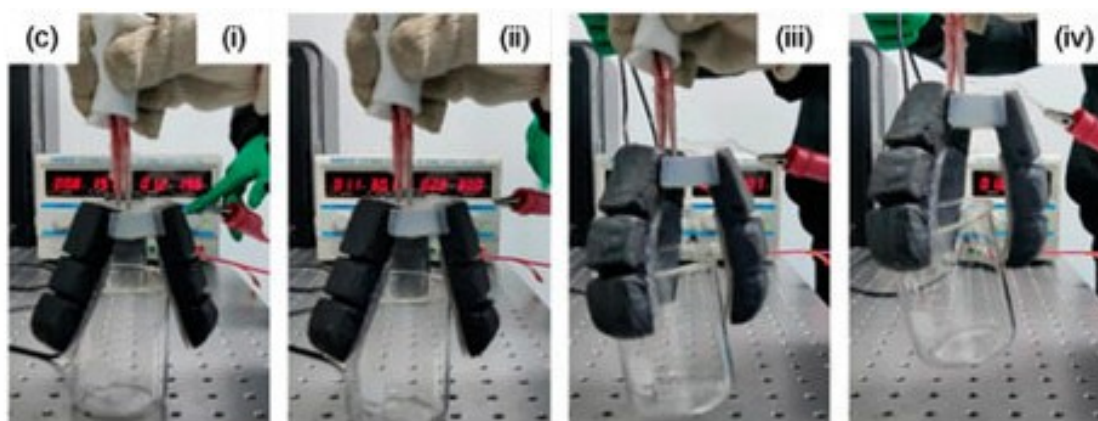


Figure 2.5: Testing of a soft gripper on a glass beaker [25]. In chemical laboratories, from-above grasping may introduce contaminants and limits what other actions can be performed during manipulation.

2.4.4 Labware Grasping

Both Yoshikawa et al. [7] and Manes et. al [24] ultimately rely on the pre-programming of grasp positions and known labware geometries to determine how to grasp objects. This fundamentally opposes the need for chemical research laboratories to be flexible and handle unstandardised labware. Although this is a key requirement of the domain, no literature as of yet proposes a method for autonomously grasping chemical labware of unknown geometry. This gives rise to a perspective shift towards to field of grasping algorithms in general, which are well-suited to grasping objects of variable shape. Chapter 4 will explore the current literature in this field and outline why the need for flexibility, orientation-constrained trajectories, and side-on grasping poses a unique challenge for grasping algorithms.

Chapter 3

Grasping Algorithms

Motivated by the lack of research into autonomous grasping robots, this chapter covers the pertinent theory and background literature in grasping algorithms. The grasping problem is first introduced, followed by a literature review of the currently available grasping algorithms. Within this literature, AnyGrasp will be highlighted as a key state-of-the-art algorithm.

After presenting AnyGrasp, its limitations when applied to the laboratory grasping problem will be identified. Subsequently the motivation for motion-aware grasping planners will be outlined and the current literature in this field summarised.

3.1 The Grasping Problem

Rather than relying on pre-programmed goals, grasping algorithms attempt to learn how to grasp objects without prior information about their geometry. Grasping algorithms can be categorised in many different ways: among others, analytic versus learning-based, 6-degree-of-freedom (DOF) versus 4-DOF, and point cloud-based versus image-based [27].

Ultimately, all grasping algorithms result in a grasp g , or set of candidate grasps $\mathcal{G}$, by which objects can be grasped with a robotic gripper. This thesis will assume that the end-effector is a parallel-jaw gripper or an end-effector with similar action; multi-fingered grippers exist but they are considered out of scope.

A grasp is defined by a 6-DOF pose $(x, y, z, \phi, \theta, \psi)$ locating it in the world, as shown in Figure 3.1. Its local coordinate system is comprised of an *approach axis* X upon which the robotic gripper approaches the object, a *closing axis* Y along which the fingers of a parallel-jaw gripper close to grasp the object, and a *normal axis* Z of

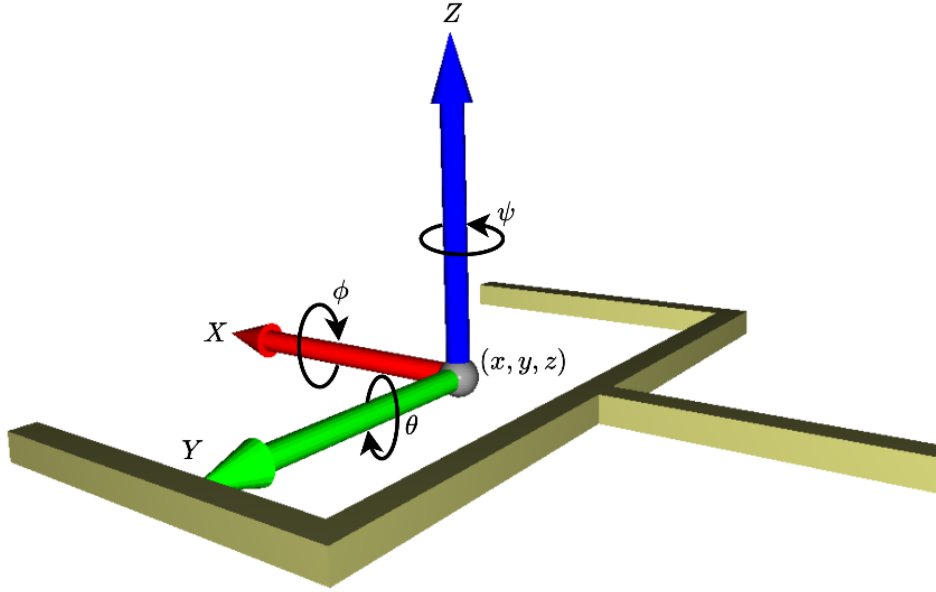


Figure 3.1: Grasps are parametrised by a translation (x, y, z) from the origin and an intrinsic rotation (ϕ, θ, ψ) about the roll, pitch and yaw axes (X, Y, Z) . X is denoted as the *approach axis*, Y the *closing axis* and Z the *normal axis*.

which is perpendicular to both. The roll, pitch and yaw of the grasp are then defined as the rotations about these intrinsic axes respectively. The grasp point, (x, y, z) , is defined as the midpoint between each finger, which contact the target object at the contact points. Newbury et al. [27] present an equivalent parametrisation of a grasp using its approach vector, grasp point and grasp orientation ϕ . This parametrisation is also commonly seen in the literature.

Some papers simplify this formulation into 4-DOF, or top-down, grasps by fixing the approach axis to be parallel to the vertical. This provides dimensionality reduction that can reduce the computational cost of many algorithms [27]. Furthermore, a completely top-down approach simplifies the vision system, often requiring only a 2D image of the scene rather than a full 3D scan. Further background into the different architectures of vision systems is provided in Chapter 4.

Lenz et al. [28] leverage both of these simplifications when proposing their grasping algorithm, demonstrating a grasp success rate of 84%-94%. However, despite their prevalence in the literature, the top-down nature of 4-DOF grasps violate the requirements of labware grasping, which requires side-on grasps. Thus, the grasping problem defined in this thesis assumes full 6-DOF grasping.

The success of the grasping problem is captured primarily by the *grasp success rate* [27]. This quantifies the proportion of attempted grasps which were able to “successfully” grasp the desired object. Although this definition of successful grasping is inconsistent throughout the literature, it typically refers to the ability to lift the object into the air or perform a specific manipulation task without dropping it. Other metrics for the grasping problem include the time required to identify grasps or the degree to which a set of candidate grasps provides coverage over a set of ground truth grasps. Occasionally, mean picks per hour (MPPH) is used to quantify the overall speed of a grasping system, considering grasping time, trajectory efficiency, and planning time [27]. However, these are much less commonly reported than grasp success rate.

3.2 Analytic Grasping Algorithms

Analytic, or geometric, algorithms are the first of two primary classes of grasping algorithms [29]. These methods were pioneered before the rise of deep learning and rely heavily on mathematical analysis of grasping forces. Many early methods deconstructed complex objects into a series of simpler shapes, upon which effective grasping poses are known [28]. An example of this is by Miller et al. [30], who deconstructed objects into primitive blobs (typically cylinders, boxes or spheres), with each primitive having a known set of grasping positions. Bowers & Lumia [31] performed a similar method. However, they incorporated the field of fuzzy logic to allow blobs to have partial membership of multiple different primitive classes. Thus, this method was more effective in grasping shapes which exist between standard primitives. After segmenting objects into primitive shapes, analytic methods typically calculated grasps according to the principal axis of each shape.

The quality of grasps in analytic algorithms largely comes from a mathematical analysis of force closure and fixturing [32]. In these methods, the contact points of a grasp are placed on the surface of the target object such that it maximises the stability of the grasp. However, this analysis is rather complex and, since analytic methods are not the focus of this thesis, they will not be described here.

When comparing with learning-based grasping algorithms (detailed in the following section), Rameshbabu et al. [33] assert that analytic methods perform better than learning-based methods. These results are based on a comparison of two learning-based and two analytic methods, using relatively simple test scenarios. In these scenarios, the analytic algorithms yielded higher success rates, however they acknowledge the potential effect of training data on the reliability of this result.

They note that learning-based methods have a higher performance potential than analytic methods but are strongly affected by the quality of training data used, especially by the variety of object shape and lighting conditions. In contrast, they found that analytic algorithms do not require large quantities of data but exhibit lower performance in cluttered environments and for uneven surfaces [33].

The observations made in this study are important considerations but do not consider cutting-edge learning-based algorithms, which provide significant improvements over earlier models. In particular, Rameshbabu et al. [33] perform their experiments in environments with lighting conditions very distinct to the data used to train the learning-based models. This is likely the primary factor in analytic algorithms providing higher performance.

To correct effects such as these, Pinto & Gupta [34] and Mahler et al. [35] both develop large-scale, openly available datasets that include adversarial conditions such as poor lighting. These datasets have improved the overall robustness of learning-based models since the time of this study. In the current day, learning-based algorithms exhibit greater robustness to variation and higher accuracy than analytic methods [27]. Ultimately, analytic grasping methods are no longer considered state-of-the-art, especially in environments such as laboratories where irregular and variable shapes exist. Therefore, they are not given much consideration in this thesis.

3.3 Learning-Based Grasping Algorithms

In comparison to analytic methods, learning-based grasping algorithms utilise machine learning techniques to generate candidate grasps and/or quantify their quality. Newbury et al. [27] present a review of these algorithms, classifying them predominantly into sampling-based and end-to-end methods.

A large portion of the review focuses on sampling-based algorithms. In these algorithms, grasps are first synthesised by a sampling-based process to form a set of candidate grasps. Subsequently, these grasps are then given a score to reflect their grasp quality. This scoring process is typically done according to a trained machine learning model [27], in comparison with the mathematical analysis conducted by analytic grasping algorithms.

The input to sampling-based grasping algorithms is typically a point cloud representation of the surface to be grasped. The requirements for this point cloud vary between different algorithms, with some requiring only the point cloud from a single view and others combining multiple viewpoints to employ grasp detection on a fully

3D point cloud. Other works leverage neural network architectures to fill in occluded regions of a partial point cloud [27]. Points from the cloud are sampled, usually according to a uniform distribution, to act as the grasp points from which to generate grasps. Some papers focus on improving this sampling method, either by sampling at equidistant points, according to a Gaussian distribution, or sampling in the latent space learnt by an autoencoder [27]. From this sampling step, grasps are iteratively built by successively defining their different parameters.

The approach axes of grasps are usually determined as the surface normal of the target object at the given grasp point [27]. This provides orthogonal grasping, which is the most intuitive and likely provides the highest change of guaranteeing force closure and grasp success. However, some methods also sample the approach axis independently of each grasp point. Finally, the grasp rotation ϕ is sampled using a grid sampling method. To ensure that the resulting grasp does not collide with the target object, most algorithms vary the grasp depth. This is defined as the offset of the grasp point from the sampled point along the approach axis. It is common to gradually increase the grasp depth using discrete steps until no collisions are detected [27].

After generating candidate grasps, each grasp is then passed into a machine learning network to calculate the grasp score [27]. This metric often explicitly or implicitly incorporates notions of stability [27], with higher grasp scores aiming to indicate more successful or effective grasps. Because many algorithms use similar sampling methods, most research is concentrated on improving these scoring networks. Following sections outlining the prominent grasping algorithms will describe various scoring networks. This section will focus on models which perform well in clutter, acknowledging that grasping in laboratories will rarely be a clutter-free environment.

In contrast, end-to-end algorithms combine this process of grasp generation and grasp scoring into one entire system [27]. These can take the form of deep reinforcement learning networks or direct regression, wherein grasps are generated and scored in a singular generated network. However, because of the exponential complexity of an end-to-end model, most methods perform dimensionality reduction to reduce the 6-DOF pose into a top-down grasp [27]. Thus, this class of methods is not considered in this thesis.

3.4 Prominent Grasping Algorithms

Mahler et al. [35] propose the grasp quality CNN (GQ-CNN), tested with a parallel-jaw gripper and a single-view RGBD image. With this set up, they achieve a

94% grasp accuracy on novel objects, albeit while considering only 4-DOF grasps. However, their main contribution to the grasping literature is the DexNet dataset, which contains 6.7 million data points on which algorithms can be trained. This milestone was later surpassed by Pinto & Gupta [34], who created a dataset surpassing 1 billion data points. Many algorithms use these datasets to ensure that their scoring models are robust to environmental variation.

Gou et al. [36] process RGB and depth images separately from a single RGBD view to build an end-to-end 6-DOF grasp synthesis algorithm. They name this RGBD-Grasp, exhibiting a grasp success rate of 91.1% in cluttered environments. They highlight the fusion of both images as a means by which their method becomes more robust to sensor noise.

Ten Pas et al. [37] is often used as a benchmark for newer models because it has a focus on practical grasping in cluttered scenarios. Though graspers can perform within the range of 75 – 95% with isolated or lightly cluttered objects [37], many models experience severe performance drop-off as the complexity of the scene increases. Ten Pas et al. [37], on the other hand, maintain a success rate of 93% even in these environments. Much like other methods, they first preprocess the point cloud by performing denoising, voxelisation, and cropping, and then sample points from the point cloud and calculate each point’s surface normal. To retain smoothness, they consider the ball of points around each point when doing this. At each point, they generate grasps from a grid sampling method, varying the translation along the y axis and the rotation about the z axis. Grasps which collide with the point cloud, or which do not contain a single point between their antipodal points, are considered invalid and are discarded. They also discuss the difference between rotation about the x and the z axis, and conclude that, for the objects that they tested on, the preferred option is about the z axis. Finally, grasp depth is adjusted until it no longer collides with the object. A 4-layer trained CNN is then used to predict the binary success of each grasp [37].

GG-CNN (Generative Grasping CNN) [38] is a prominent end-to-end algorithm which displays good performance, especially in cluttered environments. It exhibits a real-world grasping accuracy of 83% for simple environments and 81% for cluttered environments. Overall, the algorithm performs well with low computational cost, however underperforms compared to DexNet. Several extensions have been made to the GG-CNN algorithm, such as by Zhao et al. [39], to make the model more lightweight.

3.5 AnyGrasp

Fang et. al [40] present AnyGrasp, a deep learning-based algorithm designed to identify candidate grasping poses for robotic systems. Their algorithm is targeted towards use with a parallel gripper and generates a dense set of 7-degree-of-freedom poses – six degrees of freedom belonging to the positioning of the gripper and an additional degree of freedom dictating the extent to which the gripper opens.

AnyGrasp is an extension based on the Graspness-Based Sampling Net (GSNet) proposed by Wang et al. [41]. Whereas many algorithms focus on improving grasp scoring, GSNet uses a graspness metric to only retain sampled grasps which exhibit a high probability of representing useful grasps. This typically represents less than 10% of all sampled grasps, so GSNet offers faster grasp generation and denser sampling capabilities by sampling more selectively. In their paper, Wang et al. [41] use a lookahead search and force-analytic models to predict the probability that a grasp will be successful, and use this to drive their sampling process. They also integrate objectness masks to quantify the likelihood that a point will belong to a meaningful object rather than noise [41].

These two metrics are used to form a probability map from which points are sampled. Using a graspable furthest point sampling algorithm, points are selected which exhibit high graspness and are far away from already sampled points. Instead of using surface normals, GSNet uses a multilayer perceptron (MLP) to find the best approach vector for each point out of a candidate set of 300 vectors, again incorporating graspness in this decision process. Approach vectors which lie very close to each other, even for different grasp points, are then grouped cylindrically [41].

Candidate grasps are generated for this approach vector for every combination of 12 different rotations about the grasp’s z-axis and four different grasp depths. Each grasp then calculates the required gripper opening width to avoid collisions. Because of the graspness metric incorporated into sampling, there is no need to further score the grasps [41]. The architecture of GSNet is depicted in Fig. 3.2.

AnyGrasp uses GSNet to synthesise candidate grasps with a few modifications. First, an additional grasp depth is added to improve performance for small objects. Second, grasp candidates undergo collision detection and recentering during post-processing. The purpose of this recentering is to ensure that each finger of the gripper grasps at the same time, prevent any lateral sliding. Third, a stable score is computed for each grasp, defined by the perpendicular distance between the grasp pose and the centre of gravity vector of the target object. The final grasp score is

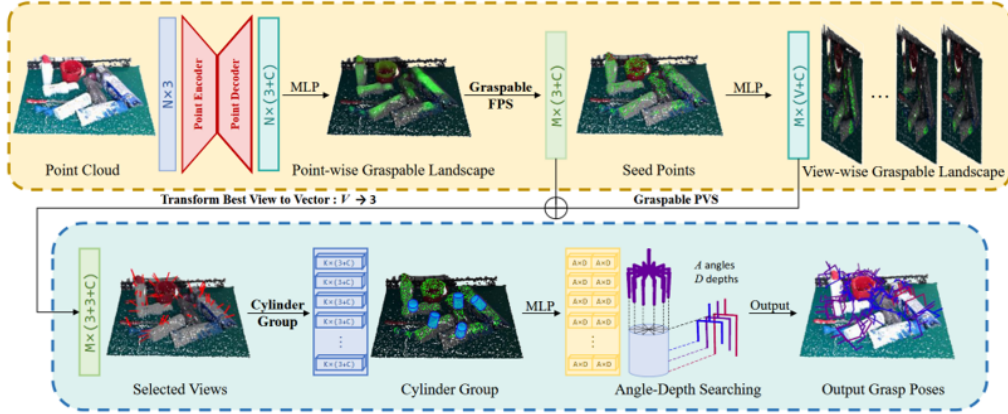


Figure 3.2: The architecture of GSNet, which employs the concept of graspness to sample more selectively than typical grasping algorithms [41].



Figure 3.3: The adversarial scenario defined by AnyGrasp in its benchmarking studies [40]. Although this scenario is adversarial in terms of grasping success, it does not reflect the complex nature of side-on grasping in cluttered scenes.

calculated by multiplying it with this stable score, such that grasps further away from the centre of gravity vector are penalised. AnyGrasp also proposes a temporal association module to enable grasp tracking for dynamic scenes, but this is considered out of scope [40].

During their testing, AnyGrasp performs some additional filtering operations on their candidate grasps. They limit to gripper orientation to those which are mostly vertical, noting that their robot struggled to configure itself for horizontal grasps. In addition, they set their planning goal to be offset of the grasp by 10cm along the approach direction for easier planning. They report an average of 93.3% success rate in real-world application, helped by training the model on real-world sensor data, beating the previous benchmark of DexNet 4.0 [40]. In adversarial environments, which they define as containing objects with smooth surfaces, AnyGrasp shows a decrease in performance to 70%. This adversarial scenario is shown in Fig. 3.3.

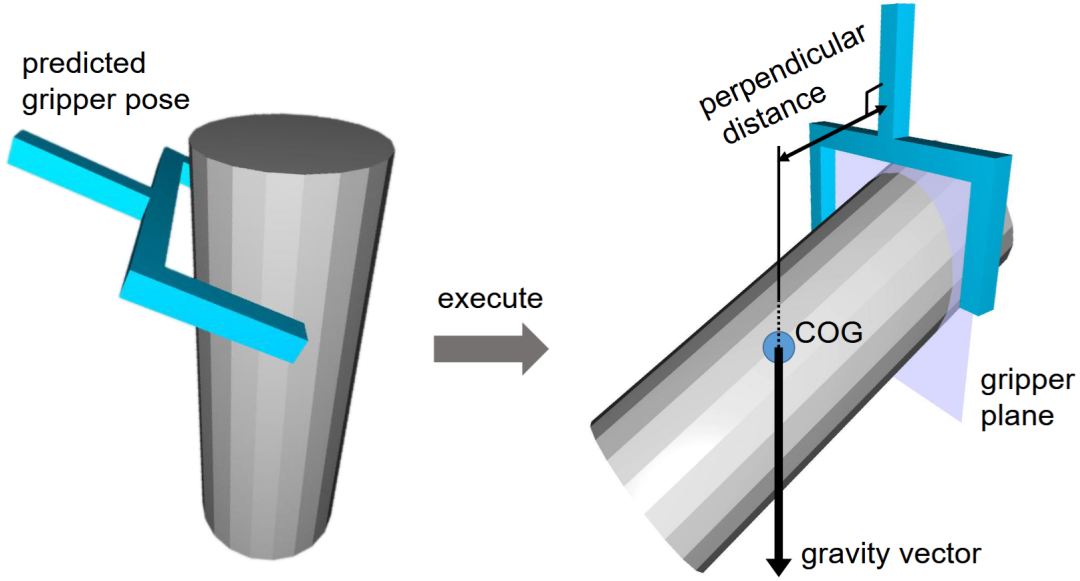


Figure 3.4: The perpendicular distance between a grasp pose and the object centre of gravity, used by AnyGrasp to define its stability score [40]. AnyGrasp assumes that the gripper will always be facing downwards during object transportation, but this is not the case in labware grasping.

3.6 Limitations of AnyGrasp for Labware Grasping

Despite its performance indicating that AnyGrasp is a state-of-the-art grasp planner, several features within its architecture may prove detrimental to the labware grasping problem.

A key contribution that AnyGrasp proposes is the inclusion of a stability score with awareness of target objects' centres of gravity (COGs). The motivation for this is inspired by studies which found that humans tend to grasp objects at the COG [40]. To incorporate this into their grasp scoring, Fang et. al calculate the perpendicular distance between each grasp and the weight vector, as shown in Fig. 3.4. They then penalise the grasp score for grasps that are far away from the COG [40]. The equation for this is shown in Eq. (3.1), using the normalised perpendicular distance as a penalty factor for the original grasp score q_0 .

$$ss_{anygrasp} = \frac{d_{\perp}}{\max d_{\perp}} \quad (3.1)$$

$$q_{anygrasp} = q_0 \cdot (1 - ss_{anygrasp})$$

In theory, the inclusion of stability-based scoring has its merits. The further away from the COG an object is grasped, the larger torque its weight generates about the gripper. As a result, these grasps may be prone to slipping. This is particularly a concern when grasping fragile objects such as labware, for which applying a stronger gripping force may not be a suitable remedy. However, AnyGrasp’s stability score explicitly makes a key assumption about the nature of the grasping problem which is not valid for labware grasping. Due to the predominately top-down nature of their benchmarks, Fang et al. [40] assume that the gripper will adopt a downfacing pose while transporting objects, as depicted in Fig. 3.4. However, in labware grasping this policy would violate the spillage-free constraint imposed. Therefore, AnyGrasp’s stability score is not an accurate representation of the labware grasping problem. In this domain, it would only serve to prioritise grasping close to the vertical centre of mass. This is also in opposition with the natural tendency to grasp labware on the neck or above the centre of mass [17]. Therefore, this feature could prove detrimental if applied to the labware grasping problem.

Second, AnyGrasp’s performance is evaluated purely in top-down scenarios, despite supporting 6-DOF grasp generation. Due to the requirements for side-on grasping in labware, the performance results posed may not be reflective of its performance in laboratories. In particular, AnyGrasp chooses its grasp depth from only five possible values: 0.5cm, 1cm, 2cm, 3cm and 4cm. This is a fairly standard process in grasping algorithms [27]. In top-down problems, this grasp depth has a minimal effect on the stability of a grasp. However, this is not the case in side-on grasping, where the depth of grasp plays a significant role in the grasp stability. Given that AnyGrasp only provides five discrete values to choose from, this may lead to difficulty identifying stable side-on grasps.

3.7 Motion-Aware Grasping

The nature of side-on and spill-free grasping presents difficulties in ensuring the feasibility of grasps. The grasping algorithms presented above are entirely decoupled from the hardware on which the grasp will be executed. Therefore, they have no means as to determine whether a grasp generated can be reached by a robotic system. In pick-and-place problem, they can also not guarantee whether it will be possible to move the object to the desired goal location given a grasp.

Existing literature tends to assume that all grasps are kinematically valid [27], with very little consideration for the efficiency of a robot which would actuate the grasping algorithm [42]. This is aided by the dominance of top-down grasping, which

exhibits a much higher rate of kinematic feasibility. Among others, Fang et al. [40] note that robotic arms struggle to configure themselves to a horizontal position, often leading to reaching robot joint limits, causing self-collisions, or colliding with the ground. Therefore, the grasps generated for side-on grasping are much more likely to be kinematically infeasible.

To compound this, the nature of side-on grasping promotes more diversity in grasp pose than top-down grasping, wherein all grasps share a common approach axis. As such, the trajectories required to grasp objects at these poses also become more diverse. This is exacerbated in complex and highly cluttered environments, where trajectories by nature are longer and many may be trajectory-infeasible. As a result, differing grasps may present very different levels of feasibility and path efficiency. Given that path efficiency may influence the throughput of chemical laboratories, there is a need for algorithms which consider grasping and motion planning jointly. The following sections will outline the current methods in this domain.

3.7.1 Reachability Awareness

One reason for which a grasp may be infeasible is if it is physically impossible for a robot to reach a grasp pose due to joint limits or self-intersection. Lou et al. [43] proposes a CNN-based network to incorporate this notion of reachability into the scoring of grasps. Using a simulated robotic arm, they create a reachability network to learn the mapping between candidate robotic arms and a reachability index. When scoring grasps, each grasp score is then multiplied by its respective reachability score.

By training the reachability network prior to operation, this algorithm allows for very fast inference. Their results indicate that this method can effectively identify feasible grasps, reporting a planning success of 96% [43]. However, the process of creating a reachability map is unique to each robot, so the network would have to be retrained for each new robot. In addition, it lacks the ability to adapt to dynamic environments or where additional obstacles exist.

Another paper proposes a new energy function for a simulated annealing-optimised grasp planner which considers both stability and reachability. However, in a similar manner, the requirement to learn the reachability space before operation limits its ability to work in generalised scenarios [44].

3.7.2 Joint Grasp and Motion Planning

Another method of motion-aware grasping used is joint grasp and motion planning. In this family of algorithms, both the grasp pose and the entire trajectory are planned simultaneously, allowing both feasibility and efficiency to be considered. Vahrenkamp et al. in [45] and [46] propose Grasp-RRT, which performs grasp pose generation, inverse kinematics and collision-free trajectory generation in a single step. RRT is used to build a biased tree from the robot’s initial position towards the target object to be grasped. For each node it determines whether it can act as a grasp pose by analytically generating a grasp from the node and then performing wrench-space analysis. By generating grasps during trajectory generation, Vahrenkamp et al. [46] argue that this method is highly efficient because only reachable grasps are generated and problem-specific constraints can be easily applied to this formulation. Fontanals et al. [47] propose a similar joint planning algorithm, with independent contact regions being exploited to guarantee force closure of grasps generated during RRT.

To extend these algorithms to pick-and-place problems, Leebron et al. [48] implement a similar strategy but with bidirectional RRTs. Two trees are grown simultaneously – one from the initial robot position and one from the desired place position – and sampling is likewise biased towards the grasp space of the object. The meeting of two trees through a valid grasp position encodes the entire pick-and-place trajectory.

However, these joint planners are limited severely in their use of analytic grasping methods rather than data-driven algorithms. By generating grasps online during the trajectory generation process, grasps must be generated analytically. Therefore, these methods cannot leverage the robustness of learning-based methods or their notions of grasp stability. Furthermore, although path-efficient grasps may be naturally found faster during RRT exploration, they do not explicitly encode any metric of path efficiency.

3.7.3 Joint Planning as an Optimisation Problem

Several papers also formulate grasp and motion planning as an optimisation problem and use trajectory optimisation techniques such as CHOMP to jointly optimise both the grasp pose and the path trajectory. Dragan et al. [49] provide foundational literature in this domain, introducing an algorithm for utilising CHOMP with a set of candidate goals rather than a fixed goal. Shao et al. [50] apply this to the grasping problem with the addition of an exploration-exploitation grasp selection process. In each iteration of CHOMP, the goal is chosen as to either explore new

grasps or exploit those which provide more optimal results. In addition to this, they incorporate a second exploitation strategy which rotates grasps about its closing axis. However, in the domain of labware grasping, where grasps must be horizontal, this is not permissible.

Wang et al. [51] provide another implementation of goal-set CHOMP in the grasping problem, alternately updating the optimal trajectory and optimal grasp across iterations. In a similar manner to Shao et al., they also perform augmentation on the original grasp set. This process is termed Configuration-Space Iterative Surface Fitting, which refines grasp according to the geometry of the robotic gripper used. Both methods show better collision avoidance and grasping success rates than non-integrated analogues.

Although CHOMP-based joint planners allow for explicit optimisation of path efficiency and smoothness, it may struggle with the orientation-constrained nature of laboratory trajectories, as this is not typically its strength [52]. Being local optimisers, they may also struggle with bad initialisations and escaping local optima, especially if different candidate grasps exhibit fundamentally different trajectories [52]. Thus, there is still a clear research gap in the motion-aware planning of grasps.

Chapter 4

Robotic Theory

This chapter outlines relevant theory in robotic systems. This is auxiliary to the main focus of this thesis but is necessary to contextualise aspects of design.

4.1 Articulated and Cartesian Robots

In this thesis, the hardware architecture of a robotic system belongs to one of two classes: cartesian robotic systems or articulated robot arms. This section will discuss the two approaches within the context of flexible laboratory systems, with an emphasis on articulated arms.

Within clinical and industrial life sciences laboratories, high-throughput processes are typically automated using cartesian robotic systems, implemented on a rigid workstation either as all-purpose tools or bespoke solutions for a particular process [2, 13]. In cartesian systems, such as Fig. 2.1, manipulators are mounted on rails which allow for planar movement around a self-contained workstation. conveyor belts are typically used to distribute samples to external workstations, after which separate systems are used for process execution. However, articulated robotic arms are emerging as viable alternatives to perform both transportation and execution within the same system [13, 53, 6]. They are used in several research papers as an alternative to automation rails. Fleischer et al. [13] use a dual-arm robot for automating pipetting actions, arguing that the use of coordinated arms (termed coordinated bimanual operation) make these robots naturally apt at replicating human tasks.

Najmabadi et al. [54] conduct a theoretical analysis of the flexibility of different architectures for laboratory automation. They quantify the flexibility of a system as being composed of functional flexibility, structural flexibility and throughput



Figure 4.1: Dual-arm pipetting system proposed by Fleischer et al. [13]

flexibility. Functional flexibility measures the capability to execute a variety of tasks, broken down into liquid-handling, transportation and interfacing with laboratory instruments. Structural flexibility refers to the physical reconfigurability of the system and how it can change its layout or physically incorporate new instruments. Finally, throughput flexibility refers to the ability for the system to parallelise tasks and is critical for high-throughput applications, as many processes have large temporal aspects [54].

Najmabadi et al. [54] use this quantification to argue that articulated robot arms possess greater theoretical flexibility than classic cartesian strategies. Arm-based strategies provide greater flexibility in all aspects except for throughput flexibility, in which cartesian systems excel with conveyor-driven transportation. The ability for a robotic arm to move freely in its surrounding space allows it to interface with many different instruments, as well as to adapt to changing environments and transport objects between distributed workstations within a limited radius. This theoretical quantification makes the role of each of these hardware architectures clear: cartesian- and conveyor-based systems excel in high-throughput cases where functional or structural flexibility is not required; robotic arms are better suited to the cases in which these requirements exist.

4.2 Vision Systems

Vision systems have significant interplay with grasping algorithms as they provide the means by which grasp planners can sense the geometry of the environment around them. Therefore, the design of the vision system can have significant ramifications on the grasping algorithm used, and vice versa.

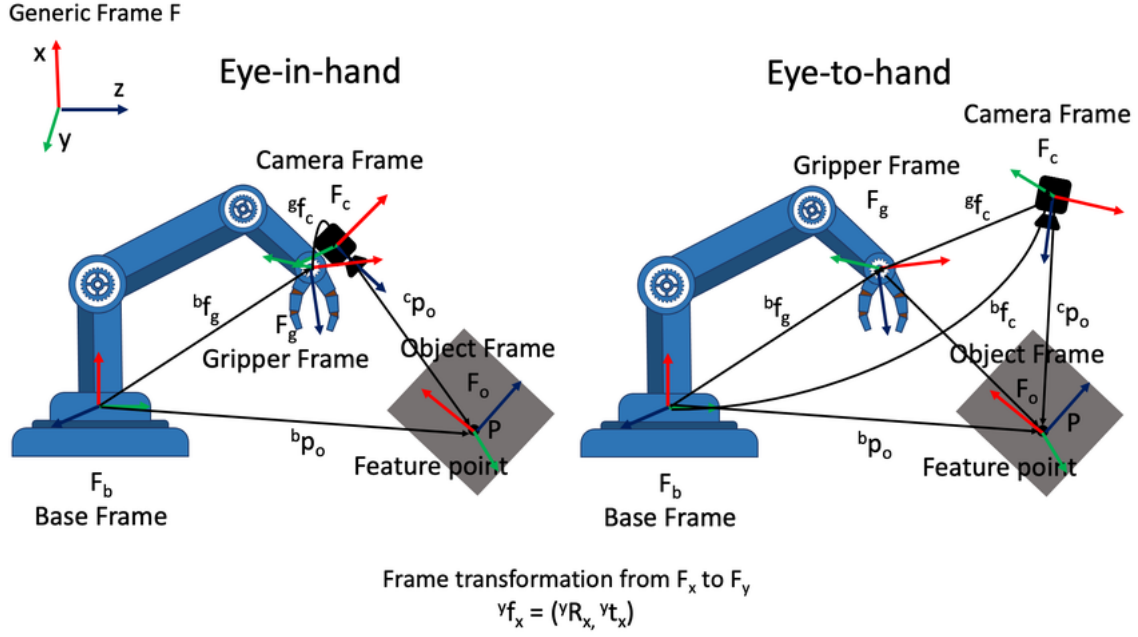


Figure 4.2: Eye-in-hand and Eye-to-hand methods [57].

4.2.1 Camera Placement

Robotic vision systems are typically classified into one of two categories: eye-in-hand methods and eye-to-hand methods [55, 56]. Systems following the eye-in-hand architecture mount their cameras rigidly to the end-effector, providing a high-resolution view of the scene at the expense of a small perception region. For robotic arm applications in particular, the eye-in-hand architecture offers an additional challenge of needing to map new areas of the environment as it moves, potentially leading to collisions with undetected obstacles. Comparatively, eye-to-hand architectures provide a more global and less precise view of the scene but require an external static or dynamic setup of a camera [55, 56]. Fig. 4.2 shows a visual comparison of these two methods.

Several papers propose hybrid architectures that use both systems in a cooperative mode. AlBeladi et al. [55] propose a hybrid vision system for object tracking which uses an eye-to-hand camera until the object is visible in the eye-in-hand camera, at which point the system switches to the latter. Other methods propose using eye-to-hand systems for positional control and eye-in-hand for orientation control [55]. Flandin et al. [56] show that such cooperative systems are mathematically viable, and especially robust when incorporating Kalman filtering or fusion with odometry.

The method of camera placement is significant because it affects the completeness and the quality of data used by the grasping algorithms. In top-down grasping,

either method works suitably well because only a single-view point cloud or image is typically needed [27]. In contrast, eye-in-hand or multi-view eye-to-hand vision systems are necessary for applications where object geometry cannot be interpreted from a single view – such as side-on grasping or grasping in cluttered environments.

4.2.2 Depth Perception

Vision systems can also be categorised according to the types of images that they sense. In a similar manner to the camera placement, some grasping algorithms are only compatible with RGD images and others with RGB-D images.

Traditional robotic vision uses classical RGB cameras that produce 2D images, and the core of this field is the problem of reconstructing a 3-dimensional scene from just 2D images. Since cameras are homographies by nature that cannot preserve depth information, images from multiple different angles are required to recover depth information, giving rise to stereo vision (or in general, multi-view) systems. As an alternative, a moving camera can also generate different perspectives, leading to structure-from-motion (SfM) problems [58]. For most robotic arm applications, the camera extrinsics (motion parameters) are measured, so the vision system can determine depth information by identifying corresponding features in different frames of the camera and then triangulating them. Especially when using extrinsics which may have small errors, robust techniques for performing this triangulation are well developed, including bundle adjustment, loop closures, epipolar geometry and RANSAC. In particular, loop closure is widely used to account for drift throughout the trajectory of a camera [58].

As an alternative to traditional vision systems, 3D sensors have been developed that can capture depth information directly. Professional laser scanning and LiDAR sensors are used industrially, but RGB-D cameras (those that capture both RGB images and depth) are both effective and less costly [59]. Rather than generating 2D images, RGB-D cameras directly generate 3-dimensional point cloud data. Point clouds from multiple perspectives can then be recovered by converting point clouds to a voxel (volumetric pixel) or surfel representation [59] and performing manual stitching of clouds. Global optimisation and refinement techniques are also common here, including RANSAC, loop closures, pose graph optimisation and sparse bundle adjustment. In these techniques, interest points can be identified to maintain local accuracy while reprojection error is evenly distributed throughout the remainder of the scene. However, vision systems for dynamic scenes remain a challenge within the literature [59].

4.3 Gaussian Processes

Gaussian processes (GPs) are the most well-recognised and widely used techniques in the family of Bayesian optimisation. Throughout a multitude of industries and applications, they have proven their excellence in optimising black-box problems [60, 61]. These black-box problems are defined as those for which an optimising agent cannot model as a function, either because of a lack of visibility into the system to be optimised or behaviour that is too complex to model mathematically. Often, these problems are also very expensive to compute, making grid-based optimisations impractical. GPs act as a surrogate model for these functions, providing an estimation that, although it may not be completely accurate to the underlying function, is cheap to compute, mathematically well-defined, and therefore conducive to being optimised [60]. Critically, GPs are highly efficient in their probing of the black boxes [60, 61], making them well-suited to optimising expensive functions.

When used in the optimisation of black-box functions, GPs employs an iterative process of two steps: surrogate modelling and black-box sampling [62]. In surrogate modelling, a GP attempts to estimate the black-box function that it must optimise. To do so, each observation made (i.e., previous evaluation of the black box) is modelled as a multivariate normal distribution [63]. The GP then estimates the function at any given point as the weighted sum of all observations, with observations being weighted higher the more they correlate with the given point.

The degree to which points correlate is defined by the kernel function and is a critical component of a GP. Fundamentally, the kernel dictates the prior assumptions about how the distance between two points on a function affects their correlation [63]. Often, the smoothness of the kernel is designed to reflect the (believed) smoothness of the black-box function. The Matérn kernel is often used to control this because changing of its hyperparameter directly controls its smoothness [63]. Other common kernels include the radial basis function, a special case of the Matérn kernel, and periodic kernels, designed for periodic spaces. In many cases, kernels are also composed together to create more complex behaviour [63].

Initially, GPs are provided with a prior belief about the mean and covariance at each observation (the Bayesian prior). To incorporate the observations into its model, the GP calculates the parameters of its *posterior* distribution, in which the mean and variance of each point in the optimisation space is updated based on the observations made. The posterior mean μ encodes the black-box estimate; the posterior variance σ^2 reflects the uncertainty associated with this prediction. For more information on the mathematical formalisation of this, Schulz et al. [63] provide a concise overview.

The decision of which point within the black-box model to evaluate is determined by an acquisition function [61, 63], whose primary responsibility is determining how to sample points in the most efficient way as to make best use of the very expensive black-box evaluation process. A common framework for determining how an acquisition function should behave is the exploration-exploitation framework. This demonstrates two competing desires: *exploiting* the areas of the optimisation problem that are already known to be good and *exploring* new areas in the hopes that they will lead to something better. A more generalised view of acquisition function presents them as methods by which to maximise the Shannon entropy or to minimise the (mathematical) regret of the system [61]. The upper confidence bound (UCB) is a standard acquisition function used, shown in Eq. (4.1) and visualised in Fig. 4.3. This method samples the point which has the highest potential value that one could be reasonably confident about. This notion of confidence is captured by the statistical confidence bound and the size of this confidence bound κ is a fundamental hyperparameter which controls the degree to which exploration is prioritised over exploitation. For small values of κ , the posterior mean $\mu(x)$ becomes the dominant factor, prioritising points which are predicted to be optimal. For large values of κ , the posterior variance $\sigma^2(x)$ dominates, putting greater emphasis on parts of the optimisation space which are highly uncertain. Given a suitable choice of κ , the UCB has been mathematically proven to reduce the system regret over time on average [63]. Under these conditions, the UCB is highly sample-efficient.

$$V_t(x) = \mu(x) + \kappa\sqrt{\sigma(x)} \quad (4.1)$$

There are several alternatives to the UCB which are used in practice. In some problems where a subset of the optimisation space is dangerous, acquisition functions must not only balance exploration and exploitation but also avoid regions of a particular extreme. A common example of this is using GPs to learn the effects of spinal chord simulation while avoiding causing a patient pain [63] or learning the environment of a mobile robot without it falling off a cliff. So-called safe optimisation approaches have been proposed for these tasks, which contain constrained acquisition functions for ensuring safety [63]. Although most commonly used when there are physical safety requirements, these methods are also applicable in black-boxes wherein a particular region is significantly more expensive to compute than others; this case can be treated as a computational safety requirement.

Much research has been conducted into determining suitable choices of κ . In some instances, κ is scheduled as a function of the size of the optimisation space and number of sampled points, adaptively lowering κ as more points are sampled [61]. This aligns with the common trend of prioritising exploration at the start of the

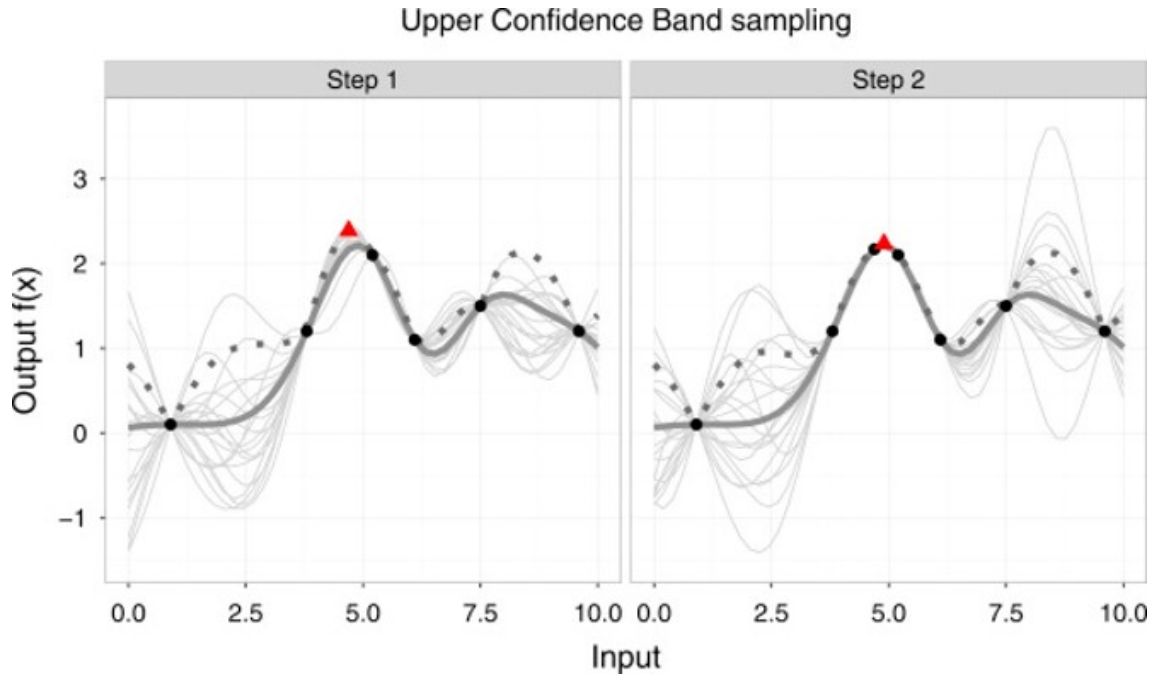


Figure 4.3: An example of sampling using the UCB acquisition function [64].

optimisation process and exploitation towards the end. Other methods draw κ from a random distribution, with parameters either tuned for the specific optimisation problem at hand and/or modelled from the size of the optimisation space and points sampled [61].

A second class of acquisition functions avoids UCB entirely and instead adopt an epsilon-greedy approach commonly seen in reinforcement learning. Candelieri [61] presents two of these methods: ϵ -PF and ϵ -RS. In each method, there is a probability $1 - \epsilon$ of choosing the point with highest predicted mean μ . Random points from the entire search space (ϵ -RS) or the Pareto Front between μ and σ^2 (ϵ -PF) are sampled with probability ϵ [61]. Ultimately, however, all methods attempt to achieve the same thing: a perfect balance between exploration and exploitation.

4.4 Trajectory Planning

Trajectory planning is a broad term which encompasses many of the motion planning capabilities of a robotic system. In this thesis, trajectory planning will refer to the process of generating waypoints through which a robotic system must move in order to execute an action. This includes motions like closing robotic grippers and transporting glassware, but any path smoothing, time parametrisation and actuation is considered out of the scope of this thesis.

The Open Motion Planning Library (OMPL) [62] contains implementations of

several commonly used trajectory planners, including Rapidly-expanding Random Trees (RRTs) and Probabilistic Roadmap Methods (PRMs). These techniques are well known trajectory planners and use a sampling-based approach to identify paths connecting two positions. OMPL also contains an implementation of RRTConnect [64]. RRTConnect extends the RRT algorithm by building a random tree from both the start and the goal position, and exhibits improved performance over standard RRT [64]. In many applications using the OMPL library, RRTConnect acts as the default planning algorithm.

An alternative approach to the sampling-based planning algorithms are optimisation-based algorithms such as Covariant Hamiltonian Optimisation for Motion Planning (CHOMP) and Stochastic Trajectory Optimisation for Motion and Planning (STOMP). First proposed by Zucker et al. [65], CHOMP leverages gradient-based optimisation and has proven to excel in smooth, path-optimal trajectories [52]. STOMP performs similarly, often identifying trajectories much faster due to its ability to escape local minima [52].

However, both of these optimisation-based methods are limited in that they are highly sensitive to the initialisations and did not perform well compared to sampling-based methods in complex environments [52]. Liu & Liu [52] attempt to address this by integrating STOMP and CHOMP into OMPL, allowing sampling-based methods to be used to find good initialisations for these methods. However, these integrations are not publicly available and tedious to implement.

Chapter 5

Problem Description

This thesis aims to contribute towards increased automation in chemical research laboratories by targeting a fundamental operation linking most processes together: the grasping of labware. Unlike commercial or clinical laboratories, which have standardised labware, research laboratories exhibit a high variability in labware shape and size. Therefore, they require flexible grasp algorithms that can identify shape topology rather than relying on preprogrammed actions. However, no such grasping algorithms exist which capture the domain-specific considerations of chemical labware grasping.

5.1 Research Gaps

To address the broader lack of labware grasping in chemical laboratories, two grasp-specific research gaps have been identified:

- **Research Gap 1:** *Current grasp planners are not tailored towards chemical labware and do not consider the side-on and spill-free constraints of labware manipulation.*
- **Research Gap 2:** *Current grasp planners do not consider path efficiency of grasp candidates during selection.*

Research Gap 1 arises from the unique geometric requirements of laboratory grasping. As discussed in the previous sections, labware must be grasped from side-on and manipulation of labware must prevent spillage. This constraint contradicts many assumptions made during grasp scoring, most notably with AnyGrasp.

The need for *Research Gap 2* arises as a corollary to this. The additional constraints applied on grasping leads to variability in the trajectory feasibility and path efficiency of grasps. Therefore, unlike standard grasping problems, choosing inefficient grasps may lead to significant losses in throughput. Currently, most motion-aware grasping algorithms focus on feasibility rather than path efficiency or rely on outdated analytic grasping methods.

5.2 Objectives

To address these research gaps, this thesis will extend AnyGrasp, a current state-of-the-art grasping algorithm, to create MALab-Grasp: a motion-aware grasping pipeline which incorporates notions of domain constraints and path efficiency. This pipeline has the following objectives:

- **Objective 1:** *Extend AnyGrasp to identify grasps which are side-on and keep labware upright to prevent spillage.*
- **Objective 2:** *Present a method of grasp scoring which identifies stable and successful grasp poses for labware.*
- **Objective 3:** *Implement a grasp selection process which selects trajectory-feasible and path-efficient grasps from a set of candidate grasps.*
- **Objective 4:** *Satisfy the flexibility requirements of chemical research laboratories by being performant under a wide range of laboratory environments and conditions.*
- **Objective 5:** *Integrate MALab-Grasp into the Franka Research 3 robotic arm and gripper, which must grasp and transport labware in simulated environments.*

5.3 Robotic System Requirements

The high-level problem that MALab-Grasp will be applied to is one of object transportation. Fig. 5.1 depicts the task that MALab-Grasp must enact in Objective 5. Within a simulated laboratory environment, a robotic arm must move from its initial position (x_i) to a grasp position (g) which it deems as path efficient. After grasping, the robotic arm must move the labware to a desired position (x_f) while ensuring it stays upright. These two movements are referred to as the pickup trajectory ($\mathcal{T}_{pickup}$) and the move trajectory ($\mathcal{T}_{move}$) respectively. Throughout this

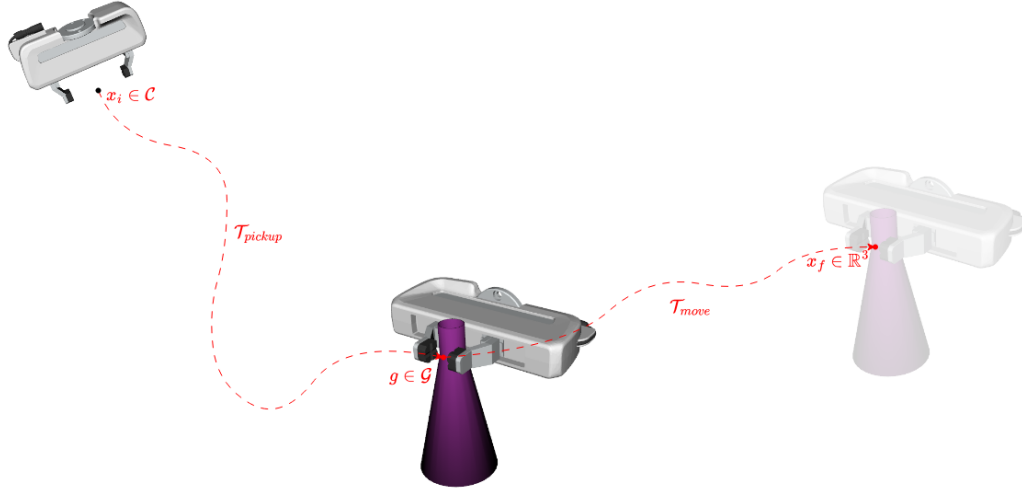


Figure 5.1: The required mission the FR3 robot must execute. The overall mission consists of an unconstrained pickup trajectory $\mathcal{T}_{pickup}$ and a constrained move trajectory $\mathcal{T}_{move}$, with the role of MALab-Grasp being to identify the best grasp pose g .

thesis, grasps are referred to as *(trajectory-) feasible/valid* if these two trajectories exist; otherwise, they are labelled as *infeasible/invalid*

Chapter 6

Preliminary Studies

This chapter conducts a series of preliminary studies used to characterise the problem space and benchmark existing algorithms. The results of these studies will be used to inform design decisions for MALab-Grasp, as outlined in Chapter 7.

6.1 AnyGrasp Benchmarking

Prior to the integration of AnyGrasp into the system, its performance was benchmarked against commonly used labware. The objective of this study was to validate whether AnyGrasp can work effectively within the context of a chemical laboratory and identify any potential areas in need of improvement.

Six tests were performed in this preliminary study, corresponding to three different types of chemical flasks in a simple (uncluttered) simulated environment. A description of these flasks and this environment is given in Chapter 9. In each scenario, the object’s point cloud was provided to AnyGrasp and the size of the candidate grasp set recorded. Grasps were also then passed through the flatness filter proposed by MALab-Grasp, as described in Chapter 7, and then finally refined using non-maxima suppression (NMS). At each of these stages, grasps were also manually classified as being on the flask neck or not. The motivation for this is that, as outlined by [17], the neck is the most natural location by which to grasp labware.

Table 6.1 summarises the number of grasps synthesised by AnyGrasp at each of these stages. Although many grasps were synthesised, the majority of these were located upon the bases of the flasks. This phenomenon can largely be attributed to two aspects of the AnyGrasp algorithm: sampling and grasp stability scoring. Because points from the point cloud are sampled to generate candidate grasps, areas with more points naturally generate more grasps. In addition, points closer to the

Table 6.1: The total number of grasps AnyGrasp synthesises before flatness filtering, after flatness filtering, and after non-maxima suppression, as well as the proportion of each of these grasps which lie on the neck of the glassware. AnyGrasp has a strong proclivity towards synthesising more grasps towards the base of flasks rather than the neck.

Object	Before Flatness Filter		After Flatness Filter		After Filter and NMS	
	# Total	# On Neck	# Total	# On Neck	# Total	# On Neck
Conic	727	120	56	6	40	5
Round-bottom	931	212	83	16	60	14
Volumetric	939	340	65	11	35	8

centre of mass of an object are given a higher grasp stability score, and so are sampled more frequently and scored better. In these cases, this region corresponds the base of each flask.

The application of flatness filtering also led to a significant reduction in the grasps synthesised by AnyGrasp. When combined with the desire to grasp labware by the neck, only five grasps could be identified in the most extreme case. In complex laboratory environments where many of these grasps may be infeasible, this may indicate that AnyGrasp’s grasp synthesis is insufficient for these scenarios.

Fig. 6.1 visualises the candidate grasp sets after flatness filtering and NMS and shows an unexpected trend. Given the symmetric nature of labware, it would have been expected that the grasp set generated by AnyGrasp would also be close to symmetric. However, very strong asymmetry was observed in the grasps generated. Although the stochastic nature of grasp synthesis does not guarantee symmetry, the lack of symmetry indicates that AnyGrasp is failing to identify the grasp set completely. The effect of this is the formation of patches of the point cloud in which there are no grasps or in which grasps are only present on one side of the labware. In complex environments with large regions of infeasibility, this lack of grasp diversity may cause AnyGrasp difficulty in identifying feasible grasps. This suggests that AnyGrasp may struggle in these complex environments.

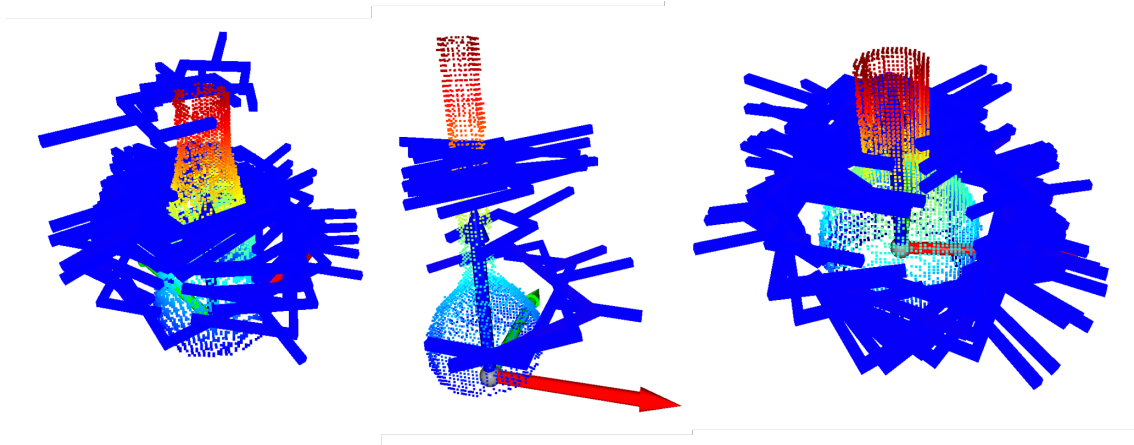


Figure 6.1: A visualisation of the candidate grasp poses computed by AnyGrasp. A strong asymmetry is observed in the grasps, despite the flasks themselves being symmetric.

6.2 Distribution of Trajectory Planning Time

Objective 4 requires MAlab-Grasp to be performant under a wide range of laboratory environments. A key factor in the efficiency of robotic systems is accurately tuning the amount of time allocated to planning of trajectories. This is especially the case in lab grasping, where many grasps are infeasible. If this allocated time is too low, then the system may struggle to identify feasible grasps. On the other hand, if the allocated time is too high, then the system may needlessly spend time on grasps which will never be feasible.

Therefore, it would be ideal to gain an understanding of the underlying distribution of the time taken for the system to plan the trajectories $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$. This study collated the planning times for each of these trajectories in a clutter-free simulated environment, using RRTConnect as the trajectory planning algorithm. 358 data points were collected across the candidate grasp sets of the conic flask, large beaker, round-bottom flask and volumetric flask in the simple environment. A description of these scenarios is provided in Chapter 9. The resulting histograms are shown in Fig. 6.2.

The distributions of both the $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ display a distinct right-skewness. This follows the expected behaviour of RRTConnect, where highly efficient trees are rare and there is a not insignificant portion of trials where it takes longer than the mean planning time. Both also exhibit a minimum planning time, accounting for computational overheads, request adapters and other baseline functions. $\mathcal{T}_{move}$ seems to be bimodal in nature. This may be because of two fundamentally different classes of trajectories – possibly one class corresponding to grasps with the robot’s wrist facing downwards and the other for the grasps with the wrist facing upwards. Ta-

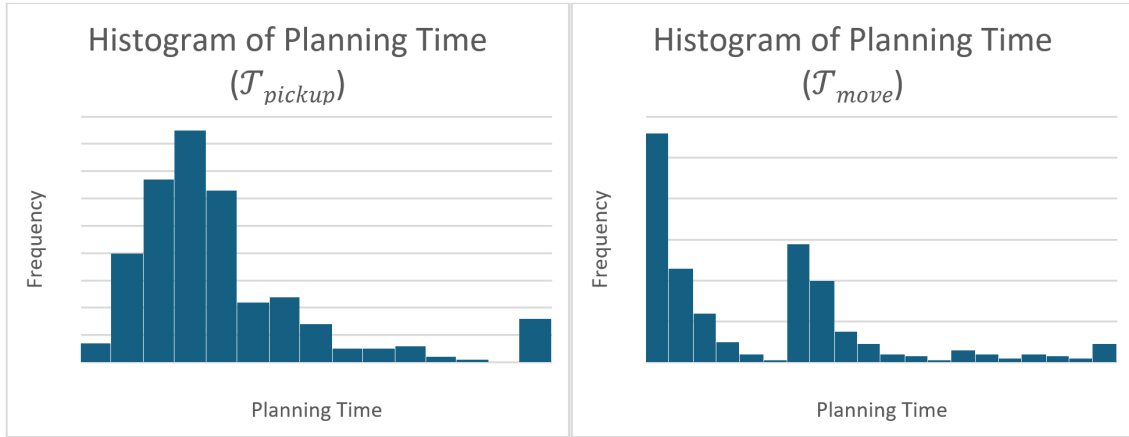


Figure 6.2: Histograms showing the empirical distribution of planning times for $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$, each with a distinct right-skewness.

ble 6.2 summarises the statistical properties of these distributions and supports these observations, reporting a high skewness factor and a mean value falling significantly above the median.

Table 6.2: Statistical properties of the planning times, showing significant positive skewness.

	Mean (s)	Median (s)	Variance (s ²)	Skewness (s ²)
$\mathcal{T}_{pickup}$	0.0759	0.0433	0.0288	6.21
$\mathcal{T}_{move}$	0.257	0.150	0.0725	2.88

Due to this heavy-tailed nature of the distribution, it was hypothesised that the lognormal distribution could model the planning times reasonably well. Fig. 6.3 shows the distribution of the logarithm of $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$. $\log \mathcal{T}_{pickup}$ is strongly indicative of a normal distribution, with $\log \mathcal{T}_{move}$ showing a bimodal normal distribution. The statistics in Table 6.3 support this with lower skewness factors and more correlated means and medians. Marginal skewness still exists, showing that the lognormal distribution is not a perfect model and likely underestimated the probability for high planning times. However, in general, the lognormal distribution seems like a suitable model for $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$.

Table 6.3: Table of the statistical properties of the lognormal distributions, showing decreased skewness factors and closer mean-median correlation than the normal distribution.

	Mean (s)	Median (s)	Variance (s ²)	Skewness (s ²)
T_{pickup}	-3.012	-3.139	0.3677	3.746
T_{move}	-1.852	-1.899	1.029	0.03946

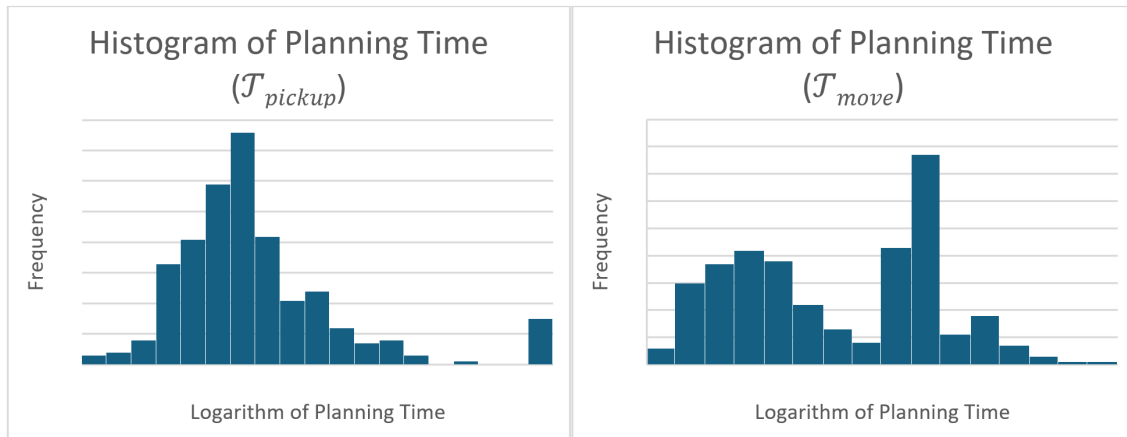


Figure 6.3: Histograms showing the empirical distribution of the logarithms of planning time for $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$. The distributions resemble Gaussian distributions, indicating that lognormal distributions may satisfactorily model trajectory planning time.

Chapter 7

MALab-Grasp Design

This section details the system design and implementation of a novel motion-aware grasping pipeline to address the problem definition outlined in Chapter 5, termed Motion-Aware Labware Grasping (MALab-Grasp). It makes three key contributions to the field of grasping:

- **Contribution 1:** *A novel grasp synthesis and scoring algorithm tailored for chemical research laboratories*
- **Contribution 2:** *A novel path-efficient grasp selection algorithm based on iterative grasp evaluation*
- **Contribution 3:** *A grasp evaluation algorithm which autonomously tunes the evaluation time to maximise pipeline flexibility.*

The architecture of MALab-Grasp is depicted in Fig. 7.1 and consists of three stages, each one linked to a key contribution. First is the **Grasp Synthesis** stage (*Contribution 1*), where a set of laboratory-tailored candidate grasps are synthesised and scored. These candidate grasps are then passed to a **Grasp Selection** stage (*Contribution 2*), which uses Gaussian process-driven sampling to iteratively compute grasps' path lengths and uses this data to select an optimal grasp once planning has terminated. Finally, this path efficiency computation is outsourced to a **Grasp Evaluation** stage (*Contribution 3*), which plans grasp trajectories with adaptive per-grasp evaluation time. Each of the following sections will detail the design of each stage respectively.

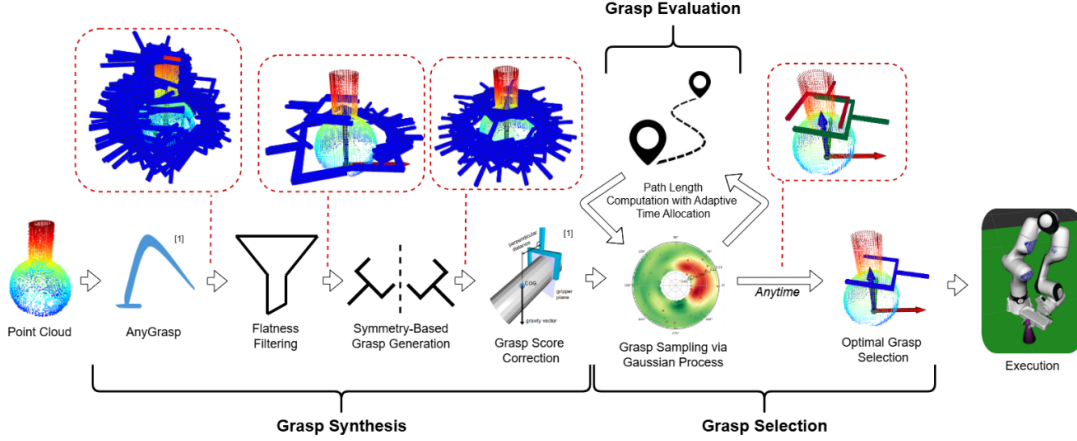


Figure 7.1: Architecture of MALab-Grasp, decomposed into Grasp Synthesis, Grasp Selection and Grasp Evaluation.

7.1 Grasp Synthesis Stage

The Grasp Synthesis stage is responsible for generating a set of candidate poses, $\mathcal{G}$, at which a robotic system can grasp a target object $\mathcal{O}$. It covers Objectives 1 & 2: extending AnyGrasp to generate grasps which satisfy domain constraints and scoring them according to a labware-specific scoring function. The follow subsections detail and justify each step.

7.1.1 Point Cloud

The primary input to MALab-Grasp is a global point cloud of the environment. In contrast to many grasp planners in the literature, MALab-Grasp requires a 3D point cloud rather than a single-view (2D) point cloud. This is in line with Objective 1, which requires side-on requirements of labware grasping. Unlike top-down grasping, the side-on grasping surface cannot be fully captured in a single view. The method used to collect these 3D point clouds is described in Chapter 9.

To prevent superfluous grasp generation, the global point cloud is cropped according to the bounding box of the target object $\mathcal{O}$, resulting in a cropped point cloud $\mathcal{P}$. This bounding box is out of the scope of MALab-Grasp and is instead provided by the vision system, described in Chapter 8.

7.1.2 AnyGrasp

MALab-Grasp uses AnyGrasp as its underlying grasp synthesiser. The motivation for this is twofold. First, MALab-Grasp leverages AnyGrasp’s state-of-the-art performance, graspness-based grasp generation, and ability to generate dense grasp sets without needing to create its own custom grasp synthesiser. Second, MALab-Grasp is explicitly decoupled from AnyGrasp and could make use of any alternate grasp synthesiser. This allows MALab-Grasp to continually improve as better state-of-the-art planners emerge.

AnyGrasp performs a dense 6-DOF grasp synthesis, computing a set of candidate grasp poses $\mathcal{G}_{anygrasp}$ with corresponding grasp scores $q_{anygrasp}$. Fig. 3.1 shows an example grasp computed by AnyGrasp and its coordinate conventions.

7.1.3 Flatness Filtering

To satisfy the side-on constraints of labware grasping as outlined in Objective 1, MALab-Grasp immediately removes any grasps which do not satisfy this constraint. In this thesis, side-on is defined as any grasp with a roll or pitch of less than 10 degrees. The implementation of this filter is shown in Eq. (7.1).

$$\mathcal{G} \leftarrow \{g \in \mathcal{G}_{anygrasp} : |\cos(\phi)| < \cos(10^\circ) \text{ and } |\cos(\theta)| < \cos(10^\circ)\} \quad (7.1)$$

Fig. 7.2 shows the candidate grasp set before and after applying flatness filtering.

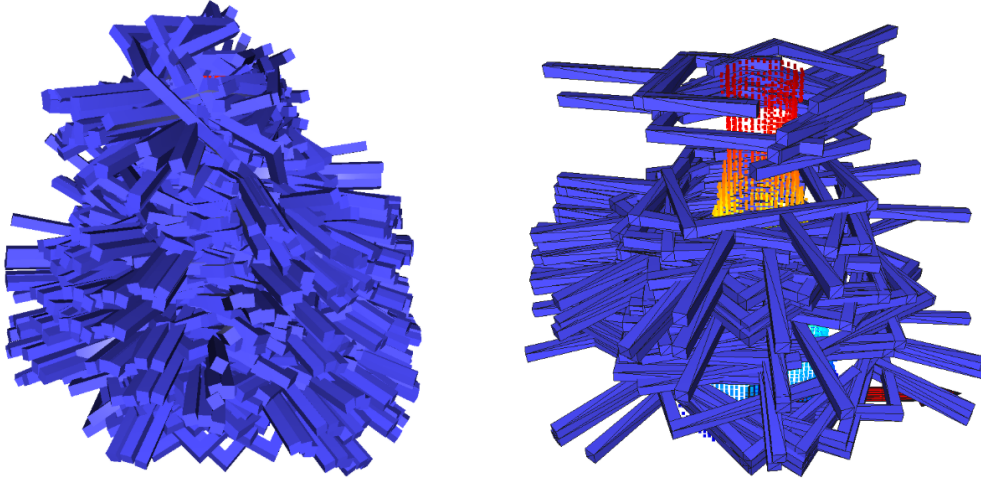


Figure 7.2: Comparison of $\mathcal{G}$ before and after applying flatness filtering. To minimise visual clutter, non-maxima suppression has been applied to each candidate set in this figure. Flatness filtering ensures labware is only grasped with a side-on approach.

7.1.4 Symmetry-Based Grasp Generation

Once the flatness filter has been applied, MALab-Grasp implements a symmetry-based grasp generation step. This is motivated by the findings in Chapter 6, which identify the lack of symmetry in AnyGrasp’s grasp set being a possible cause of poor performance in laboratories. This step leverages the symmetric nature of labware to manually generate additional grasps that AnyGrasp could not. The objective of this is to maximise the coverage of $\mathcal{G}$ over the entire grasp space. This may be critical in complex laboratory environments where there are a limited number of regions in which grasps are feasible.

The core idea of this step is to rotate grasps about the central axis of $\mathcal{O}$, provided that the point cloud is symmetric. MALab-Grasp first jointly slices the cropped point cloud $\mathcal{P}$ and the candidate grasp set $\mathcal{G}$ into a series of horizontal layers, $\{\mathcal{P}_i, \mathcal{G}_i\}$. This allows symmetry to be analysed at a local scale, so that small asymmetries in one layer (such as the lip of a beaker) do not compromise the symmetry of other layers.

After slicing, $\mathcal{P}_i$ and $\mathcal{G}_i$ are rotated in discrete intervals about the central axis of $\mathcal{O}$. For each rotation R , the original and the rotated $\mathcal{P}_i$ are compared to determine whether they are similar. This is quantified through the Chamfer distance between the point clouds. If this distance is small enough, then it provides an indication that R lies in the rotational symmetry group of this layer. That being the case, the

rotated grasps are added to the grasp set $\mathcal{G}$. Importantly, $q_{anygrasp}$ is invariant to these rotations, assuming the point cloud is perfectly symmetric. Small asymmetries may cause this assumption to be violated, leading to some cases in which $q_{anygrasp}$ is not truly accurate for symmetry-generated grasps.

This algorithm is presented in Algorithm 1, with a visual depiction of a single grasp being duplicated shown in Fig. 7.3. The grasp set before and after post-symmetry grasp generation is shown in Fig. 7.4, showing distinct improvement in coverage of the grasp space.

Algorithm 1 Post-Symmetry Grasp Generation

```

1: procedure GENERATESYMMETRYGRASPS(point cloud  $\mathcal{P}$ , grasp set  $\mathcal{G}$ )
2:    $layer\_base\_height \leftarrow 0$ 
3:   while  $layer\_base\_height < max\_height$  do
4:      $\mathcal{P}_i \leftarrow \text{CROP}(\mathcal{P}, layer\_base\_height, LAYER\_THICKNESS)$ 
5:      $\mathcal{G}_i \leftarrow \text{CROP}(\mathcal{G}, layer\_base\_height, LAYER\_THICKNESS)$ 
6:
7:     for  $R \in \{30, 60, 90, 120, \dots\}$  do
8:        $\mathcal{P}'_i \leftarrow \text{ROTATE}(\mathcal{P}, R)$  ▷ Rotate the grasps and points
9:        $\mathcal{G}'_i \leftarrow \text{ROTATE}(\mathcal{G}, R)$ 
10:
11:       if  $\text{CHAMFERDISTANCE}(\mathcal{P}_i, \mathcal{P}'_i) < \epsilon$  then
12:          $\mathcal{G} \leftarrow \mathcal{G} \cup \mathcal{G}'_i$  ▷ Add grasps to grasp set
13:       end if
14:     end for
15:      $layer\_base\_height \leftarrow layer\_base\_height + LAYER\_THICKNESS$ 
16:   end while
17: end procedure

```

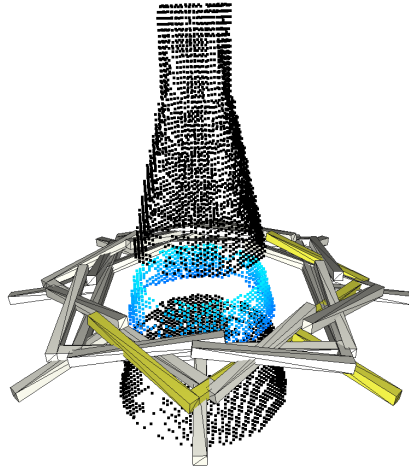


Figure 7.3: The generated symmetries of a base grasp, shown in yellow, within a single horizontal slice. The blue portion of the point cloud represents the horizontal layer.

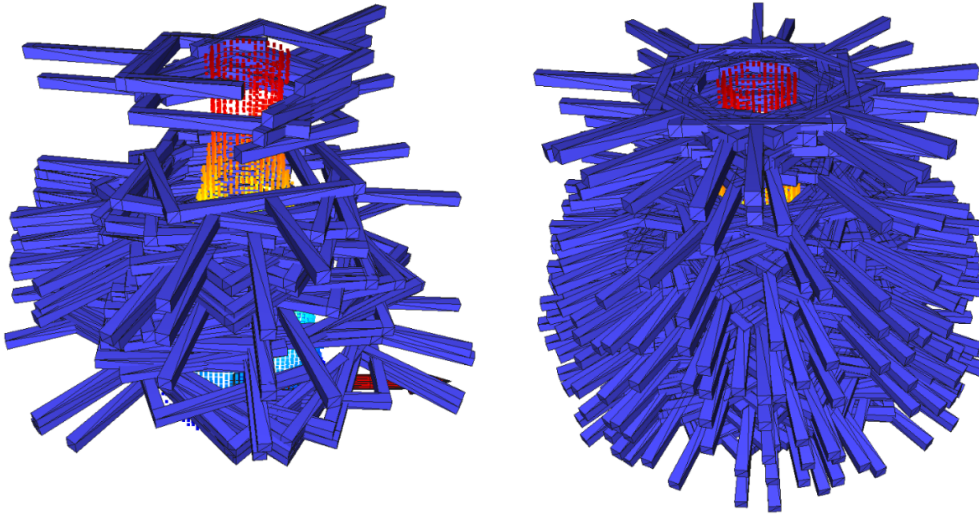


Figure 7.4: Comparison of $\mathcal{G}$ before and after applying symmetry-based grasp generation. To reduce visual clutter, non-maxima suppression has been applied to each candidate set in this figure. Symmetry-based generation increases the size of the $\mathcal{G}$ and improves its coverage across the entire grasp space.

7.1.5 Grasp Score Correction

Grasp Score Correction is a critical step in MALab-Grasp. Rather than modifying the candidate set $\mathcal{G}$, this step proposes a novel scoring algorithm to correct the shortcomings of AnyGrasp and tailor grasp scoring to laboratory environments. This targets Objective 2.

Two metrics of grasp stability are considered: static stability and dynamic stability. Static stability refers to the stability of the grasped object at an arbitrary point throughout its manipulation, disregarding any dynamic effects or external forces. This is quantified by the horizontal distance from the grasp centre to the object centre of gravity vector, as grasps further away from the centre of gravity vector will generate large torque forces on the gripper. Dynamic stability refers to the nature of the equilibrium point; either stable (if the system forms a standard pendulum) or unstable (if the system forms an inverted pendulum). This is largely dictated by the extent to which a grasp lies above or below the centre of gravity.

AnyGrasp already computes a metric of static stability, $ss_{anygrasp}$, however this assumes top-down grasping in a pick-and-place problem. To undo the effect of this, MALab-Grasp must first estimate $ss_{anygrasp}$ for each grasp. For side-on grasps, AnyGrasp's calculation of $ss_{anygrasp}$ is equal to the normalised vertical distance to the centre of mass. The method of estimating the centre of mass is not explicitly described, although it assumes the object is a solid rigid body. Therefore, the centre of mass is assumed to be the geometric centroid of the point cloud. Due to the normalised nature of the grasp score, the magnitude of this normalising factor is inherently unrecoverable. To correct for this, an additional factor of 1.5 was arbitrarily added to the normalising scaler. This estimate is shown in Eq. (7.2).

$$\hat{s}_{anygrasp} = \frac{z - z_{centroid}}{1.5 \max(z - z_{centroid})} \quad (7.2)$$

A requirement of all subsequent stability score calculations is a better estimation of the object centre of mass $COM=(\bar{x}, \bar{y}, \bar{z})$. Given the symmetrical nature of laboratory glassware, $\bar{x}$ and $\bar{y}$ are estimated as the geometric centroid of the point cloud in their respective dimensions. On the contrary, $\bar{z}$ is estimated by assuming each object is a hollow and homogeneous revolved body and discretising this as a vertical stack of concentric rings, with ring radii determined by the point cloud. This model of the labware is shown in Fig. 7.5.

This approximation will almost certainly place the centre of mass above its true position, as real-world labware will have a greater density of mass towards its base due to solids or fluids contained in the object. Therefore, the dynamic stability score

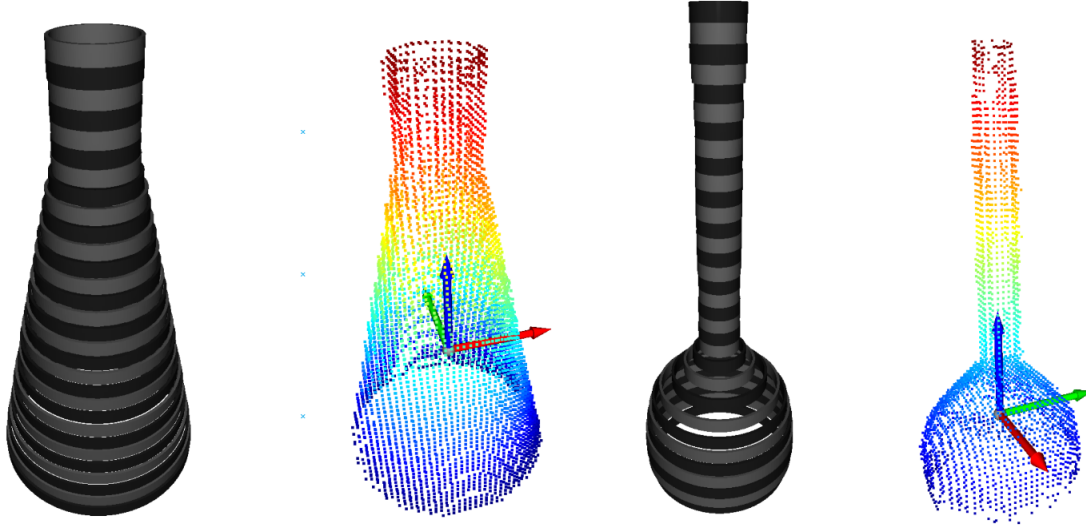


Figure 7.5: Modelling of labware using a vertical stack of concentric rings and the resultant centre of mass estimations.

detailed subsequently will behave cautiously and treat most grasps as more unstable than they truly are. This is not undesirable because it adds an additional safety margin to grasp score calculations.

From this centre of mass estimate, the corrected static stability score is calculated as the normalised horizontal distance from the grasping point to the centre of mass, as shown in Eq. (7.3). This behaves in the same manner as AnyGrasp’s stability score but correctly assumes that the gripper remains horizontal during object transportation. Fig. 7.6 provides a visual representation of calculating the horizontal and vertical distance from the centre of mass.

$$\hat{s}_{static} = \frac{\sqrt{(x - \bar{x})^2 + (y - \bar{y})^2}}{\max \left(\sqrt{(x - \bar{x})^2 + (y - \bar{y})^2} \right)} \quad (7.3)$$

The dynamic stability score is assessed using the second-order system response. The linearised dynamics of the system are modelled as a pendulum pivoting about grasp point, either as a regular pendulum (if the grasp point is above the centre of mass) or an inverted pendulum (if the grasp point is below the centre of mass). These two cases are captured in Fig. 7.7. The linearised dynamics of both these systems around $\theta \approx 0$ are shown in Eq. (7.4).

$$\ddot{\theta} + \frac{mg(z - \bar{z})}{I_{COM} + m(z - \bar{z})^2} \theta = 0 \quad (7.4)$$

This equation encapsulates both the regular pendulum case ($z - \bar{z} > 0$) and

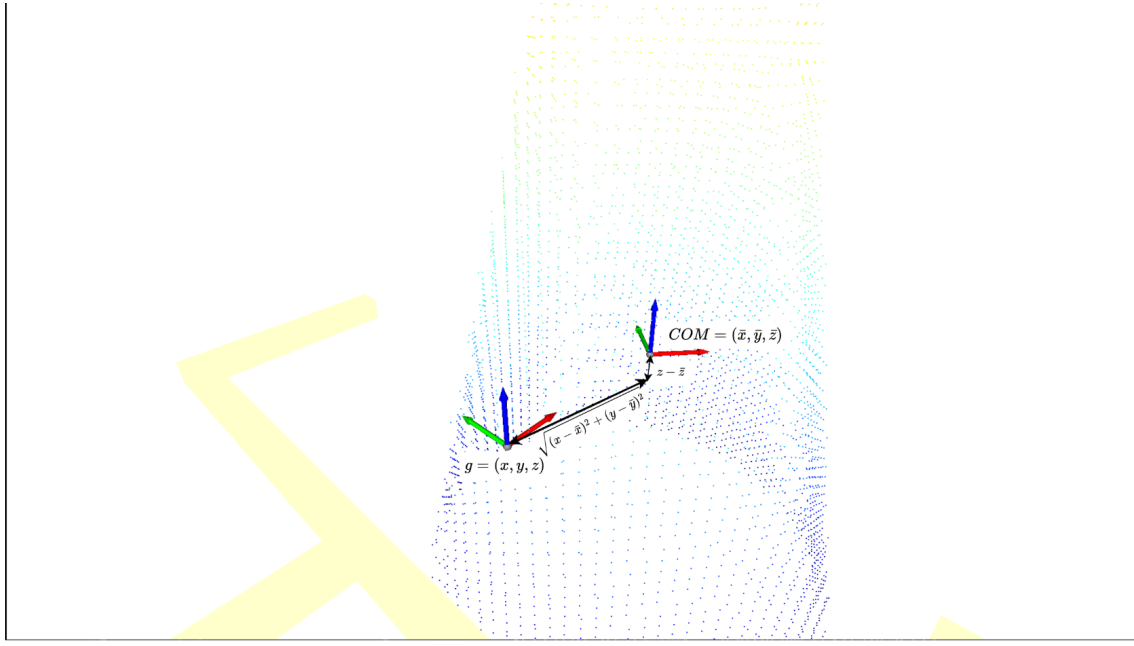


Figure 7.6: Calculation horizontal and vertical distance from a grasp with grasp point g to the centre of mass. Larger horizontal distance decreases the static stability score $\hat{s}s_{static}$ and vertical distance correlates with the dynamic stability score $\hat{s}s_{dynamic}$.

the inverted pendulum case ($z - \bar{z} < 0$). The mass and inertia of this system is calculated using the same model as the centre of mass, presented in Fig. 7.5. For small perturbations in θ , the response of this system is governed by Eq. (7.5).

$$\theta(t) = A_1 \exp(\sqrt{\lambda}t) + A_2 \exp(-\sqrt{\lambda}t), \quad \lambda = -\frac{mg(z - \bar{z})}{I_{COM} + m(z - \bar{z})^2} \quad (7.5)$$

The sign of λ dictates the dynamic stability, with grasps with $\lambda < 0$ being stable and those with $\lambda > 0$ being unstable. The magnitude of λ describes the rate at which an unstable grasp diverges or the magnitude of the restoring force for a stable grasp. Therefore, the dynamic stability score is defined as the normalised λ , with stable grasps having a score between 0 and 0.5 and unstable grasps having a score between 0.5 and 1.0. This is shown in Eq. (7.6).

$$\hat{s}s_{dynamic} = \begin{cases} 0.5 + \frac{\lambda}{2\lambda_{min}} & \lambda < 0 \\ 0.5 + \frac{\lambda}{2\lambda_{max}} & \lambda > 0 \end{cases} \quad (7.6)$$

Here, $\min \lambda$ represents the *most stable* grasp with the strongest restoring force; similarly, $\max \lambda$ represents the *least stable* grasp with the most rapid divergence. Fig. 7.8 shows a visual comparison of the static and dynamic stability scores.

MALab-Grasp incorporates the three stability scores calculated as scaling terms

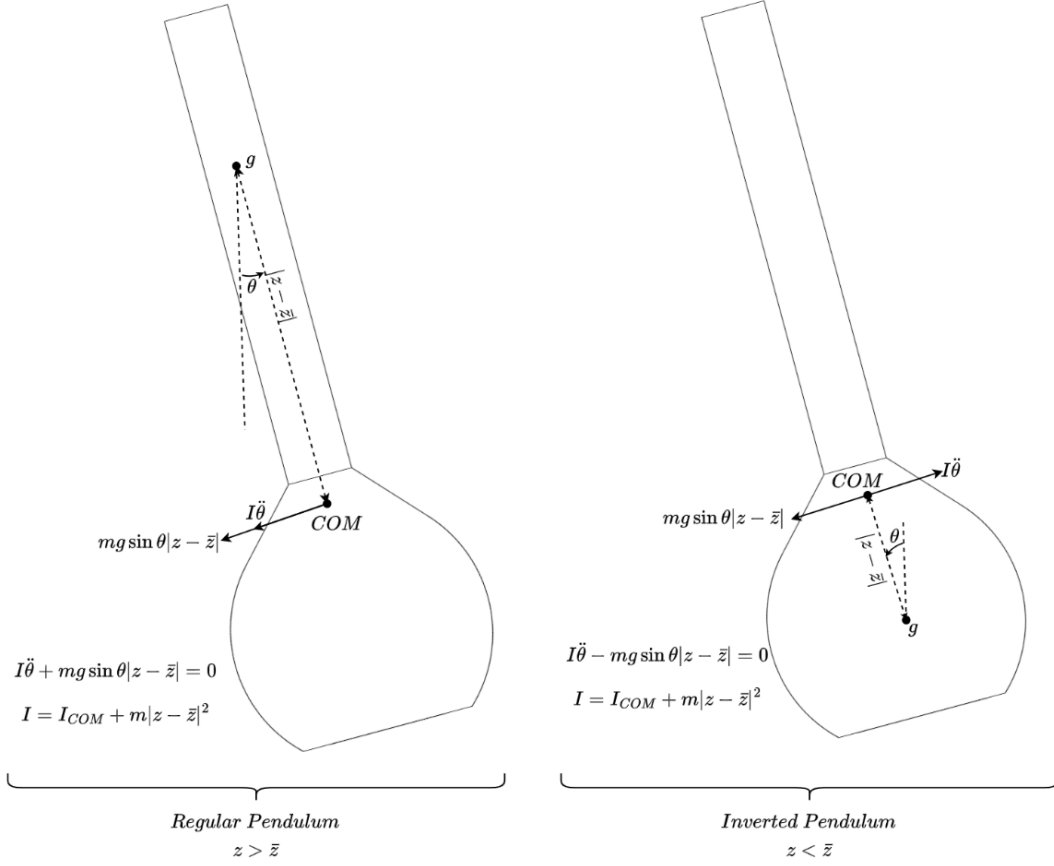


Figure 7.7: Free body diagrams of the regular and inverted pendulum cases.

to correct AnyGrasp's initial grasp score. The effect of AnyGrasp's stability score is first reverted, and then the static and dynamic stability scores are applied to culminate in a final, corrected grasp score $q_{corrected}$. This is shown in Eq. (7.7).

$$q_{corrected} = q_{anygrasp} \cdot (1 - \hat{s}_{anygrasp})^{-1} \cdot (1 - \hat{s}_{static}) \cdot (1 - \hat{s}_{dynamic}) \quad (7.7)$$

Once correcting the grasp scores for $\mathcal{G}$, MALab-Grasp performs non-maxima suppression (NMS) to identify points of local maxima with respect to the corrected grasp score. This is a standard step in all grasping algorithms to remove grasps which are not suitably distinct from one another. By reducing the size of $\mathcal{G}$, NMS also aids the efficiency of the subsequent Grasp Selection stage.

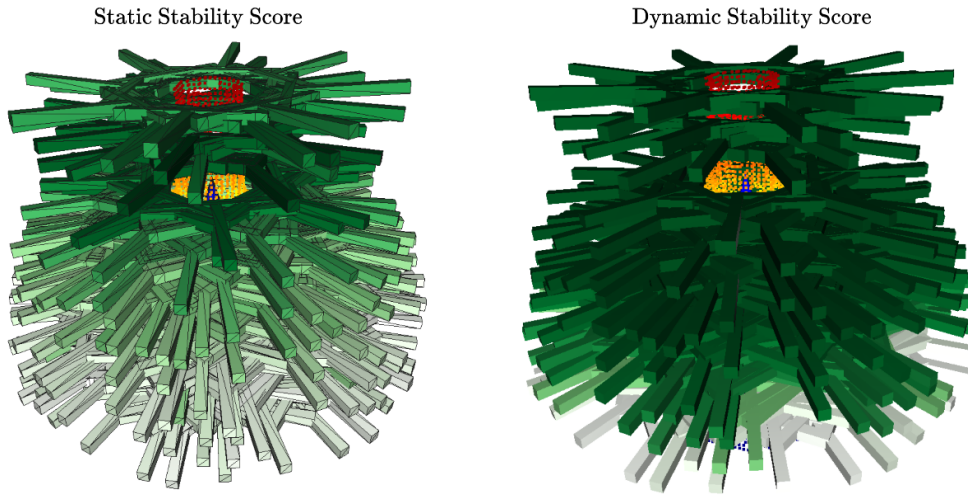


Figure 7.8: Static and dynamic stability scores for conic flask’s candidate grasp set. Grasps in lighter colours have a higher stability score and therefore are considered less stable. To reduce visual clutter, non-maxima suppression has been applied to each candidate set in this figure.

7.1.6 Coordinate Transformations

The final step in the Grasp Synthesis stage of MALab-Grasp is to apply a coordinate transform from the grasp coordinate frame to the end-effector coordinate frame. As shown in Fig. 7.9, this requires a -90° rotation about the y axis.

As part of this process, each grasp is duplicated and rotated by 180° about the end-effector’s z axis, noting that this does not change the end-effector’s geometry but may dramatically change the joint-space pose. Because of the asymmetrical nature of the robot wrist, flipped grasps may also avoid collisions more effectively than regular grasps, or vice versa. Grasps are subsequently partitioned into two groups: one where all grippers are pointing downwards (with the camera facing the ground), and the other where they are facing upwards. The objective of this partition is to reduce the likelihood of grasps which are close to each other in task space being far away in joint space. Ideally, this should retain the smoothness in grasp feasibility and/or path length during grasp selection. Both groups are then passed the grasp selection stage.

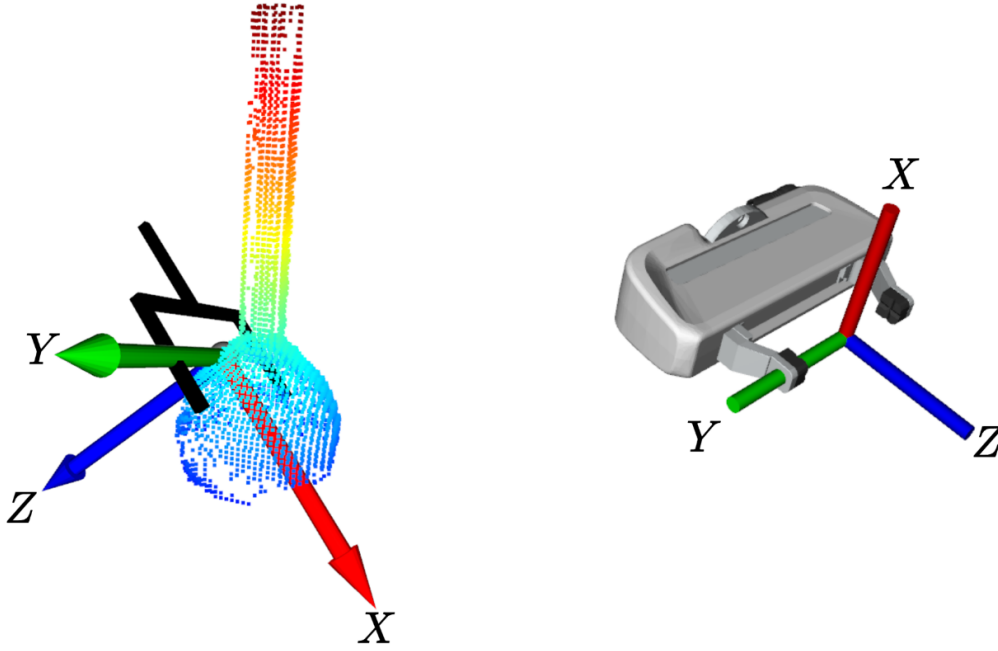


Figure 7.9: Comparison between the coordinate frames of the grasp pose and the end-effector. Mapping between these frames requires a post-transformation of -90 degrees about the y axis

7.2 Grasp Selection Stage

The Grasp Selection Stage of MALab-Grasp determines the optimal grasp, $g^* \in \mathcal{G}$, out of all candidate grasps provided by Grasp Synthesis. This stage targets Objective 3, aiming to identify both feasible and path-optimal grasps.

A naïve approach to identifying feasible grasps would be to iteratively evaluate the feasibility of grasps in a random or grasp score order. For applications where efficiency is paramount, path efficiency can be recorded for every iteration and then the optimal grasp after a given planning time be chosen.

MALab-Grasp extends this iterative approach by proposing an informed method of grasp ordering, also referred to as grasp sampling. Rather than sampling grasps in a random manner, MALab-Grasp estimates the path efficiency of grasps yet to be sampled. These estimates are then used to bias sampling towards grasps which maximise a trade-off between exploring new regions of the grasp space and exploiting regions which are known to be efficient. Within this, MALab-Grasp aims to implicitly distinguish between regions of feasibility and infeasibility, and then subsequently avoid sampling in infeasible regions.

A Gaussian process is used as the key mechanism by which MALab-Grasp estimates

path efficiency and executes informed grasp sampling is via a Gaussian process. This Gaussian process is applied on a *projected grasp space* and aims to estimate the *path score* of grasps. In each iteration, MALab-Grasp samples a grasp according to its current estimates of path lengths. After sampling, Grasp Evaluation is responsible for planning the trajectories of the grasp, identifying it as feasible or infeasible, and calculating the relevant path lengths. Once this has been completed, MALab-Grasp uses this information to update its estimates for all other grasps.

This process continues until the allocated planning time has elapsed. As such, MALab-Grasp is classed as an anytime algorithm, as it can return results within any allocated time frame. This is critical in chemical laboratories, where the allowable planning may be highly variable. For instance, MALab-Grasp may be run for an upcoming process and be given until the current process ends to find the most path-efficient grasp. MALab-Grasp is designed specifically for these situations. This aligns with the need to be compatible with a wide range of laboratory conditions, as dictated in Objective 4.

Algorithm 2 summarises the process of Grasp Selection, with the following sections outlining the key steps of this algorithm in more depth.

Algorithm 2 Grasp Selection algorithm, based on iterative grasp evaluation

```

1: function INITIALISE(grasp set  $\mathcal{G}$ )
2:    $n \leftarrow |\mathcal{G}|$ 
3:    $\boldsymbol{\mu} \leftarrow \mathbf{0}_{n \times 1}$  ▷ Path score estimates
4:    $\boldsymbol{\Sigma} \leftarrow I_{n \times n}$  ▷ Estimate uncertainties
5:    $\mathcal{G}_{eval} \leftarrow \emptyset$ 
6:    $\bar{\mathcal{G}}_{eval} \leftarrow \mathcal{G}$ 
7: end function
8:
9: function ITERATIONLOOP(grasp set  $\mathcal{G}$ )
10:   $g \leftarrow \text{GRASPWITHLARGESTUCB}(\mathcal{G})$  ▷ Eq. (7.13)
11:   $d_{path}, is\_valid \leftarrow \text{GRASPEVALUATION}(g)$  ▷ Grasp Evaluation Stage
12:   $d_{score}(g) \leftarrow \text{PATHLENGTHNORMALISATION}(d_{path})$  ▷ Eq. (7.8)
13:
14:   $\mathcal{G}_{eval} \leftarrow \mathcal{G}_{eval} \cup \{g\}$  ▷ Mark  $g$  as evaluated
15:   $\bar{\mathcal{G}}_{eval} \leftarrow \bar{\mathcal{G}}_{eval} \setminus \{g\}$ 
16:
17:   $\boldsymbol{\mu}, \boldsymbol{\Sigma} \leftarrow \text{UPDATEGP}()$  ▷ Eq. (7.9)
18: end function
19:
20: procedure GRASP SELECTION( $\mathcal{G}_{upwards}, \mathcal{G}_{downwards}$ )
21:  INITIALISE( $\mathcal{G}_{upwards}$ )
22:  INITIALISE( $\mathcal{G}_{downwards}$ )
23:  repeat
24:    ITERATIONLOOP( $\mathcal{G}_{upwards}$ )
25:    ITERATIONLOOP( $\mathcal{G}_{downwards}$ )
26:  until planning time elapsed
27: end procedure

```

7.2.1 Projected Grasp Space

A central part of the grasp selection stage is the exploiting of spatial relationships between grasps to estimate grasp path lengths without the need for explicit evaluation. To facilitate this, the candidate grasp set $\mathcal{G}$ is first projected into the (z, ψ) -space, with z being the vertical component of the grasp translation and ψ being the yaw of the grasp about the global vertical. This projected grasp space is visualised in polar coordinates as shown in Fig. 7.10, with radial direction z and angular direction ψ . Thus, the radial and angular directions are from henceforth used as equivalent terms for z and ψ respectively.

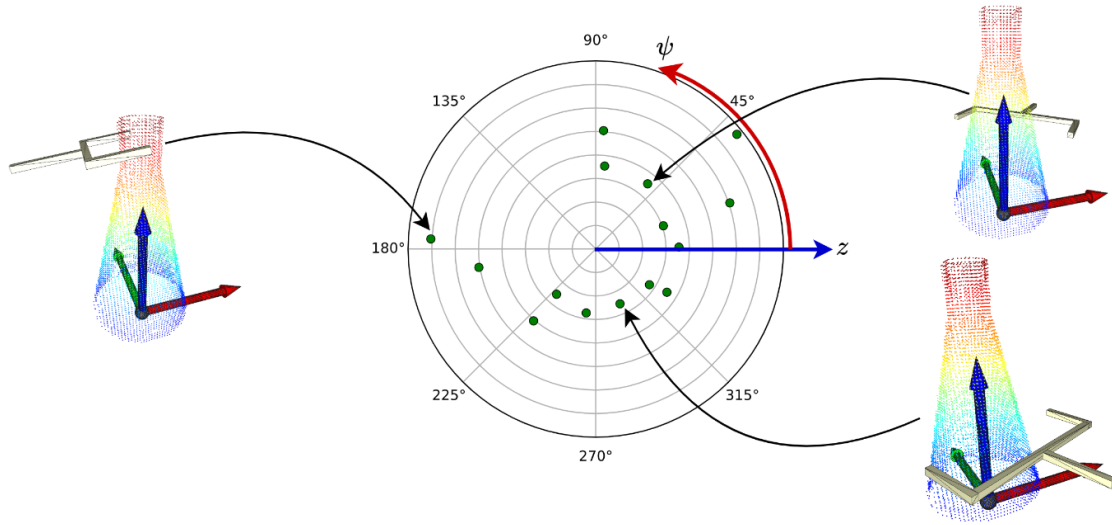


Figure 7.10: Map of candidate grasps in the (z, ψ) domain. Grasps are shown as points on a polar coordinate plane, with angle being ψ and radial distance being z . As grasps move further away from the origin in this domain, they move further upwards away from the base of the object. Grasps on the same contour line grasp the object at the same height but from a different orientation.

MALab-Grasp hypothesises that this two-dimensional space retains the majority of the spatial information about each grasp, given that roll and pitch are constrained to be close to 0 and the x and y position are predominately dictated by the shape of the object in conjunction with its yaw. Simultaneously, the dimensionality reduction from full 6-DOF poses should present a much more computationally simple model for grasp estimation.

7.2.2 Path Length Normalisation

To quantify the path efficiency of a grasp, MALab-Grasp first calculates the trajectories $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ required for a robot to first move to the grasp position and then transport the labware to its desired position. The total joint-space path length of these two trajectories is termed the *path length* of the grasp. The process of calculating the path length is delegated the Grasp Evaluation, however the results of a grasp evaluation are directly used by Grasp Selection. Each evaluated grasp is denoted as *valid/successful/feasible* if a trajectory could be planned using this grasp pose, or *invalid/unsuccessful/infeasible* if this is not the case. Valid grasps also contain their calculated path lengths; invalid grasps are assigned an infinite path length.

MALab-Grasp first normalises all path lengths received by Grasp Evaluation to be

finite-valued and zero-mean, as seen in Eq. (7.8). This is done to ensure a consistency of scale and support the use of a zero-mean Gaussian process. This normalised path length is also referred to as the path score, d_{score} . Values are constrained between -1 and 1, with invalid grasps being negative. Under this normalisation schema, values of d_{score} closer to 1 indicate a grasp has a path length closer to the optimal, whereas values of d_{score} closer to -1 indicate a grasp is invalid with a higher degree of certainty.

$$d_{score} = \begin{cases} 1 - \frac{d_{path}}{\max(d_{path})} & \text{valid grasp} \\ -1 & \text{invalid grasp} \end{cases} \quad (7.8)$$

7.2.3 Path Score Estimation

This thesis uses a zero-mean Gaussian process, $\mathcal{GP}$, to learn and exploit spatial relationships between grasps. $\mathcal{GP}$ acts in the (z, ψ) -space and estimates the path scores d_{score} of each grasp. In both of the grasp sets provided by Grasp Synthesis (downward-facing grasps and upward-facing grasps), an independent and distinct $\mathcal{GP}$ is used.

The $\mathcal{GP}$ is trained using the known path scores from the set of evaluated grasps, $\mathbf{y} = [d_{score}(g)]_{g \in \mathcal{G}_{eval}}$. Its objective is to thus learn the best estimate of d_{score} for the entire set of unevaluated grasps, $\bar{\mathcal{G}}_{eval}$. Each iteration of Grasp Selection, $\mathcal{GP}$ uses this training data to learn the parameters of its posterior distribution according to Eq. (7.9).

$$\begin{aligned} \boldsymbol{\mu}' &= \mathbf{K}(\mathcal{G}, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval}) + \text{diag}(\mathbf{1} - \varsigma)]^{-1} \mathbf{y} \\ \boldsymbol{\Sigma}' &= \mathbf{K}(\mathcal{G}, \mathcal{G}) - \mathbf{K}(\mathcal{G}, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval}) + \text{diag}(\mathbf{1} - \varsigma)]^{-1} \mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}) \end{aligned} \quad (7.9)$$

In Eq. (7.9), $\mathcal{G}$ and $\mathcal{G}_{eval}$ represent the vectorized forms of the candidate grasp set and evaluated grasp set in (z, ψ) -space. The kernel K has been chosen as a composite kernel, combining the periodic kernel for ψ and the Matérn 5/2 kernel for z . This allows the periodicity of ψ to be captured accurately and provides smoothness control for z , which may allow for better handling of any nonlinear borders between valid and invalid regions. Such borders may be caused by joint space limits or other effects which are not properly captured in (z, ψ) -space. The equations for the kernel components and the combined kernel are shown in Eq. (7.10).

$$K_{periodic}(\mathbf{u}, \mathbf{v}) = \exp\left(-\frac{2}{l^2} \sin^2\left(\frac{d(\mathbf{u}, \mathbf{v})}{2}\right)\right) \quad (7.10)$$

$$K_{matern}(\mathbf{u}, \mathbf{v}) = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(\mathbf{u}, \mathbf{v}) \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{l} d(\mathbf{u}, \mathbf{v}) \right)$$

$$K(\mathbf{u}, \mathbf{v}) = K_{periodic}(\psi_u, \psi_v) \cdot K_{matern}(z_u, z_v)$$

The final term of the formulation of Eq. (7.9), $\varsigma = [\varsigma(g)]_{g \in \mathcal{G}_{eval}}$, incorporates planning uncertainty to invalid grasps only via heteroscedastic noise. This is described in Section 7.3.

The estimated path score for the entire candidate grasp set $\mathcal{G}$ is then recovered from the posterior mean calculated in Eq. (7.9). More explicitly, this estimation is provided by Eq. (7.11).

$$\hat{d}_{score}(g) = \mathbf{K}(g, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval}) + \text{diag}(\mathbf{1} - \varsigma)]^{-1} \mathbf{y} \quad (7.11)$$

Fig. 7.11 shows the Gaussian process as a tool for predicting the path score of unknown objects, taken from an environment with no clutter or ground plane for the robot to collide with. After a number of grasps were evaluated, a large region of invalidity (negative score) was identified at $\psi = 0$, which corresponds to a region where the robotic arm would have to self-intersect with itself to reach these poses. Through iterative exploration, $\mathcal{GP}$ learns this nature, and so subsequent grasp sampling will avoid this region. The remainder of the grasp space consists of valid regions (nonnegative score), some with greater scores than others. This demonstrates the process's ability to learn more efficient and less efficient regions. Finally, the presence of outliers (invalid points within a region of estimated validity) is observed, noting that the influence of these on the surrounding grasps are minimised via the incorporation of planning uncertainty.

In addition, the $\mathcal{GP}$ encodes a metric of uncertainty in the covariance of the posterior, which resembles the confidence that the path score estimation is accurate. Each grasp is assigned an *estimation uncertainty* prescribed by the diagonal of the posterior covariance in Eq. (7.9). Explicitly, the equation for estimation uncertainty is given in Fig. 7.12.

$$\sigma_g^2 = 1 - \mathbf{K}(g, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval}) + \text{diag}(\mathbf{1} - \varsigma)]^{-1} \mathbf{K}(\mathcal{G}_{eval}, g) \quad (7.12)$$

Fig. 7.12 shows examples estimates of the path score uncertainty during the iterative grasp selection process. Five grasps have been evaluated, four being invalid and one (shown in green) being valid. The valid grasp is associated with zero uncertainty,

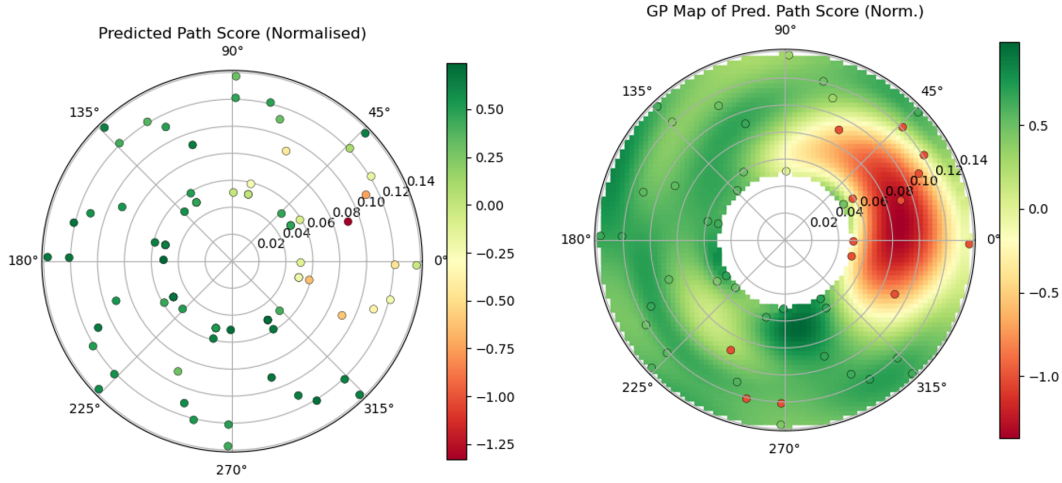


Figure 7.11: Example estimation of path score uncertainty using the Gaussian process. The left figure shows the uncertainties for discrete grasp positions in (z, ψ) space. The figure on the right generalises this to a map over the continuous domain and plots the uncertainty of evaluated grasps as scatter points.

compared to the invalid grasps, which are assigned a planning uncertainty via heteroscedastic noise to reflect the possibility that they could become valid with sufficiently longer evaluation time. In both cases, the uncertainty of evaluated grasps propagates outwards, with grasps becoming more uncertain the further they lie from evaluated grasps.

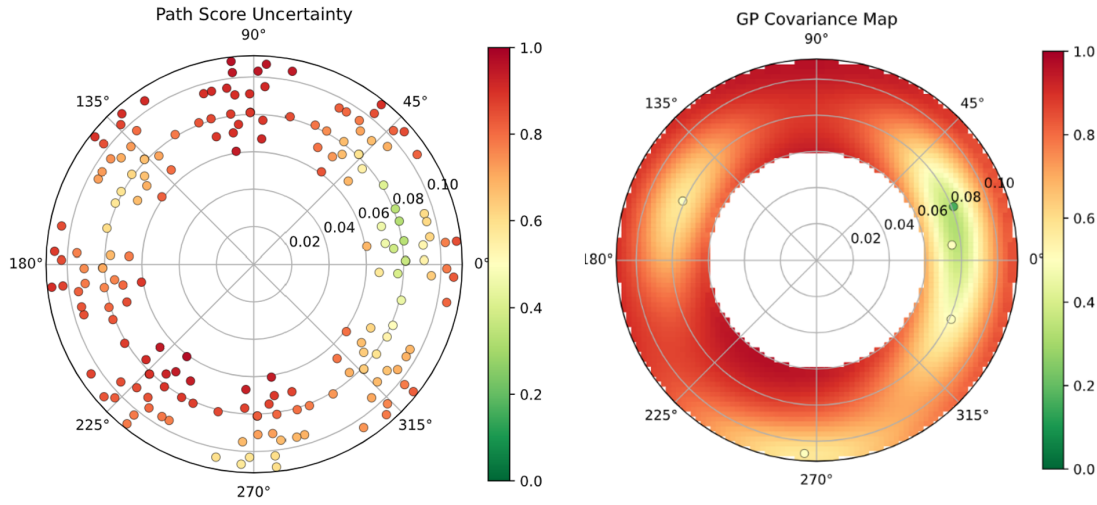


Figure 7.12: Example estimation of path score uncertainty using the Gaussian process. The left figure shows the uncertainties for discrete grasp positions in (z, ψ) space. The figure on the right generalises this to a map over the continuous domain and plots the uncertainty of evaluated grasps as scatter points.

7.2.4 Grasp Sampling

Each iteration after performing path score estimation, MALab-Grasp chooses a new grasp to evaluate accordingly. Efficient sampling of grasps to evaluate must balance *exploring* new regions of the grasp space and *exploiting* existing knowledge about where path-efficient grasps may be located. This thesis leverages the UCB acquisition function of the $\mathcal{GP}$ to do this, outlined in Eq. (4.1). MALab-Grasp uses $\kappa = 3$ as experimentation revealed this provided the desired balance between exploration and exploitation.

This biases sampling towards grasps which are predicted to be either path-efficient or that are very uncertain. The objective of this is to maximise the utility of each evaluated grasp and therefore increase the efficiency of Grasp Selection.

In addition, resampling is permitted by scaling the upper confidence bound for evaluated grasps by the planning certainty $1 - \varsigma$, as described in Grasp Evaluation. This increases the probability of grasps with very high uncertainty but prevents resampling of any grasps which are already certain. Then, the sampling weight of each grasp is given by Eq. (7.13). The grasp with the largest sampling weight is chosen each iteration for grasp evaluation. Because Grasp Synthesis provides two grasp sets, grasp sampling alternates between the two every iteration.

$$w_g = \begin{cases} UCB(g) & g \notin \mathcal{G}_{eval} \\ UCB(g)(1 - \varsigma) & g \in \mathcal{G}_{eval} \end{cases} \quad (7.13)$$

7.2.5 Optimal Grasp Choice

Once planning time has elapsed, the grasp selection stage of the grasping pipeline returns the optimal grasp. This is considered as the grasp which possesses the lowest path length while still retaining a high likelihood of grasp success. The first is captured by the normalised path score and the second by the corrected grasp score. Because of the high presence of noise in the $\mathcal{GP}$ estimates, this process is constrained purely to evaluated grasps $\mathcal{G}_{eval}$; $\mathcal{GP}$ is used solely for grasp sampling. The objective of each grasp is then given by Eq. (7.14). In this equation, the path scores are re-normalised using the maximum path length over both the downward-facing and upward-facing grasps to ensure consistency of scale.

$$J(g) = d_{path}(g) \cdot q_{grasp}(g) \quad (7.14)$$

Finally, MALab-Grasp identifies the optimal grasp as that with the highest objective over both grasp sets, shown in Eq. (7.15).

$$g^* = \underset{g \in \mathcal{G}_{eval}}{\operatorname{argmax}} J(g) \quad (7.15)$$

7.3 Grasp Evaluation Stage

The Grasp Evaluation stage of MALab-Grasp is responsible for determining the feasibility and path-efficiency of a grasp. There are two key steps in this process. First, MALab-Grasp identifies a trajectory by which to transport a target object in a spill-free manner. This targets Objective 1. Second, it adapts the amount of time given to plan this trajectory by learning the underlying distribution of planning times. This is critical in fulfilling Objective 4. The following sections detail the implementation of these two steps.

7.3.1 Trajectory Planning

To evaluate the path efficiency of a grasp, Grasp Evaluation plans the trajectory required to manipulate a target object from its current position to a specified goal position. This amounts to finding a trajectory $\mathcal{T}(x_i, g, x_f)$ for which the robot can move from an initial starting configuration-space pose $x_i \in \mathcal{C}$ to a final cartesian position $x_f \in \mathbb{R}^3$ via a given grasp pose $g \in \mathbb{R}^6$. As shown in Fig. 5.1, this trajectory is decomposed into two stages: the pickup trajectory $\mathcal{T}_{pickup} : x_i \rightarrow g$ and the move

trajectory $\mathcal{T}_{move} : g \rightarrow x_f$. Both stages impose different constraints upon the nature of the path, and so are planned separately. The algorithm for trajectory planning is described in Algorithm 3.

The planning of $\mathcal{T}_{pickup}$ follows a typical planning process, with the objective to generate a collision-free trajectory from the current robot configuration to the 6-DOF pose specified by the grasp. First, the planner attempts to generate a configuration-space pose from the provided task-space grasp pose by solving the inverse kinematics (IK) equations. Failure to identify a valid configuration indicates that the grasp pose is beyond the bounds of the robot or unreachable for another reason, allowing planning to terminate without divesting extraneous time on subsequent planning steps. After validating that the grasp pose is kinematically feasible, the planner attempts to generate a trajectory using RRTConnect within a given planning time. If a trajectory cannot be found, the grasp is evaluated as *invalid/infeasible/unsuccessful* and planning is stopped immediately. Otherwise, the algorithm proceeds to the planning of $\mathcal{T}_{move}$.

$\mathcal{T}_{move}$ represents the trajectory that transports a recently grasped piece of glassware to a desired position and is more involved because of the orientation constraints imposed on its trajectory. Throughout the entire trajectory, the grasped glassware must remain upwards, though it is free to rotate around the vertical axis. In line with this, the orientation of the goal pose is unconstrained about this vertical axis, allowing the glassware to be placed with at any orientation provided it is at the right cartesian position and fulfils the path constraints.

When assuming that labware is grasped horizontally, these orientation constraints equate to ensuring that the roll and pitch of the gripper remain constant throughout the trajectory. This is denoted in Eq. (7.16), using an orientation tolerance of $\delta = 0.4$. Although this caused slight tilting of labware during transportation, this was deemed to be cause a threat of spillage. Indeed this tolerance can likely be improved by more intelligent sampling during RRTConnect, however this is not the focus of this thesis.

$$\begin{aligned} \mathcal{T}_{move} = (\mathbf{x}_i, \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_f), \mathbf{x}_t = (x_t, y_t, z_t, \phi_t, \theta_t, \psi_t) \\ \text{subject to : } |\theta_t - \theta_i| \in \delta; |\phi_t - \phi_i| \in \delta \end{aligned} \quad (7.16)$$

Because this trajectory occurs during manipulation of labware, the gripper is first closed in the planning scene and the target object attached. This ensures that the object is also collision-free throughout the trajectory. As is the case with $\mathcal{T}_{pickup}$, IK is solved first to ensure a kinematically feasible solution, following by using RRTConnect to find $\mathcal{T}_{move}$. A grasp is evaluated as *valid* if both $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ succeed; otherwise, it is evaluated as *invalid*.

Algorithm 3 Trajectory planning / grasp evaluation algorithm

Require: maximum planning time for pickup trajectory, t_{pickup}

Require: maximum planning time for move trajectory, t_{move}

Ensure: (validity of grasp, path length)

```

1: procedure EVALUATEGRASP(grasp pose  $g$ , goal position  $x_f$ ,  $t_{pickup}$ ,  $t_{move}$ )
2:   RESETPLANNINGSCENE()
3:
4:   IKSuccessful  $\leftarrow$  SOLVEINVERSEKINEMATICS( $g$ )
5:   if not IKSuccessful then
6:     return (Invalid,  $\infty$ )
7:   end if
8:
9:   CurrentRobotPosition  $\leftarrow$  GETCURRENTROBOTPOSITION()
10:  PickupPlan  $\leftarrow$  RRTCONNECT(start: CurrentRobotPosition, goal:  $g$ , timeout:
     $t_{pickup}$ )
11:  if not PickupPlan.successful then
12:    return (Invalid: pickup,  $\infty$ )
13:  end if
14:
15:  CLOSEGRIPPERINPLANNINGSCENE()
16:  ATTACHOBJECTTOGRIPPERINPLANNINGSCENE()
17:
18:  IKSuccessful  $\leftarrow$  SOLVEINVERSEKINEMATICS( $x_f$ )
19:  if not IKSuccessful then
20:    return (Invalid,  $\infty$ )
21:  end if
22:
23:  APPLYORIENTATIONCONSTRAINTSTORRTCONNECT()
24:  MovePlan  $\leftarrow$  RRTCONNECT(start:  $g$ , goal:  $x_f$ , timeout:  $t_{move}$ )
25:  if not MovePlan.successful then
26:    return (Invalid: move,  $\infty$ )
27:  end if
28:
29:   $d_{path} \leftarrow$  CALCULATEPATHLENGTH(PickupPlan) + CALCULATEPATH-
    LENGTH(MovePlan)
30:  return (Valid,  $d_{path}$ )
31: end procedure

```

To quantify the trajectory efficiency of a grasp, the total joint-space path length of $\mathcal{T}$ is used, shown in Eq. (7.17). This provides an estimation of both the time taken to execute such a trajectory as well as energy expenditure involved. Invalid grasps are assigned an infinite path length.

$$d_{path} = \begin{cases} d(\mathcal{T}_{pickup}) + d(\mathcal{T}_{move}) & \text{valid grasp} \\ \infty & \text{invalid grasp} \end{cases} \quad (7.17)$$

The motivation for using RRTConnect is primarily due to it being well-integrated into OMPL and ROS 2, although literature suggests that it performs well in constrained environments. Ultimately, unlike existing joint grasp and motion planners, MALab-Grasp is designed to be agnostic to the trajectory planner used. Thus, RRTConnect could be replaced with more effective planners if one so desires.

7.3.2 Adaptive Evaluation Time Allocation

Most existing algorithms, including RRTConnect, manually tune the amount of time allocated for trajectory planning. However, to fulfil Objective 4, MALab-Grasp uses an adaptive allocation of evaluation (trajectory planning) time. Given that research laboratories have highly dynamic environments, the aim of this is to allow the algorithm to adapt to the changing environment without the need for manual tuning.

This thesis categorises unsuccessfully evaluated grasps into two different classes. In one class lie the grasps for which trajectory generation is impossible, possibly due to kinematic constraints or collisions with obstacles. In the second class are the grasps for which trajectory generation is possible but requiring of more time than currently allocated.

If the evaluation time is tuned to be too high, then MALab-Grasp will waste time evaluating grasps which will never become feasible. On the contrary, if the evaluation time is too low, then it will discard feasible grasps. These effects are magnified in cluttered environments, where feasible grasps are uncommon and trajectories naturally take longer to plan.

To address this, MALab-Grasp proposes an adaptive, per-grasp method for allocating evaluation time to both $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$. Rather than treating all failed planning attempts as absolute, the algorithm permits grasps to be replanned with a larger evaluation time. To accommodate this, a planning uncertainty ς is assigned to each evaluated grasp, representative of the system's confidence that the grasp has been correctly identified as valid or invalid. Valid grasps are assigned a planning

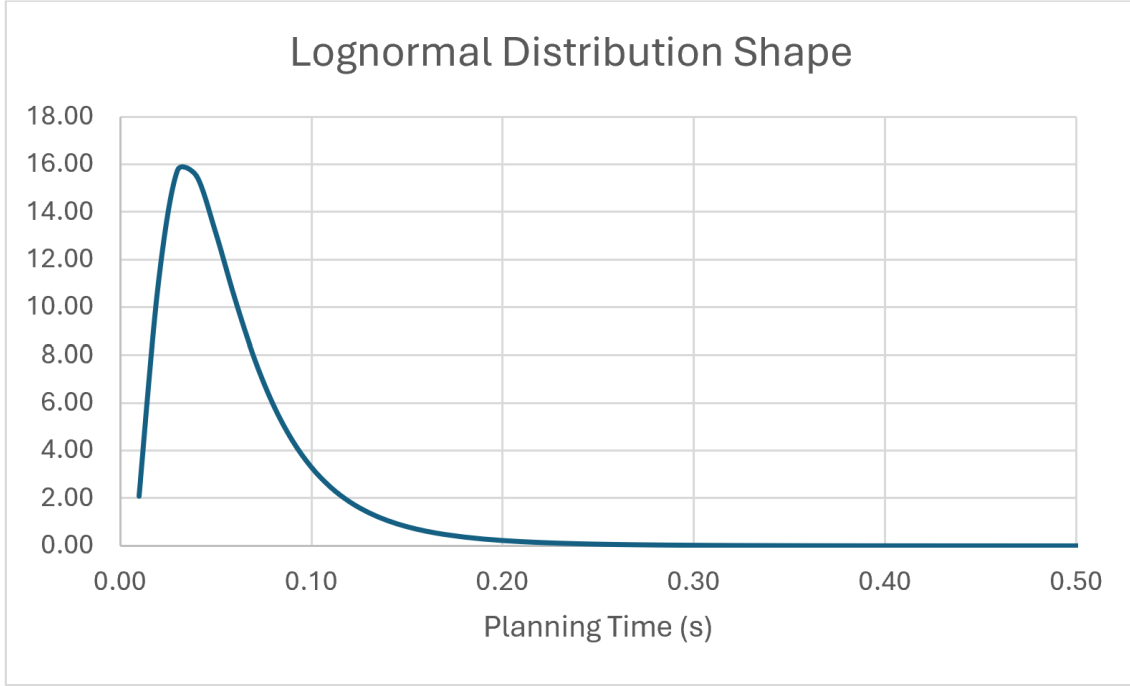


Figure 7.13: The shape of the lognormal distribution, modelled from the maximum likelihood estimates of mean and variance from evaluation time data.

uncertainty of 0, whereas invalid grasps are assigned an uncertainty according to the time which was allocated to the failed trajectory planning. Because this failure can come from either planning $\mathcal{T}_{pickup}$ or $\mathcal{T}_{move}$, there are two cases for this. The formalisation of this is given in Eq. (7.18).

$$\varsigma = \begin{cases} P(T_{pickup} > t_{pickup}) & \text{failed in pickup} \\ P(T_{move} > t_{move}) & \text{failed in move} \\ 0 & \text{valid grasp} \end{cases} \quad (7.18)$$

In Eq. (7.18), T_{pickup} and T_{move} are random variables which reflect the time taken to plan a valid pickup trajectory and move trajectory respectively. In this thesis, it will be assumed that the valid evaluation times for every grasp is drawn from the same distribution; the variation in this is well-captured in the randomness of T_{pickup} and T_{move} . As described in preliminary studies, these are assumed to be drawn from the lognormal distribution as seen in Fig. 7.13 and Eq. (7.19).

$$T_{pickup}, T_{move} \sim \text{Lognormal}(\mu, \sigma^2) \quad (7.19)$$

The choice of parameters μ and σ^2 in Eq. (7.19) are learned as the maximum likelihood (ML) estimates, calculated from the evaluation time for $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ from all successfully evaluated points. Each iteration, the parameters of the lognormal

distribution are updated to be the empirical mean and variance of the evaluation times t_i according to the ML estimates presented in Eq. (7.20).

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \log(t_i) \quad (7.20)$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (\log(t_i) - \hat{\mu})^2, \quad N \geq 3$$

This allows the algorithm to adapt to the planning environment without manual adjustment of parameters. The distributions of $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ are modelled independently, however for both of these distributions the upwards-facing and downwards-facing grasp set are combined into the same distribution.

This notion of planning uncertainty is incorporated into path score estimates via heteroscedastic noise upon each observation of the $\mathcal{GP}$. This serves to both increase the estimation uncertainty around points which have been planned with very little allocated time, as well as to penalise them less heavily during sampling. In line with this, resampling of grasps is permitted, unlike Gaussian processes. The weights of the evaluated grasps are scaled by the planning uncertainty, allowing failed grasps to be replanned with higher evaluation times. Each time a grasp is resampled, the per-grasp evaluation time specific to this grasp is increased by a factor of 2, allocating more time to attempt trajectory generation but subsequently reducing the chance that this is resampled in the future.

In addition to adjusting estimation certainty and permitting replanning, this algorithm dynamically adapts the per-grasp evaluation time for every grasp evaluation performed. Two additional Gaussian processes are used, $\mathcal{GP}_{pickup}$ and $\mathcal{GP}_{move}$, which estimate the evaluation time required for each grasp. This is grounded in the hypothesis that different environment complexities possess different expected evaluation times, and the grasps geometrically close to each other will possess similar trajectories and therefore evaluation times.

To capture the different evaluation times in different environments, each Gaussian process is given a prior mean dictated by the ML estimate of its corresponding evaluation time distribution. As the ML estimates are updated each iteration, so are the Gaussian priors. This is a key mechanism which allows the algorithm to adapt to the planning requirements of the environment without requiring manual tuning.

The notion of local sensitivity to planning time is encoded via the spatial relationships of the Gaussian processes, wherein an evaluated grasp with known evaluation times t_{pickup} and t_{move} biases the evaluation time of surrounding grasps towards these

values. The estimated evaluation time of $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$ for a given grasp is then presented in Eq. (7.21).

$$\begin{aligned}\hat{t}_{pickup}(g) &= \mathbf{K}(g, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval})]^{-1} (\mathbf{t}_{pickup} - \hat{\boldsymbol{\mu}}_{pickup}) + \hat{\boldsymbol{\mu}}_{pickup} \\ \hat{t}_{move}(g) &= \mathbf{K}(g, \mathcal{G}_{eval}) [\mathbf{K}(\mathcal{G}_{eval}, \mathcal{G}_{eval})]^{-1} (\mathbf{t}_{move} - \hat{\boldsymbol{\mu}}_{move}) + \hat{\boldsymbol{\mu}}_{move}\end{aligned}\quad (7.21)$$

Here, $\mathbf{t}$ is the vector of evaluation times for currently evaluated grasps. For valid grasps, the evaluation times are measured during the grasp evaluation process. For invalid grasps, it is assumed that the evaluation time is double the current allocated evaluation time – this aligns with the planning time allocated for this grasp when it is next resampled.

During each iteration of the grasp selection stage, $\mathcal{GP}_{pickup}$ and $\mathcal{GP}_{move}$ are updated and per-grasp evaluation time is allocated to the grasp evaluation according to the estimated evaluation time from these Gaussian processes. To be more conservative, an additional n standard deviations added to the evaluation times to account for natural variation randomness of planning. This is shown in Eq. (7.22). Grasps which are being resampled do not have these additional standard deviations, as their evaluation time has already been doubled.

$$\begin{aligned}t_{pickup} &= \hat{t}_{pickup} + n\hat{\sigma}_{pickup} \\ t_{move} &= \hat{t}_{move} + n\hat{\sigma}_{move}\end{aligned}\quad (7.22)$$

If the initialised evaluation time is larger than necessary, the algorithm aims to quickly identify the expected evaluation time and reduce the allocated time accordingly. Conversely, if the initialised evaluation time is smaller than necessary, grasps will rapidly fail and their estimated evaluation time will progressively double until a valid grasp is found and the expected evaluation time can be calculated. A summary of this adaptive planning time algorithm is shown in Algorithm 4.

Algorithm 4 Adaptive evaluation time algorithm

```

1:  $\mathcal{GP}_{pickup}, \mathcal{GP}_{move}$  ▷ GPs for tuning evaluation time
2:  $\mu_{pickup}, \sigma^2_{pickup}$  ▷ Distribution parameters of pickup planning time
3:  $\mu_{move}, \sigma^2_{move}$  ▷ Distribution parameters of move planning time
4: INITIALISEDISTRIBUTIONPARAMETERS ▷ Provided by the user
5: procedure ADAPTPLANNINGDISTRIBUTIONS(evaluated grasp:  $g$ )
6:   GETEVALUATIONRESULTS( $g$ )
7:   if  $g$  failed in the pickup trajectory then
8:      $t_{pickup} \leftarrow t_{pickup} \cdot 2$ 
9:   else if  $g$  failed in the move trajectory then
10:     $t_{move} \leftarrow t_{move} \cdot 2$ 
11:     $\hat{\mu}_{pickup}, \hat{\sigma}^2_{pickup} \leftarrow \text{COMPUTEMLE}()$  ▷ Eq. (7.20)
12:   else if  $g$  succeeded then
13:     $\hat{\mu}_{pickup}, \hat{\sigma}^2_{pickup} \leftarrow \text{COMPUTEMLE}()$ 
14:     $\hat{\mu}_{move}, \hat{\sigma}^2_{move} \leftarrow \text{COMPUTEMLE}()$ 
15:   end if
16:   if ML estimates were updated then
17:     $\mu_{pickup}, \sigma^2_{pickup} \leftarrow \hat{\mu}_{pickup}, \hat{\sigma}^2_{pickup}$ 
18:     $\mu_{move}, \sigma^2_{move} \leftarrow \hat{\mu}_{move}, \hat{\sigma}^2_{move}$ 
19:     $\mathcal{GP}_{pickup} \leftarrow \text{UPDATEPRIORWITHMEAN}(\mu_{pickup})$ 
20:     $\mathcal{GP}_{move} \leftarrow \text{UPDATEPRIORWITHMEAN}(\mu_{move})$ 
21:   end if
22:
23:   if  $g$  is invalid then
24:     $g.\varsigma \leftarrow \text{CALCULATEPLANNINGUNCERTAINTY}()$  ▷ Eq. (7.18)
25:   end if
26: end procedure
27: procedure ASSIGNEVALUATIONTIMES(grasp to be evaluated:  $g$ )
28:    $\hat{t}_{pickup} \leftarrow \text{ESTIMATEPICKUPPLANNINGTIME}()$  ▷ Eq. (7.21)
29:    $\hat{t}_{move} \leftarrow \text{ESTIMATEMOVEPLANNINGTIME}()$ 
30:   if not  $g$ .evaluated then
31:     $t_{pickup} \leftarrow \text{CALCULATEASSIGNEDPICKUPPLANNINGTIME}()$  ▷ Eq. (7.22)
32:   else
33:     $t_{pickup} \leftarrow \max(t_{pickup}, \text{CALCULATEASSIGNEDPICKUPPLANNINGTIME}())$ 
34:   end if
35:   if not  $g$ .evaluated or  $g$  failed in the pickup trajectory then
36:     $t_{move} \leftarrow \text{CALCULATEASSIGNEDMOVEPLANNINGTIME}()$  ▷ Eq. (7.22)
37:   else
38:     $t_{move} \leftarrow \max(t_{move}, \text{CALCULATEASSIGNEDMOVEPLANNINGTIME}())$ 
39:   end if
40: end procedure

```

Chapter 8

Robotic Platform Integration

In line with Objective 5, the performance of MALab-Grasp was assessed by integrating it into a high-level robotic platform. Though not the core focus of this thesis, the platform considers all aspects of an end-to-end robotic system. The exception to this is the vision system, which is considered out of scope and will only be implemented to the extent up to which other systems can function.

The robotic platform consists of four primary subsystems as shown in Fig. 8.1, each of which communicate between each other using ROS2. The central subsystem is MALab-Grasp and is described in Chapter 7. The *motion planning* subsystem plans trajectories for both moving the robotic arm and manipulating its end effector, including Grasp Evaluation for MALab-Grasp. The *execution* subsystem actuates planned trajectories on physical and/or simulated hardware and provides the interface for communication between the software and hardware layers. The *vision system* provides point cloud information to the mission planner, as well as identify obstacles for the motion planning subsystem. Finally, the *mission planning* subsystem sequences a series of planning and execution actions to effect high-level motion plans.

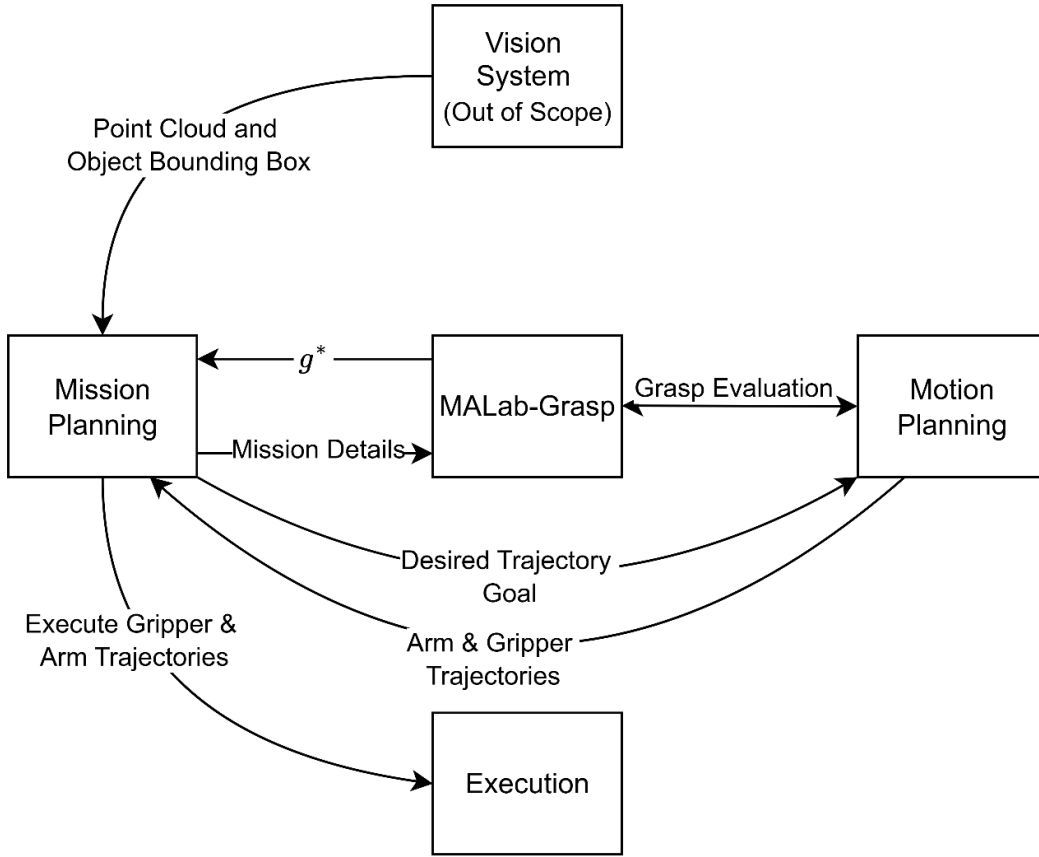


Figure 8.1: Overall system architecture, broken down into four subsystems: mission planning, grasp planning, motion planning and execution.

8.1 Mission Planning

The mission planning subsystem is designed as the primary user-facing system, providing the ability to complete the primary high-level goal of this system: to manipulate a piece of glassware from one position to another. This objective, termed a *mission* $\mathcal{P}$, can be queried directly from a ROS2 action or implemented as part of a higher-level automation pipeline. The steps completed during mission planning are summarised as below and in Fig. 8.2.

1. The user provides an object ID identifying the target object $\mathcal{O}$ and a desired cartesian position to move the (base of the) object to.
2. The bounding box $\mathcal{B}(\mathcal{O})$ is retrieved from the vision system.
3. The bounding box and environment point cloud is provided to the MALab-Grasp, which identifies the optimal grasp g^* after a user-provided planning time T .

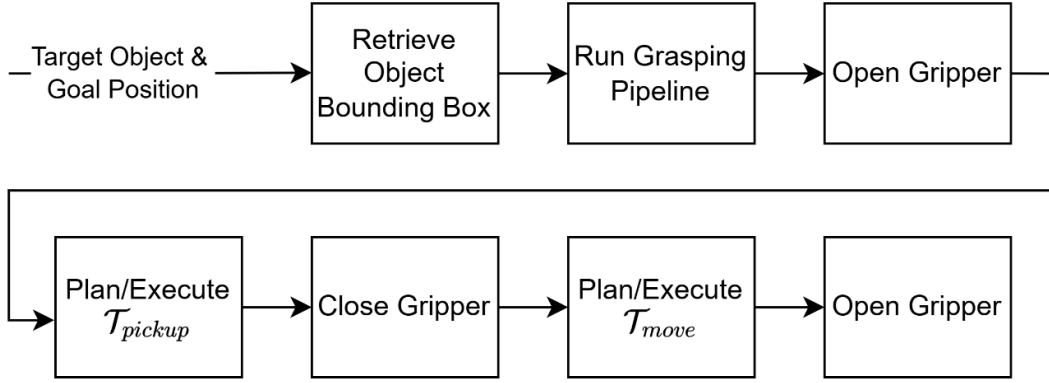


Figure 8.2: Process of planning and executing a mission.

4. The robotic gripper is opened using trajectory planning & execution.
5. The optimal grasp g^* is replanned to determine its first trajectory $\mathcal{T}_{pickup}$, which is then executed.
6. The robotic gripper is closed around $\mathcal{O}$ using trajectory planning & execution.
7. The optimal grasp g^* is replanned to determine its second trajectory $\mathcal{T}_{move}$, which is then executed.
8. The robotic gripper is opened using trajectory planning & execution.

8.2 Motion Planning

The motion planning subsystem provides trajectory generation capabilities for both Grasp Evaluation and mission execution. Grasp Evaluation is discussed in Section 7.3, with the same process being used for Steps 4 & 6 of mission planning. In a similar manner, RRTConnect is used to generate trajectories for gripper opening and closing; these are trivial motion planning problems that don't require significant design.

8.3 Execution

The execution subsystem encapsulates the hardware used in the system and the software used to actuate the hardware. A Franka Research 3 (FR3) robotic arm with a parallel-jaw gripper is used for testing and experimentation, although the proposed algorithms are designed to be robot-agnostic. This hardware is shown in Fig. 8.3.

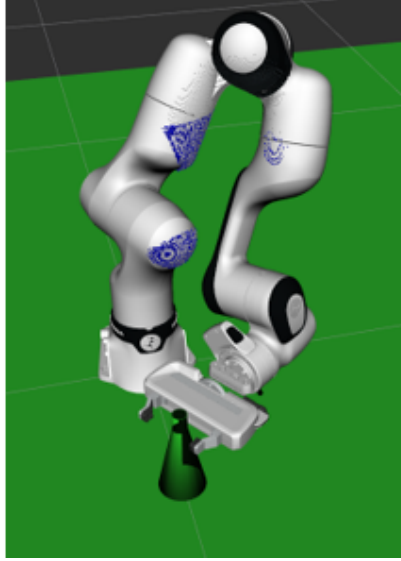


Figure 8.3: The Franka Research 3 (FR3) robot used for trajectory execution. A parallel-jaw gripper is attached as the end effector for grasping of labware.

Experiments will be conducted in a Gazebo simulation, with the FR3 modelled in the simulator using a unified robot description format (URDF) file. This file modifies the publicly available URDF file for the FR3, allowing a simulated camera to be attached to the robot wrist and overriding the default controllers in favour of custom hardware control. Execution of planned trajectories for both the FR3 and the gripper are facilitated through MoveIt2, providing a high-level abstraction of the underlying control required to actuate a robot.

8.4 Vision System

The vision system is considered out of the scope of this thesis. As such, it is only implemented to the point up to which other systems can function. Rather than maintaining a real-time point cloud of the environment, the vision system stores precomputed point clouds as files. A description of how these point clouds were generated is given in Chapter 9.

The vision system is also responsible for identifying the locations and bounding boxes of objects, both for point cloud cropping and to provide the motion planning subsystem with so that it can plan collision-free trajectories. Given that this robotic system will be tested in simulation, the vision system keeps a record of all objects it loads into the Gazebo environment. As objects are moved around the scene, the vision system updates its internal knowledge of object positions. It then forwards this onto the motion planning subsystem, which registers these changes in its planning

scene. The size of object bounding boxes are precomputed based on the known geometry of labware.

Chapter 9

Experimental Setup

To validate whether MALab-Grasp meets the objectives outlined in Chapter 5, the performance of the integrated robotic system was evaluated against a set of simulated scenarios within the Gazebo ecosystem. This chapter outlines the process of designing scenarios and the studies to evaluate MALab-Grasp’s performance.

9.1 Scenario Design

Each scenario that MALab-Grasp was evaluated in comprises of two components: a *target object* and an *environment*.

The *target object* refers to the piece of labware that MALab-Grasp is required to grasp. To be representative of real-life chemical laboratories, target objects were modelled on labware at a chemical research laboratory at the University of Queensland, Australia. The approximate measurements of a conical flask, round-bottom flask, volumetric flask, measuring cylinder and two beakers were recorded and subsequently modelled. These were then converted into STL meshes with low-fidelity (for collision detection) and medium-fidelity (for visuals) to be used in Gazebo. The original objects are shown in Fig. 9.1 and the CAD models in Fig. 9.2. These six objects were chosen to represent a wide range of shapes and sizes a labware grasper may encounter. Notably, it includes objects that have curved surfaces and beakers which lack the neck typically associated with flasks. A small beaker was also initially included in the experiments, however the robotic arm consistently failed to grasp the object without colliding with the ground. Therefore, it was excluded from further experiments.

The *environment* represents the arrangement of obstacles around a target object and, together with the target object, forms a unique scenario. Three simulated



Figure 9.1: Glassware collected from a chemical research lab at the University of Queensland, Australia, used as inspiration when designing target objects. (Measuring cylinder not shown here).

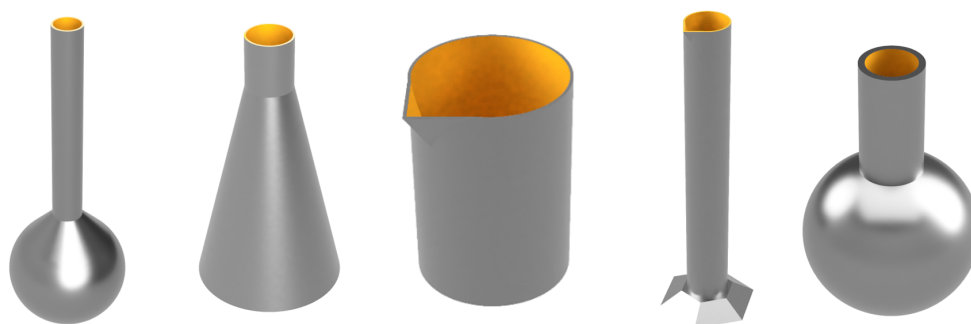


Figure 9.2: CAD Models of target objects, from left to right a 1) volumetric flask, 2) conic flask, 3) beaker in 2 different sizes, 4) measuring cylinder, & 5) round bottom flask.

environments are used in this thesis, each with differing levels of complexity and a different goal position the robotic arm must move the target object to. In each environment, the target object is placed on the ground directly in front of the robot with a position of $(0.3, 0, 0)$. The environments are detailed below, with a visual description provided in Fig. 9.3.

- *Simple Environment:* the system must move the target object to a fixed but arbitrary position of $(0.3, 0.3, 0.1)$. No cluttered objects are present.
- *Table Environment:* a large cube (“table”) is placed next to target object, and the system must place the object in the centre of the table.
- *Cluttered Environment:* the target object is placed in a row of beakers. The system must move the object into a vacant slot in a second row of beakers, avoiding surrounding obstacles.

These three environments were designed to be representative of common laboratory situations and pose different challenges to the system. The simple environment

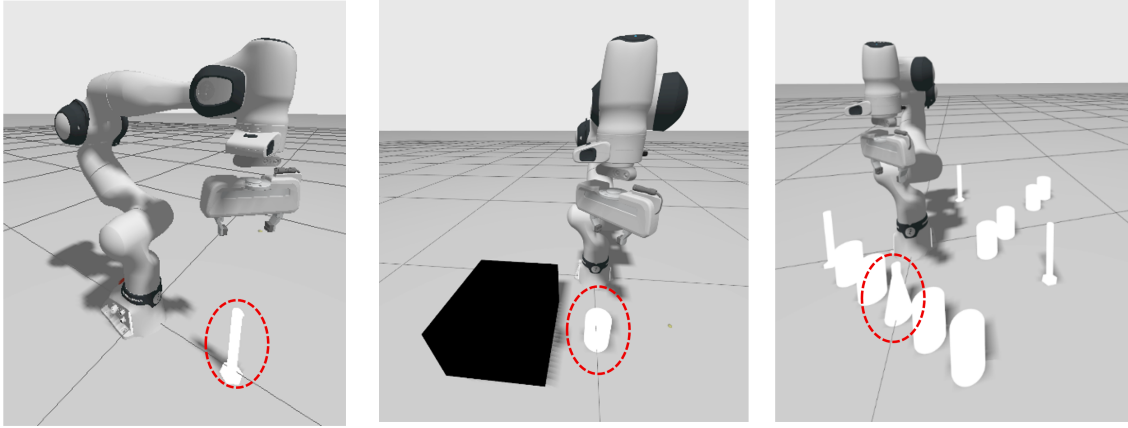


Figure 9.3: The three simulated environments used for testing, with the target object circled. Environments are referred to as the simple, table and cluttered environment respectively.

provides a baseline case where the robot must avoid colliding with itself or the ground, but where there are no other obstacles. The table environment is an intermediate case where one side of the object is blocked by an obstacle, and the robot must lift the object onto the table without colliding with it. The cluttered environment is expected to be the most difficult case, where there are collision obstacles both at the current and the goal pose of the object. In total, MALab-Grasp is tested over 18 scenarios. These are summarised in Fig. 9.4.

Experimental Scenarios						
	Conic Flask	Volumetric Flask	Round-Bottom Flask	Beaker Med.	Beaker Lrg.	Measuring Cylinder
Simple Environment						
Table Environment						
Cluttered Environment						

Figure 9.4: A matrix of scenarios MALab-Grasp is evaluated upon, comprising of a target object and an environment.

9.2 Vision Data Generation

Given that the vision system is considered out of scope, point cloud data was collected and provided manually to the robotic system. A simulated RealSense D435 RGB-D camera was attached in an eye-in-hand manner to the robot’s wrist, collecting point cloud data from the simulated environment. For each target object, the robot was manually moved around the object to collect multi-view information. Subsequently, an OctoMap was used to fuse these point clouds together into a voxelised global occupancy map. The ground plane is further segmented out to minimise map size and prevent accidental grasping on ground surfaces. The centres of the voxels in the resulting occupancy map were saved as a point cloud file, disregarding any colour information received from the sensor. The relevant file was then given to MALab-Grasp during grasp planning. In practice, a dedicated vision system would provide a constantly updating point cloud to MALab-Grasp, but during experimentation it was simpler to collect this information once and save it to a file. Examples of the point clouds generated are shown in Fig. 9.5.

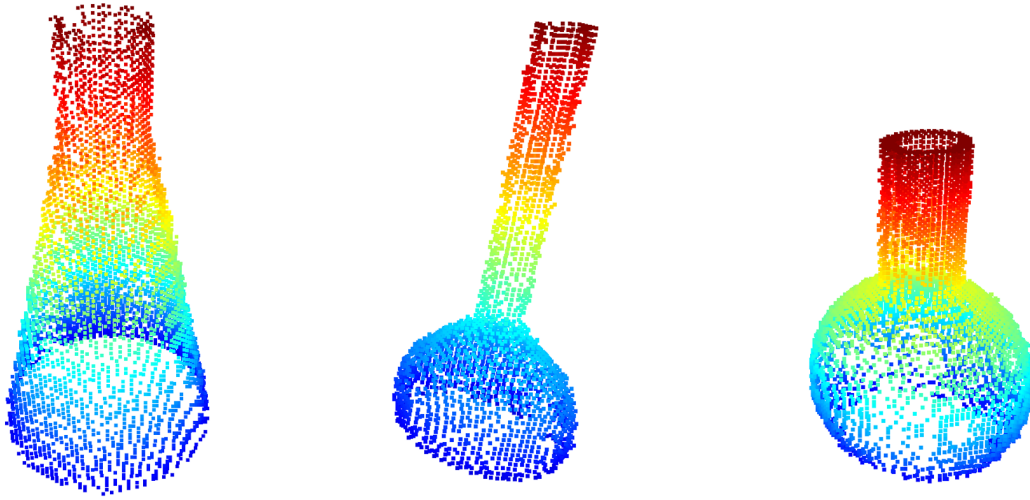


Figure 9.5: Generated point clouds of the conic flask, volumetric flask and round-bottom flask. Colour information represents vertical height and is not indicative of the true object colour.

9.3 Studies

This thesis conducts three studies on MALab-Grasp. In Study 1, the grasp success rate of the Grasp Synthesis stage of MALab-Grasp is evaluated. This is targeted towards Objective 2. In Study 2, the entire pipeline of MALab-Grasp is assessed for its ability to identify feasible and path-efficient grasps. This targets Objective 3. By evaluating performance over six types of labware and three unique environments, both of these studies also evaluate Objective 4. Finally, Study 3 examines the adaptive evaluation time algorithm proposed, and targets Objective 4. Objective 1 (identification of side-on grasps) and Objective 5 (integration into a robotic platform) are trivially validated by performing these studies.

9.3.1 Study 1: Grasp Success Rate of Grasp Synthesis

Study 1 aims to quantify whether MALab-Grasp can identify high-quality, stable grasps for labware. In this study, only the Grasp Synthesis stage is used. The key metric used in this study is *grasp success rate*, which quantifies the proportion of a set of grasps which can grasp and manipulate an object without it being dropped. This was measured in simulation using the simple environment, with a grasp being recorded as successful if it can move an object to the arbitrary position (0.4, 0.3, 0.4) while maintaining the prescribed orientation constraints.

For each target object, the grasp success rate over the five grasps with highest

corrected grasp score was measured. This was compared to a baseline method, termed *AnyGrasp*, wherein the five grasps with highest AnyGrasp grasp score are used. In this baseline, the flatness filter was still applied to ensure that the domain requirements are fulfilled, however no other parts of the MALab-Grasp were applied except for non-maxima suppression, which is standard practice in grasping literature. Grasps which are not kinematically feasible (i.e., where trajectories to grasp and move the object could not be found) were not considered.

9.3.2 Study 2: Grasp Efficiency of MALab-Grasp

Study 2 attempts to determine whether the MALab-Grasp identifies path-efficient grasps faster than the baseline method. Two key metrics were reported in this study: time-to-valid-grasp and shortest path length. Time-to-valid-grasp (TTVG) was measured as the total planning time required to identify a trajectory-feasible grasp. This metric aims to evaluate how well the grasping pipeline can intelligently choose the order in which to evaluate grasps, as to maximise the likelihood of finding a feasible grasp. Therefore, this metric is also applicable towards laboratories where there is a high proportion of infeasible grasps, even if path efficiency is not a factor. The shortest path length simply reported the joint-space path length of the grasp which results in the shortest overall trajectory from initial robot pose to desired object position.

In addition, this study characterises the anytime performance of the MALab-Grasp – that is, how the shortest path length evolves as a function of the total planning time. This behaviour, combined with TTVG, assesses how efficient and fast MALab-Grasp is. The shortest path lengths were recorded at the discrete times of 10s, 50s and 100s, as well as the time evolution throughout the entire planning phase. Finally, the time evolution of several auxiliary trends were reported for each trial, including variance of the grasps sampled, proportion of sampled grasps found as feasible, and the average evaluation time per grasp. These metrics were used to supplement the analysis of the primary metrics.

The results for 18 scenarios were analysed, comprising of all 6 target objects in all 3 environments. Due to the stochastic nature of planning and grasp selection, scenarios were repeated and their results aggregated over five trials. In each trial, the grasping pipeline was allocated 180s of planning time for the grasp selection process.

MALab-Grasp’s performance was compared to several baselines, in which grasps were ordered manually rather than applying Grasp Selection. The primary baseline is referred to as *AnyGrasp* and reflects a process typical of standard grasp planners.

The candidate grasps generated by AnyGrasp were used in this baseline, applying only flatness filtering and non-maxima suppression as in Study 1. Grasps were then iteratively evaluated in order of highest to lowest AnyGrasp grasp score until planning time has elapsed. To collect comprehensive data for arbitrary planning times, this study allocates an infinite planning time to the baseline methods; that is, they evaluated their entire candidate grasp set. A second baseline is also reported, *AnyGrasp Random*, which behaved identically except for that grasps are ordered randomly rather than according to AnyGrasp grasp score. In each of these baselines, the per-grasp evaluation time for (the trajectory planning times for $\mathcal{T}_{pickup}$ and $\mathcal{T}_{move}$) was tuned by hand.

Secondary to these baselines are three other baselines, used as auxiliary results to understand how the different stages of the grasping pipeline influence its performance. These are termed *Post-Symmetry AnyGrasp*, *Post-Symmetry Corrected* and *Post-Symmetry Random*, wherein symmetry-based grasp generation was applied and grasps are ordered according to their AnyGrasp grasp score, corrected grasp score, and in random order respectively.

9.3.3 Study 3: Adaption Evaluation Time Effectiveness

Study 3 aims to determine whether the adaptive evaluation time algorithm can effectively learn the underlying distribution of evaluation time. This was assessed by comparing the performance of MALab-Grasp with different initialisations of evaluation time to determine whether the algorithm can perform equally well regardless of initialisation. In addition, three different values of n (as described in Eq. (7.22)) are compared to analyse the effect of this hyperparameter.

In the ideal case, a full ablation study would be conducted across each scenario. However, due to time limitations of this thesis, this study was only conducted on the (*beaker large, simple environment*) scenario. For each combination of n and evaluation time initialisation, 10 trials were conducted and the median shortest path length over these trials reported. For each trial, MALab-Grasp was run for 180 seconds.

Chapter 10

Results

10.1 Study 1 – Grasp Success Rate

Fig. 10.1 compares the grasp success rate for the five highest-score grasps using AnyGrasp and MALab-Grasp. Successful grasps are also qualitatively decomposed into stable and unstable grasps. Stable grasps show no or minimal slipping or oscillation during manipulation; unstable grasps slip from their grasp position or oscillate significantly even after reaching their final position.

AnyGrasp exhibits an overall success rate of 33% (10/30) and can grasp three out of the six unique target objects. Of the 10 successful grasps, 30% were classified as stable. In comparison, MALab-Grasp demonstrates a significantly higher aggregate success rate of 63% (19/30) and can grasp all target objects except for the large beaker. Furthermore, there is a drastic increase in the stability of grasps, with 89% of successful grasps classified as stable.

On a per-object level, MALab-Grasp increased the success rate in all cases except for the measuring cylinder, in which both methods demonstrated a success rate of 100% but it increased the stability of the grasps. In 4/5 cases for the AnyGrasp, the measuring cylinder was grasped at a slight offset from the centre of mass, leading to oscillations during object manipulation. This was corrected with MALab-Grasp, wherein all grasps passed through the object centre of mass. On the contrary, the beakers were the most difficult to grasp for both methods, with neither able to grasp the large beaker and the MALab-Grasp only being able to grasp the medium beaker in one of five trials.

Two primary failure modes were identified in unsuccessful grasping, termed *dropped* and *missed*. In *dropped* grasps, the gripper could be closed around the object, but it was dropped during movement. In *missed* grasps, the grasp point lay on a slanted

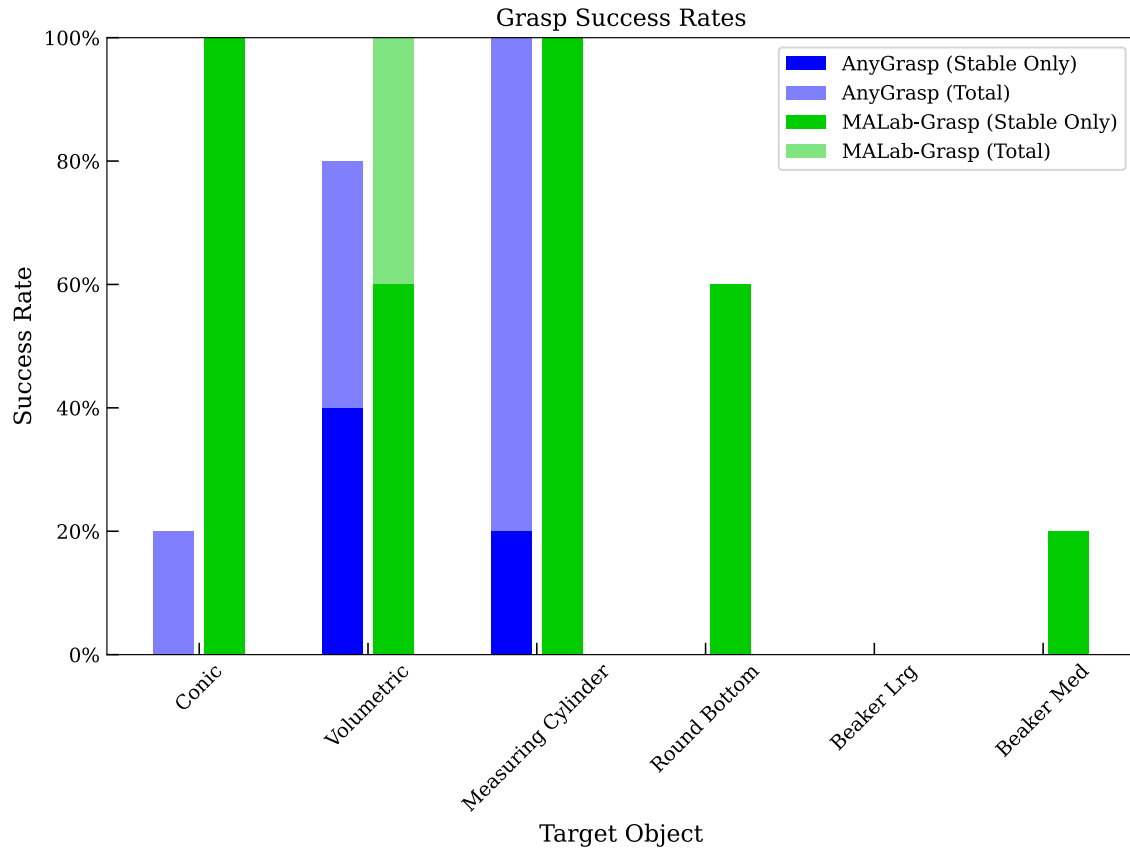


Figure 10.1: Comparison of the grasp success rates for AnyGrasp and MALab-Grasp. MALab-Grasp proposed displays significantly higher grasp success rates than the AnyGrasp baseline case.

surface, and so the action of closing the gripper pushed the object away rather than grasping it. The breakdown of unsuccessful grasps into these two categories is shown for AnyGrasp and MALab-Grasp in Table 10.1.

AnyGrasp failed due to missed grasping in all trials of the round-bottom flask, large beaker and medium beaker, with the most failures in the conic and volumetric flask being due to dropped grasps. Comparatively, MALab-Grasp displayed an equal tendency to fail due to *dropped* and *missed*. The exception to this is the large beaker, which consistently failed due to missed grasps.

Table 10.1: Decomposition of unsuccessful grasps into two failure modes: “dropped” and “missed”. The AnyGrasp displays a greater propensity for the “missed” failure mode, whereas the MALab-Grasp fails due to “dropped” and “missed” equally.

Target Object	Baseline		Grasping Pipeline	
	Dropped	Missed	Dropped	Missed
Conic Flask	75%	25%	—	—
Volumetric Flask	100%	0%	—	—
Measuring Cylinder	—	—	—	—
Round-Bottom Flask	0%	100%	100%	0%
Beaker Lrg	0%	100%	0%	100%
Beaker Med	0%	100%	50%	50%

10.2 Study 2 – Path Efficiency

10.2.1 Time to Valid Grasp (TTVG)

Fig. 10.2 compares the median TTVG over five trials for MALab-Grasp versus AnyGrasp. In 78% of scenarios, MALab-Grasp reports lower TTVG, including four cases in which MALab-Grasp identified a valid grasp where AnyGrasp could not. Two outliers exist in which AnyGrasp marginally outperforms MALab-Grasp: (*volumetric flask, simple environment*) and (*measuring cylinder, simple environment*). A third outlier also exists at (*volumetric flask, table environment*) where MALab-Grasp is significantly worse. However, in each of these cases the interquartile ranges of

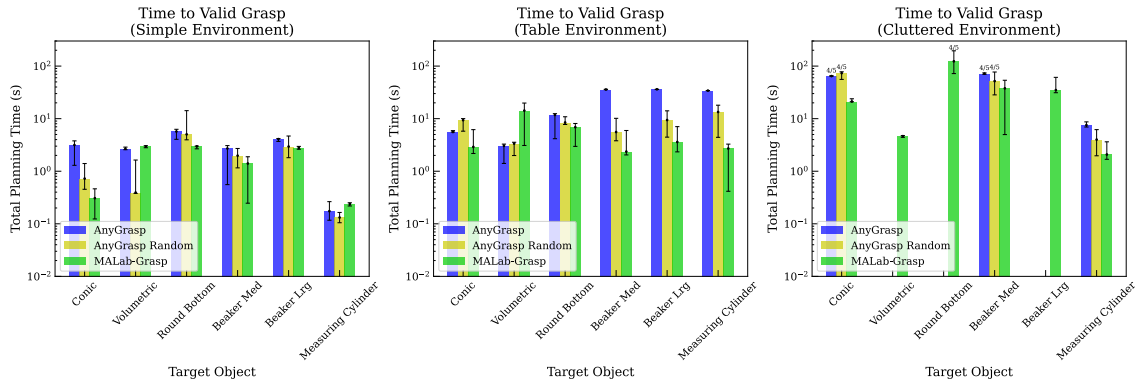


Figure 10.2: The median TTVG over five trials of a) AnyGrasp, b) AnyGrasp Random, and c) MALab-Grasp. Interquartile ranges are reported as error bars and any scenarios in which not all trials found a valid grasp are annotated as such. MALab-Grasp exhibits lower TTVG than both the AnyGrasp and AnyGrasp Random in 78% of cases.

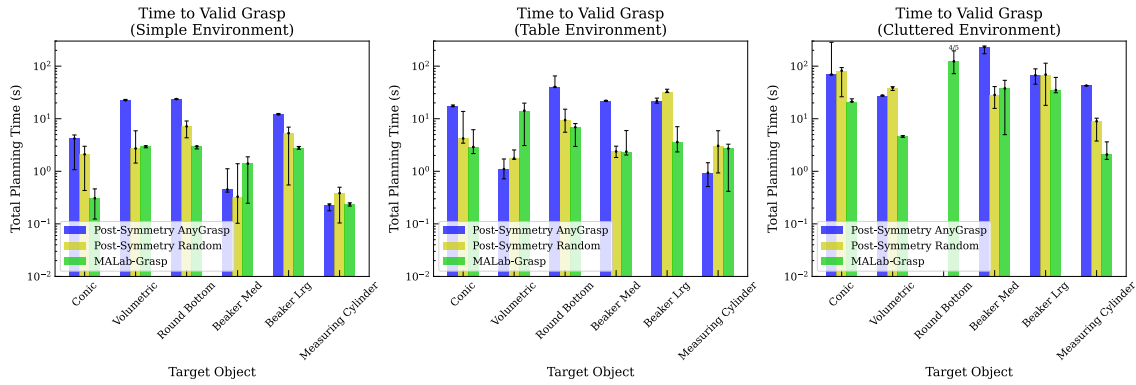


Figure 10.3: The median TTVG over five trials of a) Post-Symmetry AnyGrasp, b) Post-Symmetry Random, and c) MALab-Grasp. Interquartile ranges are reported as error bars and any scenarios in which not all trials found a valid grasp are annotated as such. Results for the cluttered environment supports the findings made from Fig. 10.2.

MALab-Grasp and AnyGrasp overlap, indicating minimal statistical significance.

Fig. 10.2 also includes the *AnyGrasp Random* baseline, wherein grasps are ordered randomly rather than via the AnyGrasp grasp score. No clear trend is shown between the two baselines, with AnyGrasp performing better than random in some cases and worse than random in others. The largest discrepancies are observed in (*beaker med*, *table environment*), (*beaker lrg*, *table environment*), and (*measuring cylinder*, *table environment*), wherein AnyGrasp ordering performs egregiously bad compared to random ordering. MALab-Grasp exhibits lower TTVG than *AnyGrasp Random* in 78% of cases, identical to *AnyGrasp* but with a different set of outliers. Two of these outliers occur with the volumetric flask, wherein MALab-Grasp is also outperformed by *AnyGrasp*. Both of these outliers are statistically significant based on the clear separation between interquartile ranges. The third outlier, (*round bottom flask*, *table environment*), shows no evidence of statistical significance.

Because AnyGrasp failed to identify a valid grasp in five of the six cluttered scenarios without post-symmetry grasp generation, there is no baseline to compare the performance of MALab-Grasp within these cases. To rectify this, the post-symmetry baselines *Post-Symmetry AnyGrasp* and *Post-Symmetry Random* were used. Fig. 10.3 shows the same comparisons made in Fig. 10.2 with these new baselines. MALab-Grasp exhibits a lower TTVG in all cluttered scenarios when compared to *Post-Symmetry AnyGrasp*, and all but one when compared to *Post-Symmetry Random*. The results for the remainder of the scenarios support the trends shown in Fig. 10.2.

Fig. 10.4 performs the same comparison with grasps ordered by the corrected grasp score (*Post-Symmetry Corrected*). In the simple environment, there is no clear trend between the *Post-Symmetry Corrected* and *Post-Symmetry Random*, with corrected

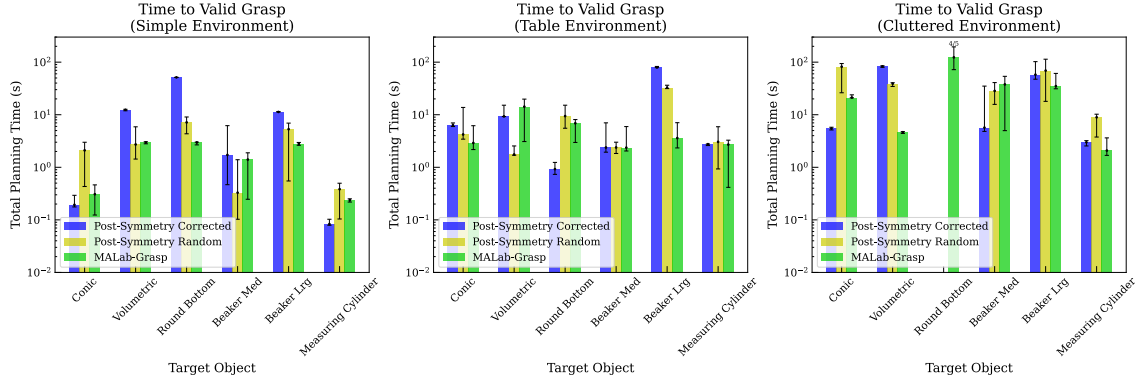


Figure 10.4: The median TTVG over five trials of a) Post-Symmetry Corrected, b) Post-Symmetry Random, and c) MALab-Grasp. Interquartile ranges are reported as error bars and any scenarios in which not all trials found a valid grasp are annotated as such. Post-Symmetry Corrected intermittently displays higher performance than MALab-Grasp and baseline methods.

grasp scoring improving TTVG in some cases and worsening it in others. In the more complex environments (table and cluttered), corrected grasp scoring improves TTVG significantly in most scenarios.

Fig. 10.5 summarises the results of this section by presenting the percentage improvement that MALab-Grasp offers over each of the baseline methods. Scenarios are aggregated by their environment and the median improvement reported over all target objects. The *Post-Symmetry AnyGrasp* baseline performs the worst out of all baseline cases, with MALab-Grasp consistently improving the TTVG by around 80% in all environments. All baselines ordered by AnyGrasp grasp score also perform worse than their randomly ordered counterparts. In comparison, in the table and cluttered environments *Post-Symmetry Corrected* performs significantly better than all other baselines. In the table environment, its performance is commensurate with that of MALab-Grasp. In all other cases, MALab-Grasp improves TTVG upon baseline methods by between 18 and 80 percent. Because the results for each environment are aggregated over only six objects, the interquartile ranges in Fig. 10.5 are substantial.

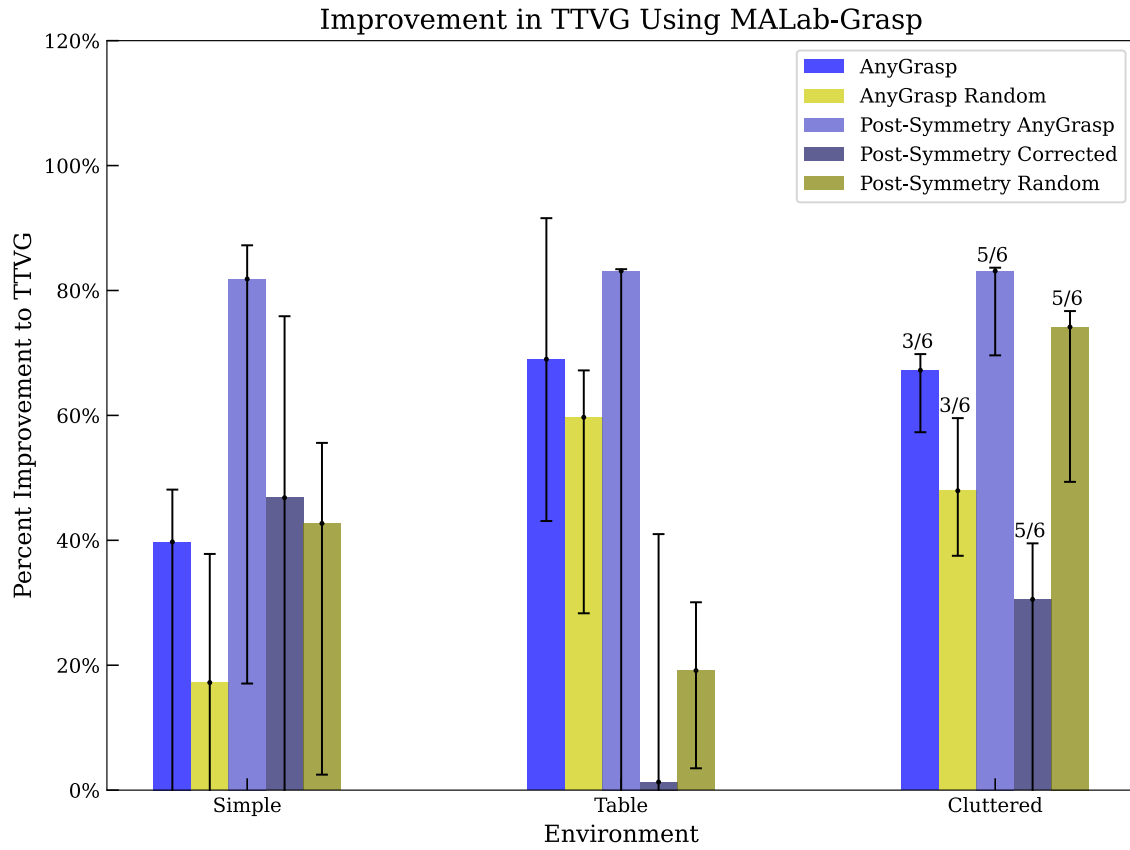


Figure 10.5: The median percent improvement in TTVG, aggregated over all target objects within a single environment. The percent improvement is reported as the degree to which using MALab-Grasp reduces the TTVG when compared to each baseline method. In most cases, MALab-Grasp reduces the TTVG by between 18% and 80%.

10.2.2 Identifying Valid Grasps

The results for Section 10.2.1 show a clear difference between MALab-Grasp and AnyGrasp not only in the TTVG, but the proportion of trials in which a valid grasp was identified. Table 10.2 summarises these results. Although each method identified a trajectory-feasible grasp in 100% of simple and table environment scenarios, MALab-Grasp significantly improves upon AnyGrasp’s ability to identify valid grasps in adversarial scenarios. Out of the 30 trials conducted over six target objects, MALab-Grasp identified a trajectory-valid grasp in 96.7% of trials in the cluttered environment. In comparison, AnyGrasp only identified a valid grasp 43.3% of cases. The application of post-symmetry grasp generation increased the rate to 83.3%, but it was still not sufficient to identify any valid grasps in the (round bottom, cluttered) scenario.

Table 10.2: Proportion of trials in which a valid grasp could be identified, grouped by environment.

Method	Simple	Table	Cluttered	All
MALab-Grasp	100%	100%	96.7%	98.9%
AnyGrasp	100%	100%	43.3%	81.1%
Post-Symmetry AnyGrasp	100%	100%	83.3%	94.4%

10.2.3 Shortest Path Length

Fig. 10.6 compares the median shortest path length found by MAlab-Grasp and AnyGrasp over the entirety of the planning process. Because these results are reported over the entire planning process, the order in which grasps are evaluated are not relevant; therefore, the random baselines are not used.

MAlab-Grasp identified a shorter path than AnyGrasp in 89% of scenarios, including three in which AnyGrasp could not identify a valid path. Two outliers occurred, with one (*beaker med, simple*) being statistically significant. Of all environments, the table environment exhibited the largest decrease in path length when using MAlab-Grasp.

As in Section 10.2.1, the performance of the MAlab-Grasp was also compared with the post-symmetry baseline *Post-Symmetry AnyGrasp*, shown in Fig. 10.7. This baseline identified shorter path lengths than AnyGrasp in all scenarios, including two scenarios in which AnyGrasp was not able to identify a valid grasp. When compared to this post-symmetry baseline, MAlab-Grasp identified a shorter path in 67% of cases.

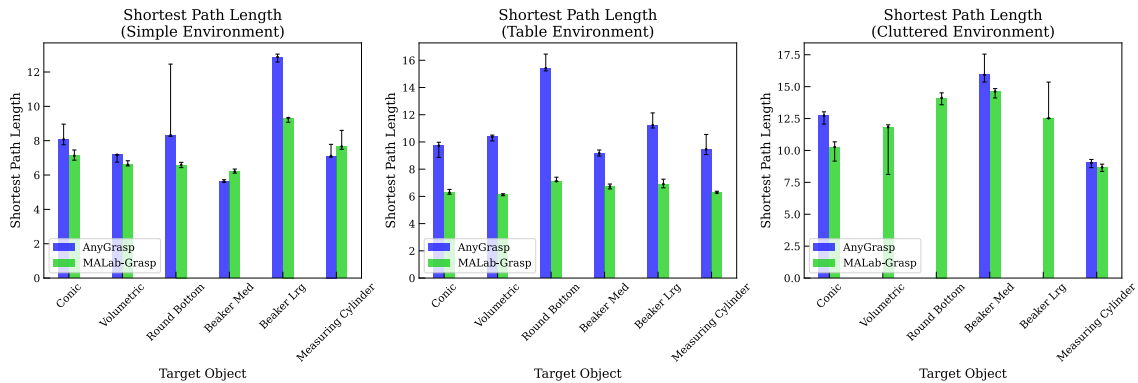


Figure 10.6: The median shortest path length over five trials of a) AnyGrasp and b) MAlab-Grasp. MAlab-Grasp identifies grasps with shorter path lengths than the baseline method in 83% of trials.

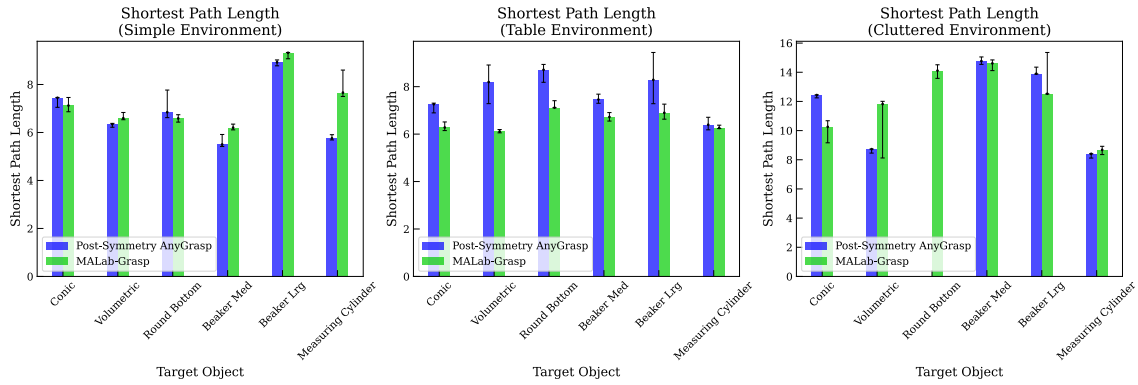


Figure 10.7: The median shortest path length over five trials of a) Post-Symmetry AnyGrasp and b) MALab-Grasp. The grasping pipeline identifies grasps with shorter path lengths than the baseline method in 83% of trials.

Fig. 10.8 shows the percentage improvement that MALab-Grasp offers over the baseline methods in each environment. The results support those found in Fig. 10.6 - Fig. 10.7, indicating that MALab-Grasp performs significantly better than the baselines in the table environment. This improvement is approximately 35% compared to *AnyGrasp*, and 15% compared to *Post-Symmetry AnyGrasp*. Across all environments, applying post-symmetry accounted for the majority of the path length improvement. To this end, MALab-Grasp offered no statistically significant improvement in shortest path length over *Post-Symmetry AnyGrasp* in the simple and cluttered environments.

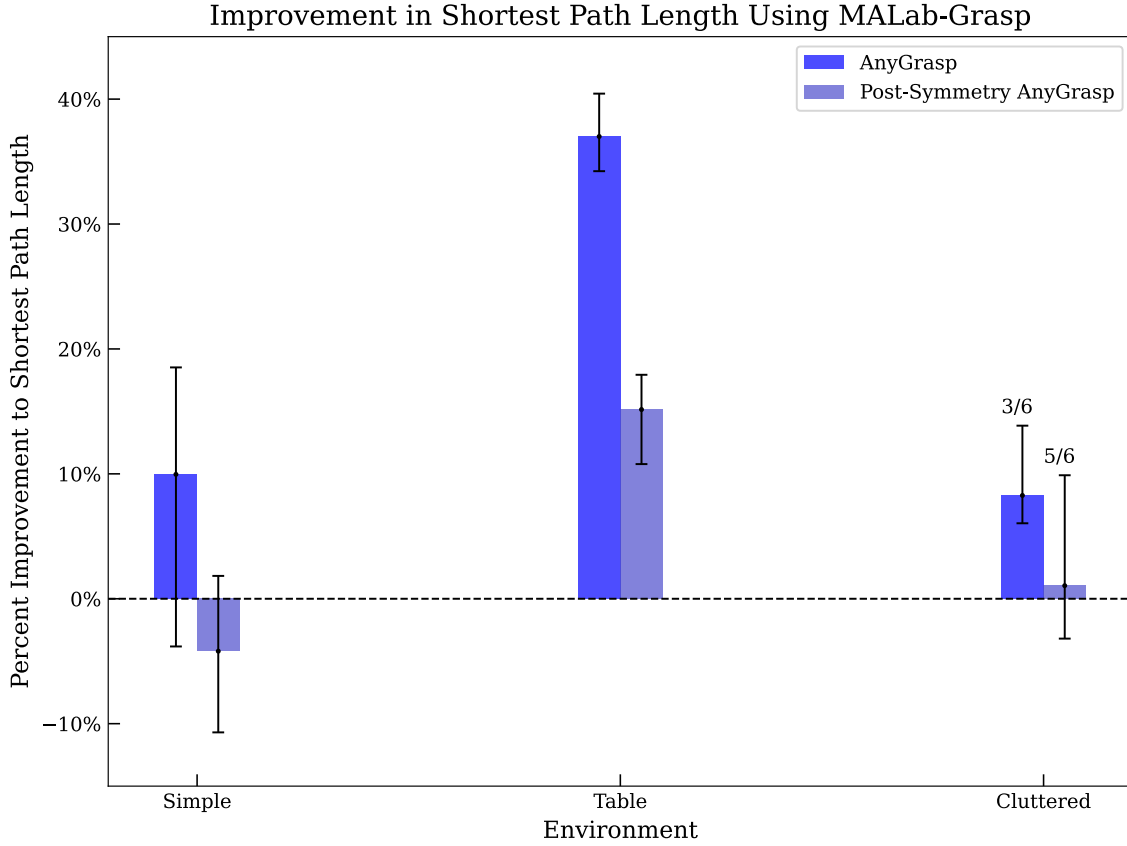


Figure 10.8: The median percent improvement in shortest path length, aggregated over all target objects within a single environment. The percent improvement is reported as the degree to which using MALab-Grasp reduces the shortest path length when compared to each baseline method. The table environment shows the most significant improvement, with MALab-Grasp reducing the shortest path length by over 35% (AnyGrasp) or 15% (AnyGrasp Random).

10.2.4 Anytime performance

Fig. 10.9 captures the anytime performance of MALab-Grasp compared to AnyGrasp and *AnyGrasp Random*. It displays the progression of shortest path lengths throughout planning for the conic flask in each of the three environments, with the results for other scenarios shown in Appendix A. In the conic flask scenarios, the grasping pipeline consistently identifies grasps with lower path lengths compared to the baseline methods irrespective of planning time. This disparity is most noted in the table environment, where there was an over 25% decrease in the path length at all stages of the planning process. *AnyGrasp Random* clearly outperformed AnyGrasp in the conic flask scenarios. This is evidenced by the *AnyGrasp Random* curve decreasing faster in terms of path length, despite both baselines reaching the same final path length. Equivalently, the area between the two curves is significant. The only case in which MALab-Grasp was outperformed in the conic flask scenarios was in the initial

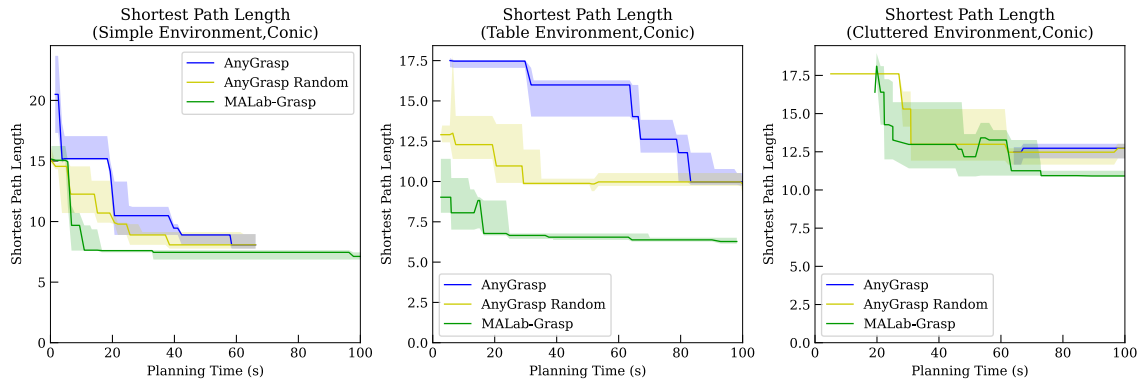


Figure 10.9: The time evolution of shortest path length for the AnyGrasp, AnyGrasp Random, and MALab-Grasp for the conic flask scenarios. The median results over five trials are reported with the shaded regions displaying the interquartile ranges.

stages of the cluttered environment, where MALab-Grasp took approximately 20s longer than *AnyGrasp Random* to identify a valid grasp. After this time, however, it identified shorter path lengths faster than the baseline.

The results for other scenarios, shown in Appendix A, largely support these results. Out of the 15 scenarios in which both baseline methods and MALab-Grasp find valid solutions, MALab-Grasp displays better anytime performance in 10 cases (71%). This was qualitatively determined by assessing whether the trendline for MALab-Grasp lay below the baselines for the large majority of the planning time. When considering the outliers to this statistic, in only one case did one of the baseline methods clearly outperform MALab-Grasp throughout the entire planning process: in the (*cluttered environment, measuring cylinder*) case.

To capture the effect that applying symmetry has on the results, Fig. 10.10 displays the same results but after symmetry-based grasp generation has been applied. The results are less clear with these baseline methods, with symmetry-based grasp generation improving upon AnyGrasp baselines. In particular, the shortest path lengths at very low planning times are very similar for many methods, with MALab-Grasp not displaying a noticeable improvement.

When considering the remainder of scenarios in Appendix A, more outliers are present than when considering the AnyGrasp baselines. Many of these are already discussed in previous results and are due to baselines identifying lower path lengths than MALab-Grasp or exhibiting lower TTVG. In the situations at which all methods had similar TTVGs and lowest total path lengths, MALab-Grasp displayed a similar curve to the *Post-Symmetry Random* baseline, with the path lengths of both methods decreasing by similar rates. Two examples of this are the (*round-bottom, simple environment*) and the (*measuring cylinder, table environment*). In the former,

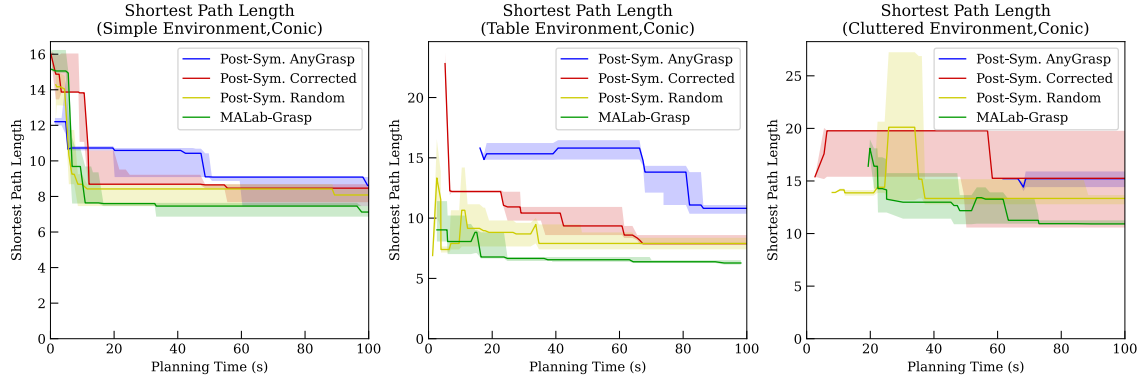


Figure 10.10: The time-evolution of shortest path length for Post-Symmetry AnyGrasp and MALab-Grasp for the conic flask scenarios. The median results over five trials are reported with the shaded regions displaying the interquartile ranges.

MALab-Grasp displayed better anytime performance than the baselines; in the latter, it performed worse.

To provide a better quantification of the anytime performance, the shortest path lengths are recorded at three discrete times of 10s, 50s, and 100s, shown in Fig. 10.11. When compared to AnyGrasp, MALab-Grasp shows better anytime performance in the simple and table environment, identifying shorter path lengths irrespective of planning time. In the cluttered environment, it also performs well for short and long planning times, but is significantly worse when given an intermediate (50s) planning time. The trends are less clear when comparing to *Post-Symmetry AnyGrasp*, with large statistical deviations for the simple and cluttered environments. However, MALab-Grasp does show better anytime performance in the table environment. A notable trend in this environment is that MALab-Grasp shows a more significant improvement as planning time shortens.

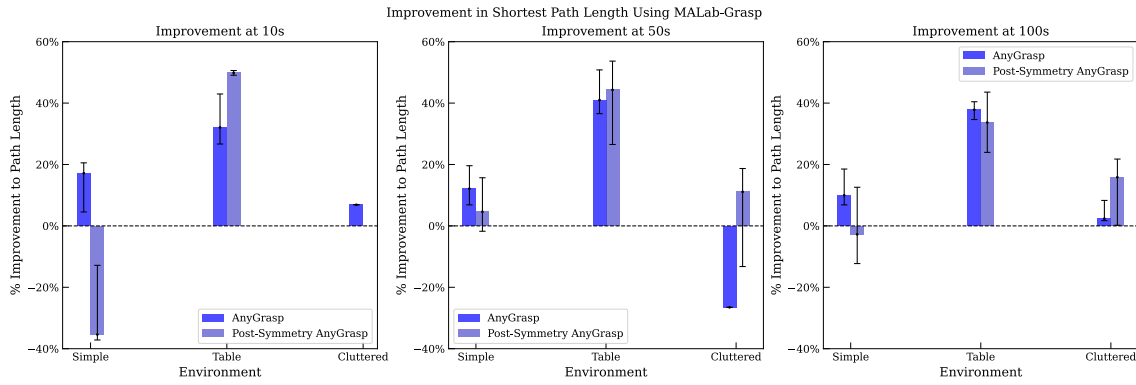


Figure 10.11: The median percent improvement in shortest path length for three discrete planning times, aggregated over all target objects within a single environment. The percent improvement is reported as the degree to which using MALab-Grasp reduces the shortest path length when compared to each baseline method. The table environment shows the most significant improvement, with MALab-Grasp reducing the shortest path length by over 35% (AnyGrasp) or 15% (AnyGrasp Random).

10.2.5 Exploration of Grasp Space

To assess each method’s exploration of the grasp space, the radial (vertical height) variance of the evaluated grasp set was recorded and displayed as a function of planning time. This metric provides an indication of the diversity of grasps which are being evaluated. Fig. 10.12 and Fig. 10.13 display these variances for the conic flask testcase in each environment. Results for other objects are reported in Appendix A. In all trials, MALab-Grasp displays high radial variance in the initial stages of grasp selection, consistently followed by a decrease in variance as planning time progresses. This is in direct contrast to AnyGrasp, which in almost all cases displays significantly lower radial variance in the initial stages of planning, with a slow growth over time. Randomly ordered baselines typically show a variance better than AnyGrasp but worse than MALab-Grasp. *Post-Symmetry Corrected* shows the least predictable behaviour, sometimes performing commensurate with MALab-Grasp, but in most scenarios possesses lower initial radial variance than MALab-Grasp. One significant outlier is noted in Appendix A: AnyGrasp performs equally to MALab-Grasp for the volumetric flask.

Similar experiments were performed with the angular (yaw) variance in the projected state space; however, since all methods displayed similar behaviour with no clear trend, these were not reported.

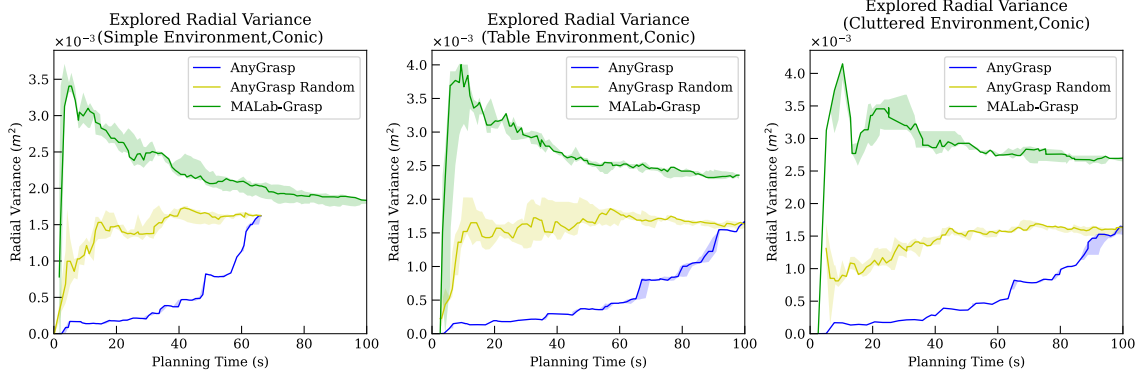


Figure 10.12: The time evolution of radial variance for AnyGrasp, AnyGrasp Random, and MALab-Grasp in the conic flask scenarios. Results depict the median variance over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp promotes a higher variance throughout the planning process, especially towards the start of planning.

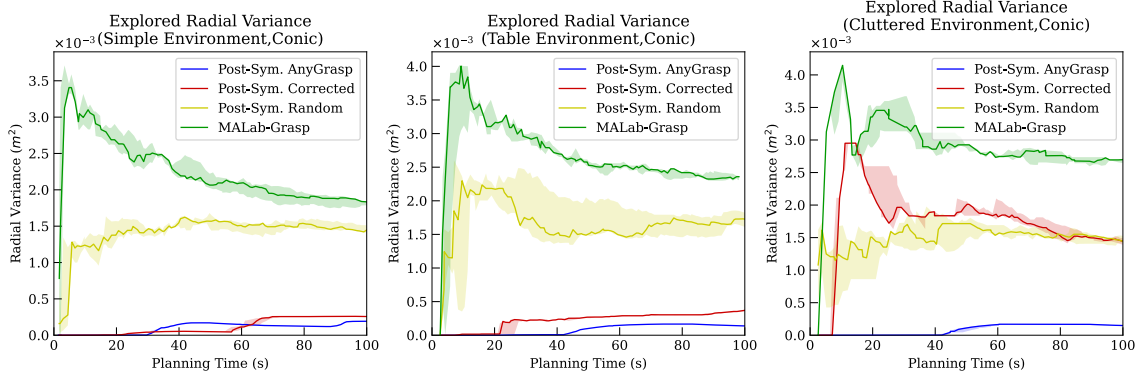


Figure 10.13: The time evolution of radial variance for MALabGrasp and Post-Symmetry AnyGrasp, Random and Corrected in the conic flask scenarios. Results depict the median variance over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp promotes a higher variance throughout the planning process, especially towards the start of planning.

10.2.6 Identification of Feasible Regions

The ability for MALab-Grasp to identify feasible and infeasible regions of the grasp space was quantified by recording the proportion of successfully evaluated grasps (those identified as valid). Fig. 10.14 and Fig. 10.15 show these results over the planning time duration for the conic flask in each environment, with the results for all other objects contained in Appendix A. MALab-Grasp outperformed the AnyGrasp in the large majority of testcases, sampling a higher proportion of valid grasps. In some simple cases – (*volumetric, simple*), (*volumetric, table*) and (*measuring cylinder, simple*) – this was not the case, with AnyGrasp able to identify more valid grasps. The *Post-Symmetry Corrected* shows no clear relationship with the MALab-Grasp,

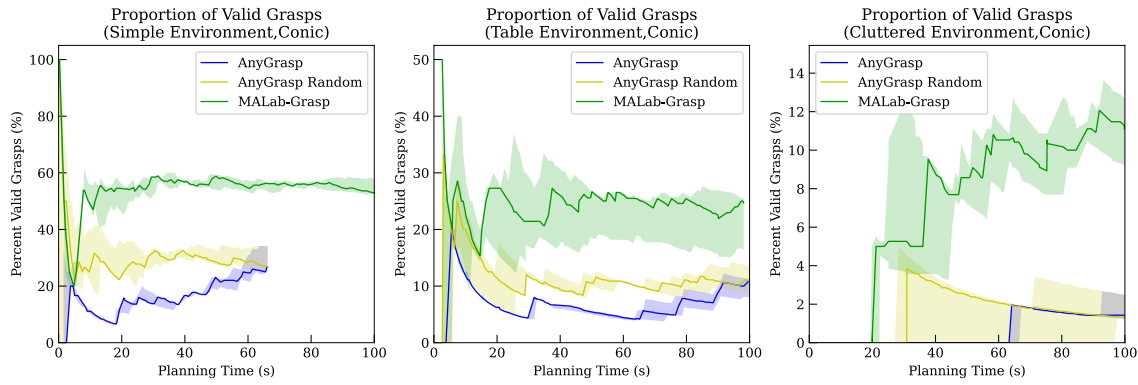


Figure 10.14: The proportion of evaluated grasps found to be valid using MALab-Grasp compared to AnyGrasp and AnyGrasp Random in the conic flask scenarios. Results depict the median proportion over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp samples valid grasps more frequently than the baseline methods.

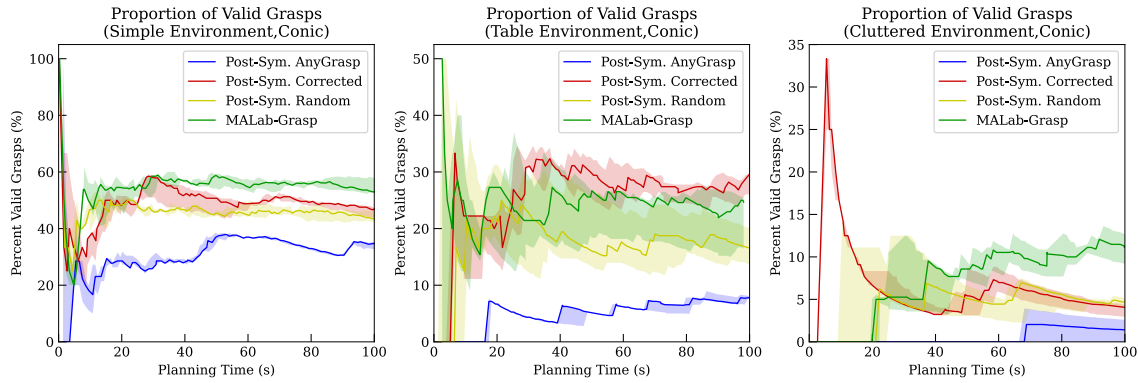


Figure 10.15: The proportion of evaluated grasps found to be valid using MALab-Grasp compared to Post-Symmetry AnyGrasp, Corrected and Random in the conic flask scenarios. Results depict the median proportion over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp samples valid grasps more frequently than the baseline methods.

with both outperforming the other in certain scenarios.

10.3 Study 3 - Adaptive Evaluation Time

Fig. 10.16 shows the effect that changing initialisations and values of n has on the performance of MALab-Grasp in the (*beaker large, simple environment*) scenario. Although only showing one metric in a single scenario, the results show no statistically significant difference between any of the test cases. Therefore, these results suggest that MALab-Grasp identifies the same shortest path lengths irrespective of how much time is initially allocated for trajectory planning.

To support the findings in this study, Fig. 10.17 shows the average time taken to

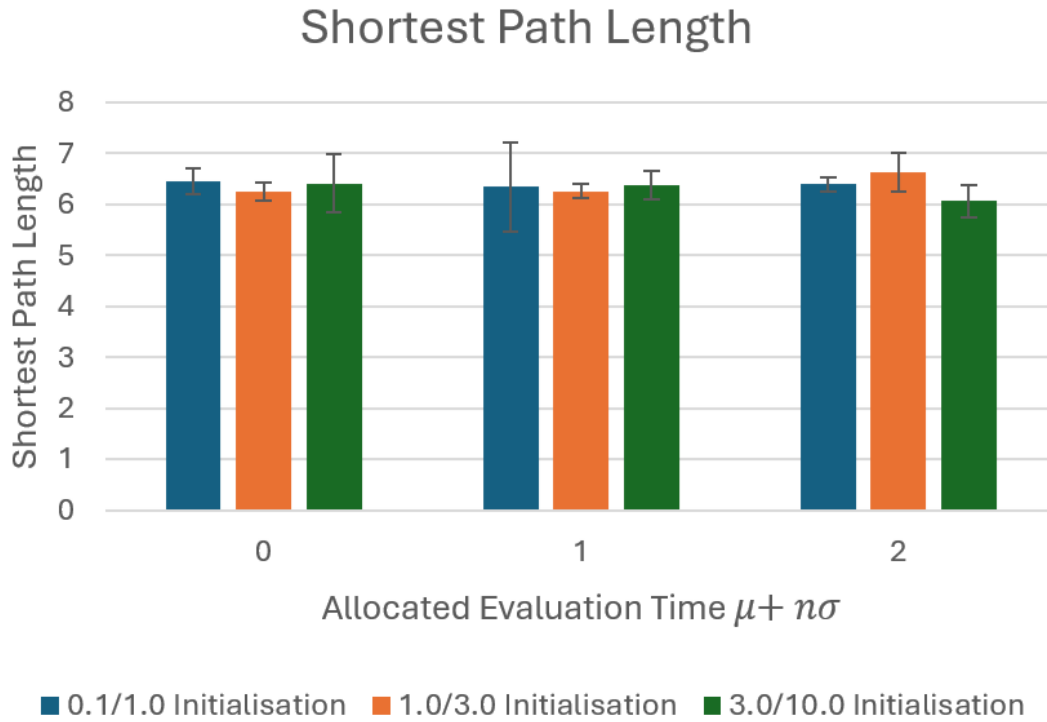


Figure 10.16: A comparison of the shortest path length found for different evaluation time initialisations and values of n . The median path length is reported over ten trials in the (beaker large, simple environment) scenario, with error bars showing the interquartile ranges. Initialisations are notated by the default allocated time for pickup and move (i.e., 0.1/1.0 indicates 0.1s allocated for pickup trajectory planning and 1.0s allocated for move trajectory planning). The results show no statistical significance in changing initialisations or values of n .

evaluate a grasp in each scenario from Study 2. In general, MALab-Grasp shows a substantially lower evaluation time in most scenarios. Three outliers exist to this trend: (*beaker med, simple environment*), (*beaker med, table environment*), and (*round-bottom flask, cluttered environment*).

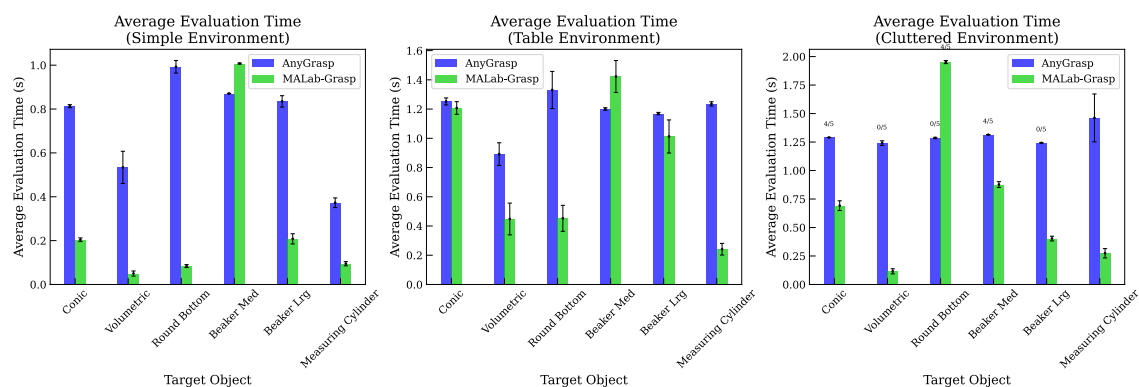


Figure 10.17: The average evaluation time of MALab-Grasp and AnyGrasp (taken from Study 2). Error bars show variance over five trials and bars are annotated in any scenario where a valid grasp could not be identified in all five trials.

Chapter 11

Discussion

This section conducts an analysis of the key findings of this thesis, limitations of this work and directions for further research.

11.1 Key Findings

The following sections present an analysis of the key findings collected in Chapter 10.

11.1.1 MALab-Grasp Improves Grasp Success Rate

The results of Study 1 show that AnyGrasp struggles significantly with stably and successfully grasping labware. During their benchmarking, AnyGrasp reports an average success rate of 93.3% for top-down grasping, with this reducing to 70% for adversarial (smooth-surfaced) objects [40]. Despite this, AnyGrasp was only able to successfully grasp labware in this study 33% of the time. Two causes for this decrease in performance are recognised. First, improper scoring of grasps prevents stable grasps from being correctly chosen. AnyGrasp stability metrics assume objects are transported with the gripper in a downfacing position, but this fails to translate to labware grasping where the gripper remains side on. In a similar manner, AnyGrasp does not incorporate the surface slantedness into its metrics, failing to identify the difficulty of this. Second, AnyGrasp chooses the depth of a grasp from one of five discrete values. This is effective in top-down grasping when depth has minimal impact of grasp quality. However, with the side-on nature of labware grasping, depth becomes a key factor in the stability of a grasp. In labware grasping, the desired depth for greatest stability (i.e., that which positions the grasp point at the centre of mass) may lie between two discrete options. This is seen mostly clearly in the

large beaker, where a grasp depth surpassing the maximum (40mm) was required. In this case, all grasping attempts grasped the slanted side of the beaker rather than the centre, causing grasping failure.

In comparison, MALab-Grasp improved grasp success rates dramatically, with an overall success rate of 63% shown in Fig. 10.1. Furthermore, successful grasps were shown to be more stable, exhibiting less swinging or slipping than when using AnyGrasp. When at least one successful grasp was found by AnyGrasp, MALab-Grasp performed perfectly out of 15 trials. This indicates that, provided AnyGrasp can synthesise at least one stable grasp, MALab-Grasp can identify it and assign it a high grasp score. This validates that MALab-Grasp fulfils Objective 2.

The primary difference between MALab-Grasp and AnyGrasp is that MALab-Grasp grasped objects around the centre of mass. Not only did this increase the stability of chosen grasps, but due to the geometry of labware it also prevented grasping on the slanted sides of labware. This is also likely aided by the symmetry-based grasp generation, which is able to duplicate grasps while retaining their stability, therefore turning one stable grasp into many. This is not a flaw in the measurement process, as each duplicated grasp offers a distinct candidate for grasping.

However, the pipeline still struggled with the same objects as AnyGrasp. In particular, it failed to successfully grasp the large beaker in any case. The likely cause of this is AnyGrasp limiting its grasp depth to 40mm, which is insufficient to grasp the beaker around its centre of mass. A similar phenomenon is seen in the medium beaker and round-bottom flask to a lesser extent, with many grasps successfully grasping the object but dropping it throughout the trajectory due to instabilities. This demonstrates that, although the grasping pipeline can score stable grasps highly, they are unable to synthesise stable grasps in the case where AnyGrasp does not identify any. This may suggest the performance of MALab-Grasp in identifying stable grasps may be ultimately limited by AnyGrasp. In light of this, more investigation into labware-specific grasp synthesisers is warranted. One compelling reason for this is that grasp sampling can directly take into account the side-on constraints of labware grasping without needing a separate filtering step.

When considering the nature of failed grasps being *dropped* or *missed*, AnyGrasp showed a greater tendency to miss objects completely, while MALab-Grasp was more frequent in successful grasping objects but dropping them in flight. However, the reason for these two modes of failure is suspected to be due to the geometry labware, rather than one mode of failure being more egregious than the other. Observations of the grasping process indicated that *dropped* grasps occurred typically when attempting to grasp a flask on a downwards-slanted surface. In this case, the

closing force of the gripper pushes the object into object down. This seemingly generates enough friction to prevent the flask from being pushed away from the gripper. On the other hand, *missed* grasps typically were caused by grasping a sideways-slanted or upwards-slanted surface, such as the side of a beaker or underside of a round-bottom flask. In this case, the closing force pushes the object slightly upwards, counteracting any friction forces. Thus, in these cases the objects are more likely to be pushed away when being grasped.

The results in Table 10.1 support this, with beakers typically causing *missed* grasps and flasks typically causing *dropped* grasps. The most significant difference between MALab-Grasp and AnyGrasp is in the round-bottom flask, with MALab-Grasp failing due to dropping and AnyGrasp failing due to missing. This suggests that MALab-Grasp is attempting to grasp from the upper half of the spherical base and AnyGrasp is attempting to grasp from the lower half. This aligns with the expected way that both methods score grasps. Ultimately, this mode of failure does not provide any strong indication that MALab-Grasp is failing in a better way than AnyGrasp.

11.1.2 MALab-Grasp Identifies a Valid Grasp in More Scenarios

The results throughout Study 2 indicate that MALab-Grasp significantly improves upon AnyGrasp’s ability to identify trajectory-valid grasps in adversarial scenarios. As evidenced by Table 10.2, AnyGrasp struggles to identify valid grasps in the cluttered environment, wherein there are large regions of infeasibility caused by surrounding objects preventing the robot from grasping at a particular angle. The underlying reason for this lack of performance is attributed to AnyGrasp failing to achieve coverage of the entire grasp space during grasp synthesis, especially due to the additional flatness constraints applied to the grasp set. This is supported by the findings in the preliminary studies in Chapter 6.

These results for AnyGrasp contrast significantly with those presented by Fang et. al [40] when benchmarking the algorithm. These benchmarks are wholly top-down problems, wherein objects are grasped from above in a pick-and-place style manner as in Fig. 3.3. Infeasible grasps in these top-down problems are wholly caused by gripper collisions with surrounding obstacles, for which AnyGrasp visually detects and avoids during grasp synthesis. This yields a candidate grasp set which is predominately feasible. Comparatively, infeasibilities in the side-on grasping problem are much more varied, often coming from joint limits, collisions throughout the entire robot arm, or cluttering along the entire trajectory rather than solely the grasping point. Given that AnyGrasp cannot encode any of these factors during grasp synthesis, the

grasp set in these scenarios may be highly infeasible, especially in cluttered scenes. Ultimately, the benchmarking studies of AnyGrasp are specific to top-down grasping and not reflective of the complexities of side-on labware grasping.

Table 10.2 shows that MALab-Grasp identifies valid grasps in cluttered environments for 96.7% of cases, which is dramatically higher than AnyGrasp’s 43.3%. Symmetry-based grasp generation played a significant role in this, with this step increasing AnyGrasp’s identification of valid grasps from 43.3% of trials to 83.3%. This provides an indication that this step effectively increases the coverage of the grasp space, filling the regions in which AnyGrasp did not generate a flat grasp. It also indicates that the symmetry-based method generates grasps which are both feasible and unique from the existing grasp set. This is significant because it acts as a purely geometric post-processing step, which is highly time-efficient and not dependent on the underlying grasp synthesiser used. Therefore, by exploiting the symmetric nature of labware, it provides a means by which state-of-the-art grasp synthesiser can be tailored towards laboratories. This prevents the need for developing a domain-specific grasp synthesis algorithm and allows laboratories to leverage new grasping algorithms as they emerge. Furthermore, it may be possible to leverage this same symmetry-based grasp generation to create complete grasp sets from a single-view point cloud by rotating grasps about the labware’s centre. This would offer sizable benefit as systems would not need to dedicate time before grasping for point cloud generation.

The remaining 13.3% of cases in which MALab-Grasp identifies a valid grasp, but the symmetry baselines cannot, all come from the (*round-bottom flask, cluttered environment*) scenario. In this scenario, MALab-Grasp was the only method which could identify a trajectory-valid grasp. Given that symmetry baselines evaluate every grasp that MALab-Grasp does, this performance improvement is attributed to the adaptive evaluation time algorithm. This is discussed in further sections.

11.1.3 MALab-Grasp Identifies Valid Grasps Faster than AnyGrasp

In addition to identifying valid grasps in more scenarios, MALab-Grasp decreases the time taken to identify a valid grasp (TTVG), especially in the cluttered and table environment, when compared to randomly ordered baselines. The application of symmetry-based grasp generation is not a significant factor in this, with no clear trend shown between the performances of the *AnyGrasp Random* and *Post-Symmetry Random* baseline methods. Instead, this indicates that the $\mathcal{GP}$ -based sampling process of MALab-Grasp is effective in choosing an ordering in which to evaluate

grasps.

The cause of this improvement is attributed to two mechanisms. First, MALab-Grasp leverages the acquisition function of $\mathcal{GP}$ to sample grasps which maximise the variance of the evaluated grasp set $\mathcal{G}_{eval}$, especially in the initial stages of grasp planning. This is evidenced in Fig. 10.12 and Fig. 10.13, which show that MALab-Grasp exhibits significantly higher radial variance than all baselines. As a result, it explores the grasp space as maximally as possible, increasing the likelihood of sampling in a feasible region. In contrast, randomly ordered baselines do not explicitly maximise exploration. If a region of the grasp space contains more grasps, the random baselines may oversample these regions due to sampling density biases. If these dense regions correspond also to infeasible regions, then the random baselines spend significantly more time in these regions than MALab-Grasp does. This is most clearly the case for the conic flask, where preliminary studies found that a large portion of grasps were on the base of the flask – the majority of which is infeasible due to the robot colliding with the ground plane. Accordingly, the conic flask shows a significant worsening in TTVG when using random baselines compared to MALab-Grasp.

Second, MALab-Grasp leverages the estimated path lengths computed via $\mathcal{GP}$ to learn infeasible regions of the grasp space. Subsequently, the exploitation term of the acquisition function penalises regions with high rates of infeasibility, therefore leading to lower sampling in these regions. The efficacy of this method is shown in Fig. 10.14 and Fig. 10.15, which reports that MALab-Grasp samples more feasible points than random ordering. In the early stages of planning (before a valid grasp has been found), this manifests itself as avoiding regions which have been sampled multiple times or those who have been given elevated evaluation time and have still failed to find a feasible grasp. These two mechanisms have greater influence when the grasp space contains large regions of infeasibility. Accordingly, Fig. 10.5 show that the largest improvement in TTVG comes from the cluttered environment and table environment when compared to *Post-Symmetry Random* and *AnyGrasp Random* respectively. This is shown in Table 10.2, acknowledging that the limited number of target objects used leads to relatively large and overlapping interquartile ranges.

These results provide an indication that GP-based grasp ordering can identify valid grasps more efficiently than random ordering of grasps, especially in environments where large regions of infeasibility exist. This provides significance beyond laboratory environments and into any grasping problem with complex environments where finding a feasible grasp is not trivial, even if optimising the path length is not a requirement. In addition, GP-based methods can be implemented independently of grasp synthesis and, unlike existing methods, do not rely on online grasp synthesis,

deep neural architectures, or knowledge of the robot’s kinematics. Therefore, this novel approach to feasibility detection shows great promise and prompts further research.

11.1.4 Grasp Score Orderings are Biased Baselines

In this chapter, time-dependent metrics for the performance of MALab-Grasp are benchmarked against baselines in which grasps are evaluated in random order, rather than in order of AnyGrasp grasp score. This is because the findings indicate that ordering grasps by AnyGrasp grasp score presents a worse-than-random performance in terms of TTVG. The primary cause of this is likely the nature in which AnyGrasp applies its stability score. Within the laboratory grasping problem, the effect of this is to promote (increase the relative grasp score) grasps which lie close to the vertical centre of mass and penalise those which lie further away. As discussed in previous sections, this does not accurately reflect the notion of stability with the laboratory grasping domain. However, when ordering grasps according to this score, it introduces two key effects. First, it implicitly encodes an ordering that is dependent on the vertical position of the grasp. Therefore, the radial (vertical) variance of evaluated grasps grows more slowly than a random ordering. This is supported by Appendix A, which shows that the AnyGrasp-ordered baselines exhibit much lower variance than random baselines in almost all cases. Second, grasps close to the object centre of mass are promoted in the grasp ordering. For many of the objects – particularly the three flasks – this region is highly infeasible, overlapping with the region in which the robot cannot reach without colliding with the ground plane. As a result, this infeasible region will likely be oversampled, leading to inefficiencies in the grasp selection process. This is supported by A, which shows that AnyGrasp-ordered baselines identify a lower proportion of valid grasps than randomly ordered baselines. Outliers to these two trends exist because the stability score is only one of two factors within the AnyGrasp grasp score; the other factor represents graspness and other metrics of grasp quality.

The culmination of this is that ordering grasps by AnyGrasp score is detrimental to the efficiency of grasp selection, despite it being a natural ordering to use. Fig. 10.5 captures this well, with the *AnyGrasp* baseline performing significantly worse than its random counterpart. Furthermore, the introduction of symmetry-based grasp generation amplifies this effect. *Post-Symmetry AnyGrasp* performs the worst out of all baseline methods, taking on average 80% longer than MALab-Grasp. Because AnyGrasp grasp score is invariant under rotation for symmetric objects, each generated grasp is assigned the same grasp score as its originator. Therefore, an AnyGrasp

grasp score-based ordering will evaluate each symmetric grasp in sequence. Since all symmetric grasps have the same vertical position, this will exacerbate the existing issues of low radial variance and oversampling near the centre of mass.

On the other hand, ordering grasps via the corrected grasp score provided unexpectedly high performance, often performing better than random ordering. In 44% of trials, it also outperformed the MALab-Grasp with regards to the TTVG and displayed near-equivalent performance for the table environment. The reasons for this are suspected to be the accidental alignment between infeasible regions and unstable regions as dictated by the corrected grasp score. The corrected grasp score heavily penalises grasps which lie below the centre of mass, as these form unstable equilibria during manipulation. As a result, these grasps are undersampled. However, this also aligns with the large region of infeasibility corresponding to the robot intersecting with the ground plane. Correspondingly, as seen in A, *Post-Symmetry Corrected* accidentally samples a higher proportion of valid grasps than both the randomly ordered baselines and MALab-Grasp. Unlike *Post-Symmetry Corrected*, MALab-Grasp must first sample from this region to learn that it is infeasible, leading to comparatively slower results in some cases.

Although corrected grasp score ordering does display good performance, it lacks generalisability because it relies on properties of the environment and labware that are not guaranteed to always be true. In some cases, labware may be elevated on pedestals, removing the effects of the ground plane. In other cases, surrounding clutter may form an infeasible region around the top of labware, which corrected grasp scoring heavily promotes. Therefore, these ordering methods are not recommended for robust applications.

11.1.5 MALab-Grasp Identifies Grasps with Shorter Path Lengths

A key feature of MALab-Grasp is that it identifies optimal grasps with shorter path lengths than AnyGrasp. Fig. 10.8 shows the improvement that MALab-Grasp provides in each environment, with the table environment showing the largest median improvement over AnyGrasp by approximately 35%. Given the large interquartile ranges, there is only enough statistical evidence to suggest that MALab-Grasp performs better in table and cluttered environment. This also suggests that MALab-Grasp only identifies more optimal grasps than AnyGrasp in complex environments with large regions of infeasibility.

Symmetry-based grasp generation again plays a significant role in these findings,

accounting for the majority of improvement when compared to AnyGrasp. This also justifies the lack of improvement in the simple environment, where symmetry generation is less necessary. These findings further validates that symmetry generation identifies high-quality grasps.

The comparison between MALab-Grasp and *Post-Symmetry AnyGrasp* shows that the inclusion of GP-based grasp selection also further reduced the optimal path length found on average, although to a lesser extent than symmetry-based grasp generation. Of all three environments, only the table environment displays a statistically significant improvement. The cause of this improvement in optimal grasp is attributed to the adaptive planning algorithm, analysed in later sections.

11.1.6 MALab-Grasp Does Not Exhibit Meaningful Anytime Performance Increase

Leading into this experimentation, it was hypothesised that MALab-Grasp would present better anytime performance than the baseline methods. This was grounded in the expectation that a $\mathcal{GP}$ -based sampling method would be able to accurately estimate the path lengths of grasps yet to be evaluated and use this to bias sampling towards more optimal grasps. Such an effect would be seen in the time progressions in A, with MALab-Grasp expected to depict a lower area under curve (AUC) than randomly ordered methods. In some scenarios, MALab-Grasp shows a smaller AUC due to either identifying a valid grasp faster or identifying a more optimal grasp. The table environment showed the desired performance the most frequently, with the *beaker medium* and *measuring cylinder* objects showing better anytime performance with MALab-Grasp than random ordering. In these scenarios, both methods identified a valid grasp at similar times and found a similar optimal grasp, but curve for MALab-Grasp decreased much faster. This may indicate that in these scenarios MALab-Grasp identifies regions of high path efficiency. However, this trend is not consistent across all trials. In scenarios such as (*beaker med, simple environment*) and (*measuring cylinder, cluttered environment*), the anytime performance of MALab-Grasp and *Post-Symmetry Random* is largely indistinguishable from one another.

These results are quantified in Fig. 10.11, which shows the shortest path length at the three discrete times of 10s, 50s and 100s. The table environment shows a statistically significant improvement when using $\mathcal{GP}$ -based sampling at all three times, with the performance increase being greater at lower times. However, this trend is not replicated with the other environments.

The suspected cause for $\mathcal{GP}$ -based selection not improving the simple environment

lies within the simplicity of the scenario, with it actually making the anytime performance worse. Without obstacles to obfuscate the trajectories of grasps, the path lengths for each grasp are likely to be homogeneous. This may cause very little variation in path length with the nondeterministic effects of RRTConnect dominating.

On the other hand, $\mathcal{GP}$ -based selection not improving the cluttered environment indicates that MALab-Grasp struggled to accurately estimate the path lengths of unevaluated grasps. Several factors are believed to be at play here. First, the inherent randomness of trajectory planning may make the grasp space landscape non-smooth and highly irregular, making it difficult for the $\mathcal{GP}$ to learn an accurate map of path lengths. Second, throughout the planning process it was observed that many infeasible grasps were found even in feasible regions of the grasp space. This indicates that the pose of the grasp in the projected grasp space (z, ψ) does not fully define the feasibility of a grasp, challenging the assumption that feasibility is a smooth function in this projected grasp space. As a result, single infeasibilities may corrupt the estimated path length of a given region. Finally, it is also possible that the path length also does not form a smooth function in this space. Ultimately, more studies are needed to determine the exact cause of this, but there is a strong indication that a 2-dimensional GP cannot fully capture the complexities of the system.

11.1.7 The Adaptive Evaluation Time Algorithm Provides Indications of Success

While acknowledging that Study 3 was not as comprehensive as others, the results provide an indication that MALab-Grasp is able to effectively identify the underlying planning distributions of the trajectory planning problems. Fig. 10.16 suggests that MALab-Grasp is invariant to the initial trajectory planning times allocated, providing the same performance with regards to shortest path length irrespective of very high or very low initial evaluation times. This would be a significant result because it would remove the need for MALab-Grasp to be carefully tuned as the complexity of a given laboratory environment evolves. However, studies with a greater range of scenarios and reporting a more comprehensive set of metrics are needed to validate this.

The results from Study 2 also support the hypothesis that the adaptive time allocation algorithm is effective. The presence of this algorithm was the only factor leading to finding feasible grasps in the *(round-bottom flask, cluttered environment)* scenario. In this scenario, it is likely that the trajectory planning times manually tuned were not sufficient to identify a grasp, as it is one of the most complex

environments. This is evidence to show that MALab-Grasp is able to identify when these trajectory planning times are too low and increase them accordingly. This is supported by Fig. 10.17, which shows that MALab-Grasp increased its average planning time dramatically in this scenario.

The same figure also shows the converse: in less complex environments, MALab-Grasp is able to decrease the evaluation time allocated to improve the time-efficiency of the system. In 77.8% of scenarios, it showed a statistically significant decrease in the average evaluation time. This is suspected to be due MALab-Grasp decreasing allocated evaluation times upon detecting that they had initially been overallocated. However, there are other factors affecting average evaluation time – such as proportion of valid grasps – that makes this impossible to definitively prove. Ultimately, more studies are required to determine the full effect that the adaptive evaluation time algorithm has on the performance of MALab-Grasp.

11.2 Limitations and Further Work

11.2.1 Nondeterministic Planning

Though MALab-Grasp is intended to be compatible with arbitrary trajectory planners, the use of RRTConnect introduces substantial nondeterminism which may influence both the results of studies and the performance of MALab-Grasp. This high degree of randomness is evident throughout the results in Section 10.2, both in the path lengths reported by grasp evaluation and the time taken to execute. Baseline methods display large interquartile ranges despite ordering their grasps in identical fashion. In the extreme case of this, the baseline methods in a number of trials failed to plan a valid grasp in some instances, yet were able to in others. This randomness was mitigated by reporting median values over five instances for each trial, to achieve statistically significant results. In further work, the results collected in this thesis should be validated with a higher number of instances per trial.

In addition to introducing high variation in the data collected, the nondeterminism may directly influence the grasp selection process. Because RRTConnect does not perform any path refinement, grasp evaluation computes path lengths on trajectories which are not necessarily optimal. As a result of this, grasps which are nearly identical and possess nearly identical optimal trajectories may be reported to have vastly different path lengths. This serves to decrease the smoothness of the grasp search space, thereby making it more difficult for the GP to compare the path efficiency of different regions and optimise the path efficiency. These effects are

shown in Section 10.2. In particular, this effect is likely exacerbated in complex environments, where paths may be longer therefore differences in path lengths are exaggerated. Future work should apply path refinement strategies such as STOMP in light of this, with MALab-Grasp predicted to be more effective in this case.

11.2.2 Relative Grasp Scoring

A key limitation of the grasp scoring method used in both AnyGrasp and MALab-Grasp is its relative nature. In both cases, stability scores are applied relative to the most and/or least stable grasp. Although this preserves the ordering of grasps, safety-critical or delicate applications – such as transporting glassware – would benefit from a grasp scoring which provides a definitive binary measure for whether a grasp would succeed or not. This is best shown in the large beaker case, where no stable grasps existed but MALab-Grasp was not able to recognise this. This situation could lead to a multitude of potential hazardous scenarios when applied to real-world scenarios.

Therefore, an absolute grasp scoring metric which provides binary success is worthwhile future work – that is, one that can definitively say whether a grasp is successful or not. This is a nontrivial problem because grasp success is intricately linked to the type of gripper hardware used, such as its length and whether it is a rigid or a compliant gripper.

11.2.3 Combining Path Efficiency and Grasp Quality

In a similar manner, MALab-Grasp proposes both a method for identify high-quality grasps and path-efficient grasps. These two factors were combined via an objective function to determine the optimal grasp. However, this was not a strong focus of this thesis and these two factors were ultimately studied separately. Given that there is no clear-cut way to combine these, more research needs to be done on the best practices in these situations. This is especially important in chemical research laboratories, where low stability could have significant safety ramifications.

11.3 Alignment with Project Plan

This thesis was originally titled “*Automated Glassware Handling in Chemical Research Laboratories*” and proposed an end-to-end system for glassware handling. As such, the focus of this thesis was initially planned to be on the co-design of

transparency-aware vision systems, compliant grippers, and laboratory grasping. A two-stage project plan was initially proposed, with the first half of the thesis focussing on implementing a baseline system and the second half proposing an improved system. However, the project encountered unexpected roadblocks in integrating the components of the baseline system together. As a consequence at the midpoint of this thesis it was identified that the scope of the project was too large to make any meaningful contributions towards the current research base. Therefore, the project scope was narrowed to focus solely on grasping, discarding the planned work around transparency-aware vision systems and soft gripping. This proved a wise decision, allowing this thesis to be finalised on schedule and present meaningful results.

Chapter 12

Conclusions

This thesis proposed MALab-Grasp, a novel grasping pipeline which extends AnyGrasp to be tailored towards labware grasping in chemical research laboratories. The motivation for this arises from the lack of literature surrounding labware grasping, owing primarily due to the lack of standardisation in chemical labware. Simultaneously, current grasping algorithms do not capture the domain-specific constraints of laboratories.

First, MALab-Grasp performs a novel grasp scoring strategy, which was shown to significantly improve the stability labware grasping when compared to AnyGrasp. However, it remains limited in that it cannot synthesise new grasps beyond what AnyGrasp provides.

Second, MALab-Grasp addresses the high variation in grasp feasibility and path efficiency by presenting a new paradigm of grasp selection. Rather than relying on random ordering of grasps, a Gaussian process-based sampling strategy is employed. This approach identified feasible grasps faster, in a wider range of environments, and with more optimal path lengths than AnyGrasp. A key contributor in this performance increase is the use of symmetry-based grasp generation, which provides high coverage of the grasp space and shows potential in applications with only partial point clouds. The Gaussian process performed well in identifying infeasible regions but struggled in distinguishing between regions of high and low path efficiency. This may be due to factors influencing path efficiency and feasibility that are not well captured by solely the grasping pose.

Finally, MALab-Grasp learns the underlying distribution of trajectory planning times to autonomously tune the trajectory planner. Although more studies are needed to validate this, MALab-Grasp shows a strong indication that it was able to adapt its planning times to suit the complexity of the environment without needing

manual tuning.

In summary, MALab-Grasp successfully achieved all objectives that were outlined, and improves upon the performance of state-of-the-art grasping planners within the labware grasping domain. To continue advancing the state of automation within chemical research laboratories, further research validating these results, developing custom grasp planners and investigating grasp scoring metrics which can guarantee stability is recommended.

Appendix A

Time Evolution of All Scenarios

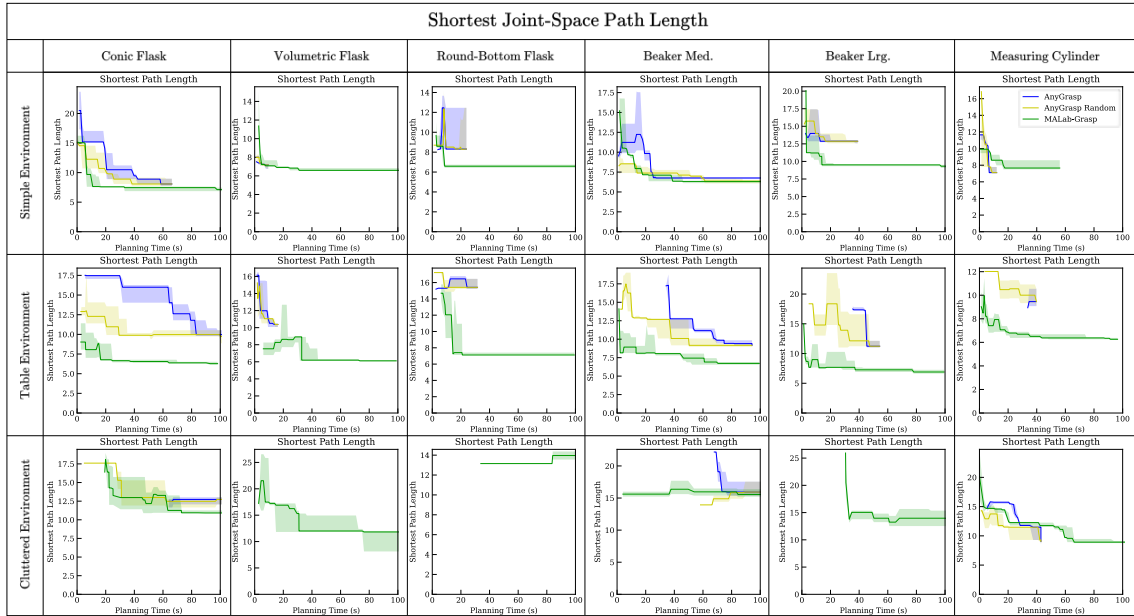


Figure A.1: The time evolution of shortest path length for the AnyGrasp, AnyGrasp Random, and MALab-Grasp in all scenarios. The median results over five trials are reported with the shaded regions displaying the interquartile ranges.

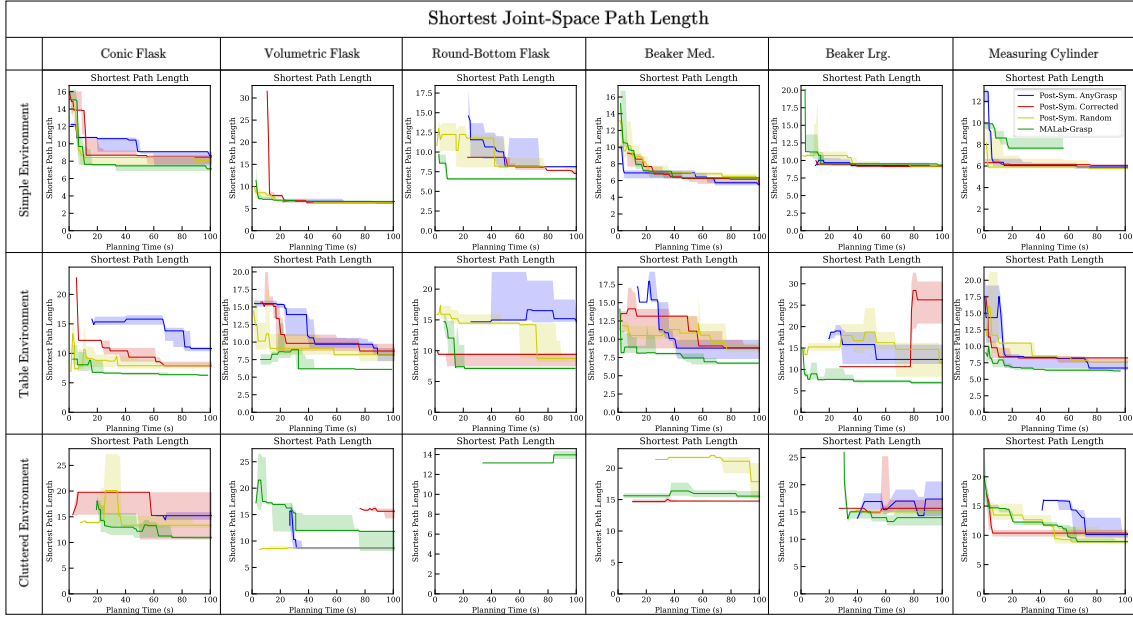


Figure A.2: The time evolution of shortest path length for Post-Symmetry AnyGrasp and MALab-Grasp in all scenarios. The median results over five trials are reported with the shaded regions displaying the interquartile ranges.

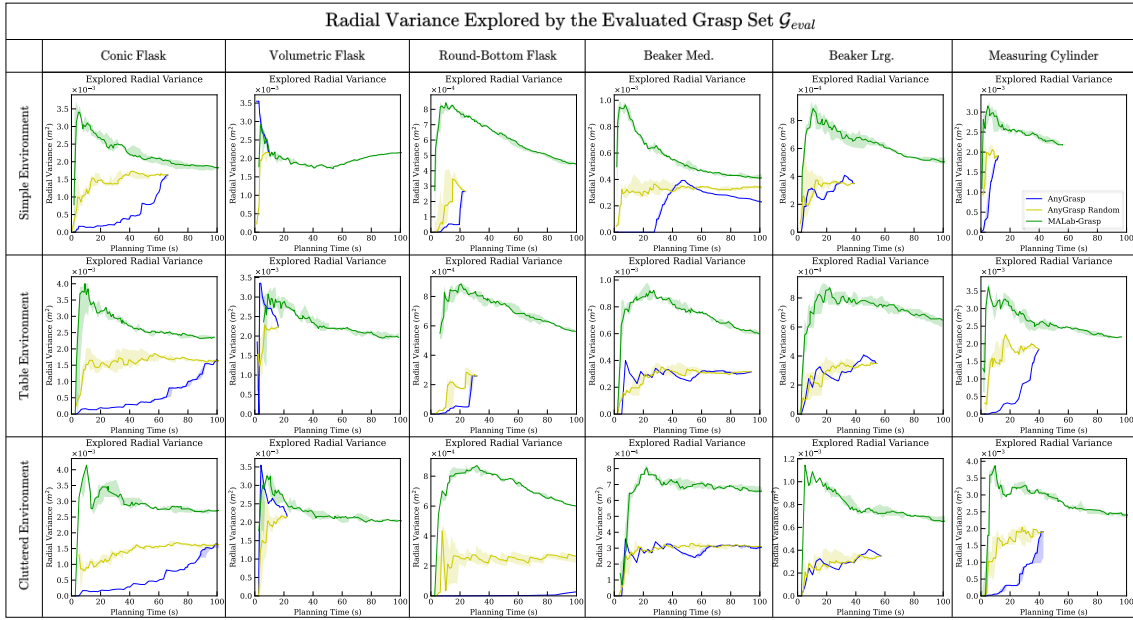


Figure A.3: The time evolution of radial variance for AnyGrasp, AnyGrasp Random, and MALab-Grasp in all scenarios. Results depict the median variance over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp promotes a higher variance throughout the planning process, especially towards the start of planning.

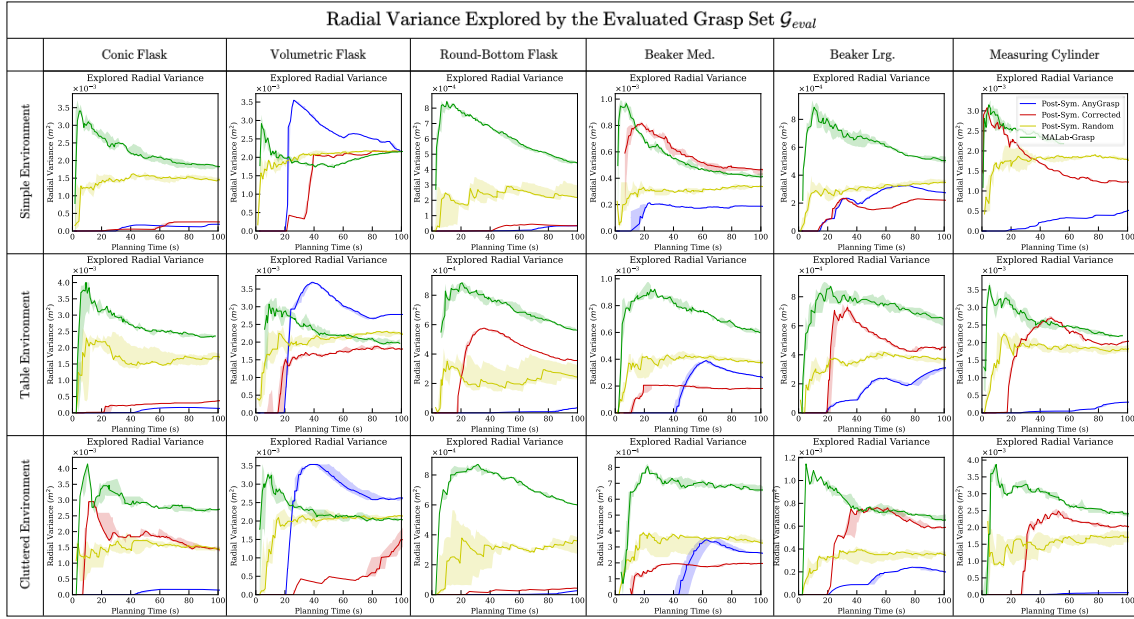


Figure A.4: The time evolution of radial variance for MALabGrasp and Post-Symmetry AnyGrasp, Random and Corrected in all scenarios. Results depict the median variance over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp promotes a higher variance throughout the planning process, especially towards the start of planning.

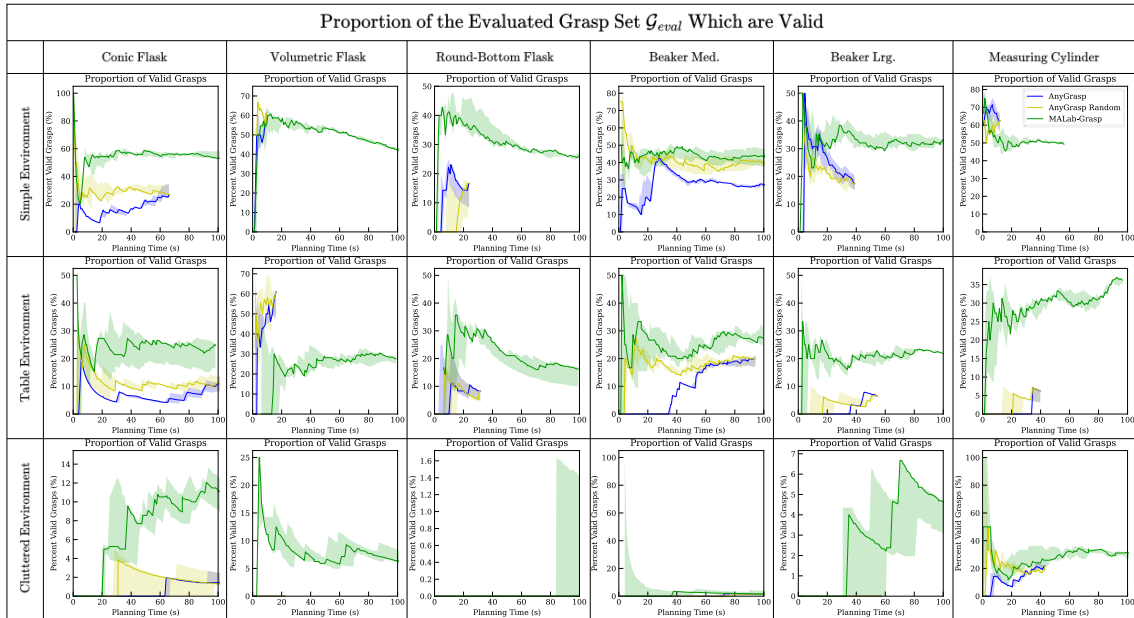


Figure A.5: The proportion of evaluated grasps found to be valid using MALab-Grasp compared to AnyGrasp and AnyGrasp Random in all scenarios. Results depict the median proportion over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp samples valid grasps more frequently than the baseline methods.

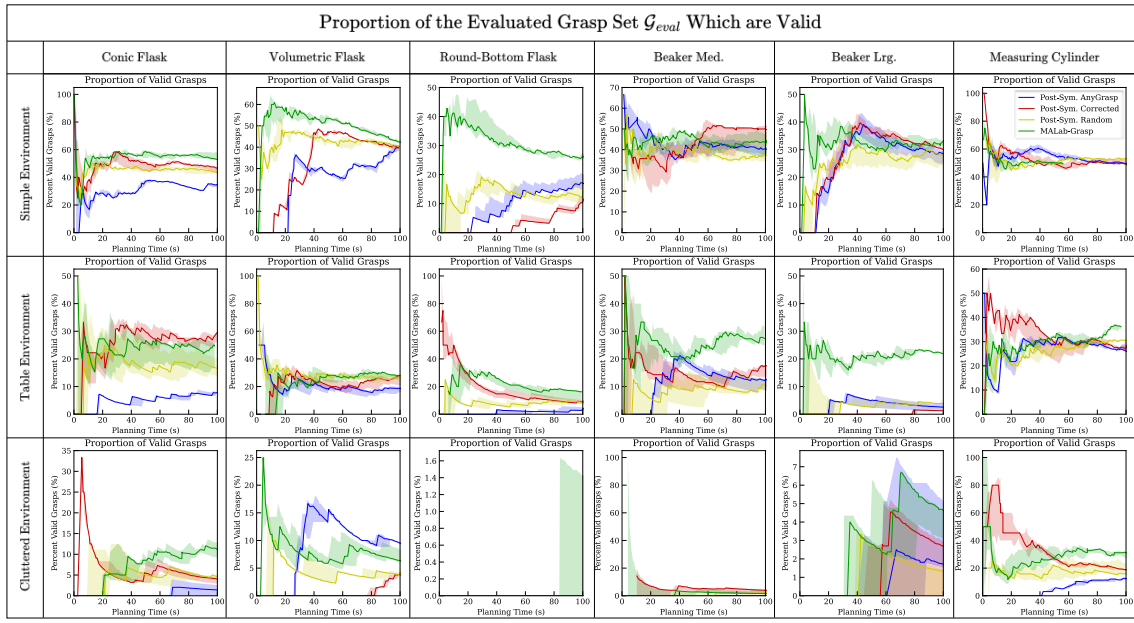


Figure A.6: The proportion of evaluated grasps found to be valid using MALab-Grasp compared to Post-Symmetry AnyGrasp, Corrected and Random in all scenarios. Results depict the median proportion over five trials, with the shaded regions depicting the interquartile ranges. MALab-Grasp samples valid grasps more frequently than the baseline methods.

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